A Comprehensive Survey of Machine Learning Applied to Resource Allocation in Wireless Communications

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***Abstract*—Telecommunications play a pivotal role in shaping today's interconnected world by fostering global development, supporting seamless information exchange across vast distances, and revolutionizing the interactions between individuals, businesses, and governments. Accessible and reliable communication networks transcend geographical barriers, promoting economic growth, the dissemination of knowledge, and societal connectivity. The integration of artificial intelligence into telecommunications has been transformative, revolutionizing the entire industry by improving system efficiency, allowing new services, and reducing complexity. By leveraging machine learning algorithms, telecommunication operators analyze vast data sets to gain insights into customer behavior, network performance, and market trends. This data-driven approach enhances service efficiency, leading to optimized network deployment, improved customer experience, and targeted marketing strategies. Machine learning's impact extends to resource allocation optimization. Intelligent management of network resources reduces latency, congestion, and downtime, ensuring enhanced user experiences and increased overall network capacity. This optimization is vital for integrating emerging technologies like the Internet of Things and future generations of mobile systems and promoting sustainability by reducing energy consumption, contributing to a greener future. As technology evolves, the synergy between telecommunications and artificial intelligence will pave the way for a more connected, intelligent, and prosperous future. Given the relevance of this research topic, this paper presents a comprehensive survey of machine learning techniques applied to resource allocation in wireless communication systems. The objective is to guide the scientific and industrial community in the optimized selection of machine learning techniques according to network demands and network resource allocation to be refined. Additionally, it aims to encourage an in-depth discussion regarding the limitations presented in the current literature and future challenges for researchers.**

***Index Terms*—Artificial intelligence, machine learning, optimization, resource allocation, wireless communications systems, telecommunications networks.**

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I. Introduction

TELECOMMUNICATIONS have transformed global connectivity, supporting a diverse range of essential services beyond traditional communication, including computing, sensing, and localization [1]. In advanced networks such as Fifth Generation of Wireless Communication Technology (5G) and the forthcoming Sixth Generation of Wireless Communication Technology (6G), this expanding functionality introduces new challenges in managing and allocating resources across multiple service domains, each with unique demands. However, communication remains a foundational layer among these services, requiring reliable, high-throughput, and low-latency data transmission to support real-time applications, such as Internet of Things (IoT) [2]. Consequently, efficient resource allocation in communication services is critical for optimizing both performance and user experience, making it a priority in modern network design.

Resource allocation optimization refers to strategically managing network resources, such as frequency, time, computational capacity, spatial distribution, and power, to maintain high-quality service while minimizing operational costs. By optimizing these resources in wireless communications, network operators can reduce congestion, minimize interference, and ensure consistent service quality, even as demand for bandwidth and throughput grows, which consequently enables the provision of other types of services in 5G and beyond networks with greater quality and efficiency [3], [4]. Given the projected increases in connected devices and data consumption, resource allocation optimization offers a sustainable approach to maximizing the potential of existing infrastructure, deferring the need for new investments and contributing to overall network efficiency and sustainability, consequently stimulating economic and social development [5].

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Recent advancements in Machine Learning (ML) provide powerful tools for tackling the complex resource allocation demands within telecommunications. ML enables adaptive, data-driven optimization that exceeds conventional methods by dynamically adjusting to real-time conditions and fluctuations in network demand [6]. With capabilities for processing large volumes of data and identifying complex patterns, ML-based approaches can allocate resources based on detailed insights into user behavior, geographic location, and environmental factors, improving network performance and operational efficiency. This approach represents a shift towards more intelligent and agile resource management that aligns with the evolving requirements of modern wireless networks. Hence, given the ongoing research into ML-based solutions for resource allocation optimization in wireless communications, characterized by increasingly sophisticated and advanced algorithms, alongside the evolution of traditional networks into more software-defined, programmable and decentralized architectures, the management and optimization of computational resources also become essential to elevate the quality of other services offered by the network.

A. Motivation

Resource allocation optimization has become increasingly relevant due to the unique challenges posed by the vast array of data transmission demands in 5G and anticipated 6G networks. While emerging networks also support additional services—such as sensing, and positioning—the allocation of resources within communication and computing services remains a primary concern. Specifically, advancements in communication performance metrics, such as bandwidth management, Spectral Efficiency (SE), and latency reduction, are vital for ensuring a reliable and high-quality user experience. Simultaneously, the increasing complexity of computational requirements, including the need to support advanced optimization algorithms and the transition to programmable and decentralized network architectures, highlights the necessity for efficient computational resource management.

By narrowing our analysis specifically to the optimization of communication and computational resources, rather than adopting a broader approach encompassing other service types and resources, we aim to provide a deeper and more detailed exploration of a targeted topic. This focused approach allows for a deeper technical discussion and a more consolidated understanding of the subject. Given the high stakes and fast-paced evolution of communication and computing services, a concentrated analysis allows researchers and industry professionals to apply ML solutions that address these pressing needs with targeted efficiency. While sensing and localization services are also critical, optimizing communication and computing resources often has a cascade effect, indirectly enhancing the capacity and performance of other services that rely on the foundational infrastructure of wireless networks. In summary, the focus driven by the critical need to address challenges in both communication and computational resource management becomes essential, as these are fundamental to the efficient operation of modern wireless networks and to supporting additional services.

ML has emerged as an indispensable tool in addressing these complex resource allocation challenges in communication services, evolving significantly over the past decade. Initially, ML approaches in resource allocation optimization were limited to basic prediction tasks. However, advancements in Deep Learning (DL), Reinforcement Learning (RL), and Federated Learning (FL) now enable more granular, real-time, and adaptive allocation strategies that surpass conventional optimization techniques in both precision and scalability. For instance, RL techniques allow networks to autonomously adapt resource allocation based on fluctuating network demands, geographical variations, and user mobility patterns. Additionally, DL models process vast amounts of data to detect and respond to nuanced patterns, making it possible to optimize resources dynamically as network conditions change.

In this paper, we provide a focused survey that reviews how modern ML approaches are uniquely suited to resource allocation optimization within communication and computing services. By examining the effects of ML on critical network Key Performance Indicators (KPIs) and results obtained through efficiency gains, including energetic, spectral, computational, latency, and packet delivery, we offer insights into both the effectiveness and limitations of current methods. This approach allows us to address specific challenges and opportunities in wireless communication, making the case that targeted ML-driven optimization in this domain holds the potential to revolutionize next-generation networks. Finally, we outline future directions, paving the way for ongoing research that can extend the benefits of ML-driven resource allocation optimization across additional network services and scenarios.

B. Review of Existing Surveys and Gap Analysis

The existing surveys and tutorials on the use of ML for resource allocation in wireless communications are reviewed in Table I.

Table I provides a structured comparison of surveys addressing the application of ML techniques for resource allocation optimization in wireless networks. It includes key attributes of each work: (i) the description of the paper, (ii) the number of surveyed papers for the state-of-art, (iii) the publication period for ML approaches, (iv) the range of wireless network applications covered, (v) the focus on resource allocation aspects such as power, spectrum, and computational resources, (vi) the classification of ML paradigms, including Supervised Learning (SL), Unsupervised Learning (UL), RL, and other solutions, (vii) the extent of discussions on distributed learning approaches, and (viii) the number of distinct ML algorithms reviewed. This comprehensive overview highlights the focus and contributions of each survey while offering a comparative perspective on their methodologies and scopes.

Our survey demonstrates notable advancements compared to prior works by covering a broader publication period (2019-2024), focusing on more recent developments. This recency is particularly important in the context of AI and ML, fields that evolve rapidly with the introduction of advanced hardware and increasingly complex algorithms.

TABLE I
COMPARISON OF THIS SURVEY WITH EXISTING ONES AND TUTORIAL PAPERS

Ref.	Title of the paper	No. of Surveyed Papers	Period of Publication	Coverage of Wireless Applications	Focus on Resource Allocation	Classification of Surveyed ML Solutions					No. of Different ML Algorithms Surveyed
						UL	SL	RL	Others	Discussion on Distributed Learning	
[7]	Machine and deep learning for resource allocation in multi-access edge computing: A survey	68	2014-2022	MEC Networks	Offloading, scheduling, and joint resource allocation.	✓	✓	✓		It discusses FL, highlighting its role in preserving device data privacy and speeding up edge training.	22
[8]	Artificial intelligence-empowered resource management for future wireless communications: A survey	20	2015-2019	Future B5G and 6G networks	RRM and joint 3C resource management.	✓	✓	✓	Bio-inspired algorithms (e.g., Ant Colony Optimization, PSO, GA)	The article briefly addresses cooperative AI schemes involving agent collaboration for resource management.	11
[9]	Applications of machine learning in resource management for RAN-slicing in 5G and beyond networks: A survey	88	2014-2022	RAN-slicing in 5G and beyond networks	Radio, power, computing, caching memory, storage, and transport bandwidth.	✓	✓	✓	Evolutionary algorithms (e.g., GA) and other distributed techniques, mainly in RL algorithms.	It categorizes distributed techniques, such as FL and TL, focusing on RAN slicing, distributed training, and ensuring data privacy.	21
[10]	A survey on resource allocation in vehicular networks	39	2014-2019	Vehicular networks	Radio resource allocation schemes based on DSRC, cellular, and heterogeneous vehicular networks.			✓		It mentions the use of distributed learning approaches for dynamic and autonomous decision-making, specially in DRL.	4
[11]	The frontiers of deep reinforcement learning for resource management in future wireless HetNets: Techniques, challenges, and research directions	76	2017-2021	Future wireless heterogeneous networks	Radio resource types (e.g., power, spectrum & access control, rate, joint of radio resources).			✓		It addresses distributed learning techniques, specifically the role of multi-agent models in DRL on resource management and the challenges in heterogeneous networks.	12
[12]	Application of machine learning in wireless networks: key techniques and open issues	88	2010-2019	Wireless networks	Resource management (e.g., power, spectrum, backhaul, beamformer, computation resource, cache), mobility, localization, and networking.	✓	✓	✓	Complementary learning methods (e.g., Joint Utility Strategy Estimation Based Learning, Collaborative Filtering, TL)	It highlights distributed learning in wireless networks to reduce signaling overhead and enable autonomous decision- making for resource allocation.	22
[13]	Artificial neural networks-based machine learning for wireless networks: a tutorial	37	1990-2019	Wireless networks	Spectrum resources, channel access and selection, caching and computation resources, and power and energy management.	✓	✓	✓		It highlights the challenge of distributed ANN learning, algorithms, emphasizing FL as a solution for data analytics across BSs with limited backhaul.	7
[6]	Machine learning for resource management in cellular and IoT networks: potentials, current solutions, and open challenges	59	2005-2019	Cellular and IoT networks	Scheduling and random access, power allocation and interference management, resource discovery, cell and channel selection, traffic analysis and mobility prediction, clustering and data aggregation.	✓	✓	✓		It discusses distributed learning, including FL, as a promising approach for privacy-preserving and decentralized resource management, addressing challenges related to context and privacy of IoT data.	28
[14]	ML-based radio resource management in 5G and beyond networks: a survey	31	2012-2022	5G and beyond networks	RRM sub-problems (e.g., KPIs prediction, user, sub-carrier and power allocation), with respect to different network metrics (e.g., QoS, QoE, throughput).	✓	✓	✓	ML approaches of FL, including majorize, semi-definite relaxation and Gaussian randomization, and distributed learning approaches with MEC in different NNs.	It categorizes distributed approaches, including FL and MEC in the surveyed papers, and it highlights distributed solutions as promising to reduce training time, optimize computational resources and minimize latency in 5G/B5G networks.	20
Our survey	A comprehensive survey of machine learning applied to resource allocation in wireless communications	130	2019-2024	Wireless communications, mainly 5G/6G networks.	Radio, communication and computing resources (e.g., power, frequency, time, computational resources, and spatial).	✓	✓	✓	Evolutionary (e.g., GA and NSGA-II) and other population-based algorithms (e.g., MA and EDA), FL-based algorithms (e.g., FL, A3FN, FL-GR, and FedAvg), and other distributed and collaborative learning approaches.	It provides a literature review on distinct distributed learning approaches, discussing the suitability in different scenarios by combining them with other ML paradigms in future networks, outlining main benefits.	77

These advancements enable our survey to capture more up-to-date and comprehensive insights. Additionally, while some surveys tend to focus on specific types of wireless networks or applications (e.g., Mobile Edge Computing (MEC) networks or Radio Access Network (RAN) slicing), our work covers both current and future wireless networks more broadly, including a wide variety of use cases relevant to 5G and 6G technologies.

Regarding resource allocation, Table I illustrates that "resource" can have varying meanings depending on the context, with each survey selecting specific themes to explore. In our case, the focus is on the allocation of radio, communication, and computational resources, providing a unified perspective across these domains. Furthermore, our survey stands out by addressing a wide range of ML paradigms, including SL, UL, RL, and other advanced and evolutionary paradigms. It also provides an in-depth discussion on distributed learning approaches, analyzing their integration with other ML paradigms and encouraging their application in future works. Lastly, our survey examines 77 distinct ML algorithms, more than any other reviewed work presented in the table. This higher number of algorithms across different ML techniques, combined with our analysis of a larger number of papers and more recent research, allows us to identify innovative solutions and emerging trends in the field, providing valuable insights for researchers and practitioners.

To conduct the search strategy and selection of papers for our survey, reaching the numbers previously described in Table I, we used inclusion-exclusion criteria focusing on articles about AI, specifically ML techniques applied to resource allocation optimization in wireless communications. Relevant papers were located in IEEE Xplore, Science Direct, and SpringerLink, each offering unique filtering methods. Inclusion criteria targeted keywords like "Artificial Intelligence", "Machine Learning", and "Resource Allocation" in titles, abstracts, or keywords, with filters tailored to each platform's focus: generalized searches for IEEE Xplore and more specific filters for AI, telecommunications, and wireless networks in Science Direct and SpringerLink. The generalization in IEEE Xplore is justified by its explicit focus on technology, engineering, and computer science, with journals, conference papers, books, and technical standards specific to these areas. In contrast, Science Direct and SpringerLink cover broader disciplines such as natural sciences, social sciences, and medicine, requiring more targeted searches to ensure relevance to our study. Exclusion criteria ensured relevance by limiting publication years (2019-2024), omitting review articles and tutorials, and emphasizing original research in ML for wireless communications. Limiting the time frame to recent publications is particularly beneficial, as the field of ML evolves rapidly alongside technological advancements, driving constant innovation and necessitating

up-to-date articles to reflect the latest solutions and algorithms.

C. Our Survey Scope and Contributions

Compared with the existing surveys, as discussed in the previous section, this paper presents a comprehensive survey of the latest ML approaches applied to resource allocation in wireless communications for improving network KPIs and efficiency performance gains. Over 100 papers from 2019 to 2024 have been reviewed in depth, including those focusing on wireless communications applications, optimizations addressed in network KPIs, the taxonomy of ML algorithms classified by different learning paradigms and their suitability on specific scenarios, assessment metrics, and efficiency performance gains.

This paper presents valuable and significant insights to provide a deeper understanding of ML-based resource allocation optimization in wireless communications. In this context, our main contributions are summarized as follows:

- A more up-to-date and comprehensive survey of current ML research in resource allocation for wireless communications, covering 130 papers and 77 ML algorithms from 2019 to 2024, including monitoring of different KPIs, exploring diverse learning paradigms and distributed approaches, and discussing trade-offs for multi-resource optimization and network balancing, as detailed in the state-of-the-art presented in Section III;
- A taxonomy of ML algorithms for resource allocation in wireless communications, presented in Section IV, identifying leading approaches, their motivations, advantages, limitations, and implementation challenges;
- A review of experimentation frameworks and dataset usage, as outlined in Section V, analyzing the most common simulation tools and data samples in ML-based resource optimization. This contribution also highlights potential standardizations and benchmarks to advance wireless communication solutions;
- An evaluation of ML algorithm performance across various wireless communication scenarios and resource allocation problems, as proposed in Section VI-A. This study highlights their limitations, adaptability, and strengths, exploring hybrid approaches to integrate complementary techniques for addressing complex challenges effectively;
- A review of simulation and training parameters in ML-driven wireless communication systems, as discussed in Sections VI-B and VI-C. This analysis addresses the need to align experimental conditions closer to real-world scenarios and highlights commonly under-reported algorithm parameters. Furthermore, it promotes best practices for the transparent reporting of ML parameters in future research, aiming to enhance reproducibility and enable fair comparisons;
- A comprehensive analysis of the main limitations, challenges, and future directions for employing ML techniques in optimizing resource allocation in wireless communications, combined with emerging mobile networks and trending technologies. This discussion explores how such integration can enable dynamic multi-resource optimization, extending 5G to address the potential services, use cases, and demanding requirements of 6G networks, as detailed in Section VII.

These contributions may provide the scientific community with insightful analysis of suitable algorithms and metrics for evaluating the performance of ML-embedded systems according to different network features and scenarios. This can lead to advancements in the field, driving progress in research and practical application of these solutions.

Furthermore, the findings of this article may facilitate the identification of current knowledge gaps, stimulate new research, and develop more effective approaches to optimize resource allocation in wireless communications systems. In this manner, this paper seeks to provide valuable insights to the research society in enhancing performance, efficiency, and innovation in long-distance communication networks, contributing to ongoing advancements in the field of telecommunications and Artificial Intelligence (AI).

D. Structure of This Survey

The survey is organized into eight main sections, systematically addressing the critical aspects of ML-based resource optimization in wireless communication systems, as shown in Figure 1. Section II provides the necessary background, including an overview of considered resources, KPIs, their interrelations, and the foundational ML paradigms and distributed learning approaches. Section III explores the state-of-the-art applications of ML in resource allocation, categorizing the surveyed papers by ML paradigms and distributed approaches while examining KPI trade-offs. Section IV introduces a taxonomy of ML algorithms, focusing on leading algorithms across unsupervised, supervised, reinforcement, and evolutionary learning paradigms. Section V details the experimentation framework and datasets used in ML-based resource allocation optimization, discussing simulation practices, dataset usage, and reproducibility. Section VI delves into the solutions proposed by leading ML algorithms across various resource allocation problems and scenarios, examines the study of simulation parameters for ML-driven wireless communication systems, and addresses the reproducibility challenges in ML through a review of parameter documentation in the selected works. Section VII discusses current limitations, challenges, and future directions, emphasizing transparency, available technological resources, interdependence of wireless communications and ML employment, emerging trends, and concluding remarks. Section VIII concludes the survey, summarizing key insights and providing final remarks. Finally, the Appendix presents the list of acronyms included in our survey for clarity and further study.

II. Background

5G and beyond networks aim to deliver not only communication services but also advanced services in computing, sensing, and positioning. This expansion arises from the need to support a growing diversity of applications demanding high

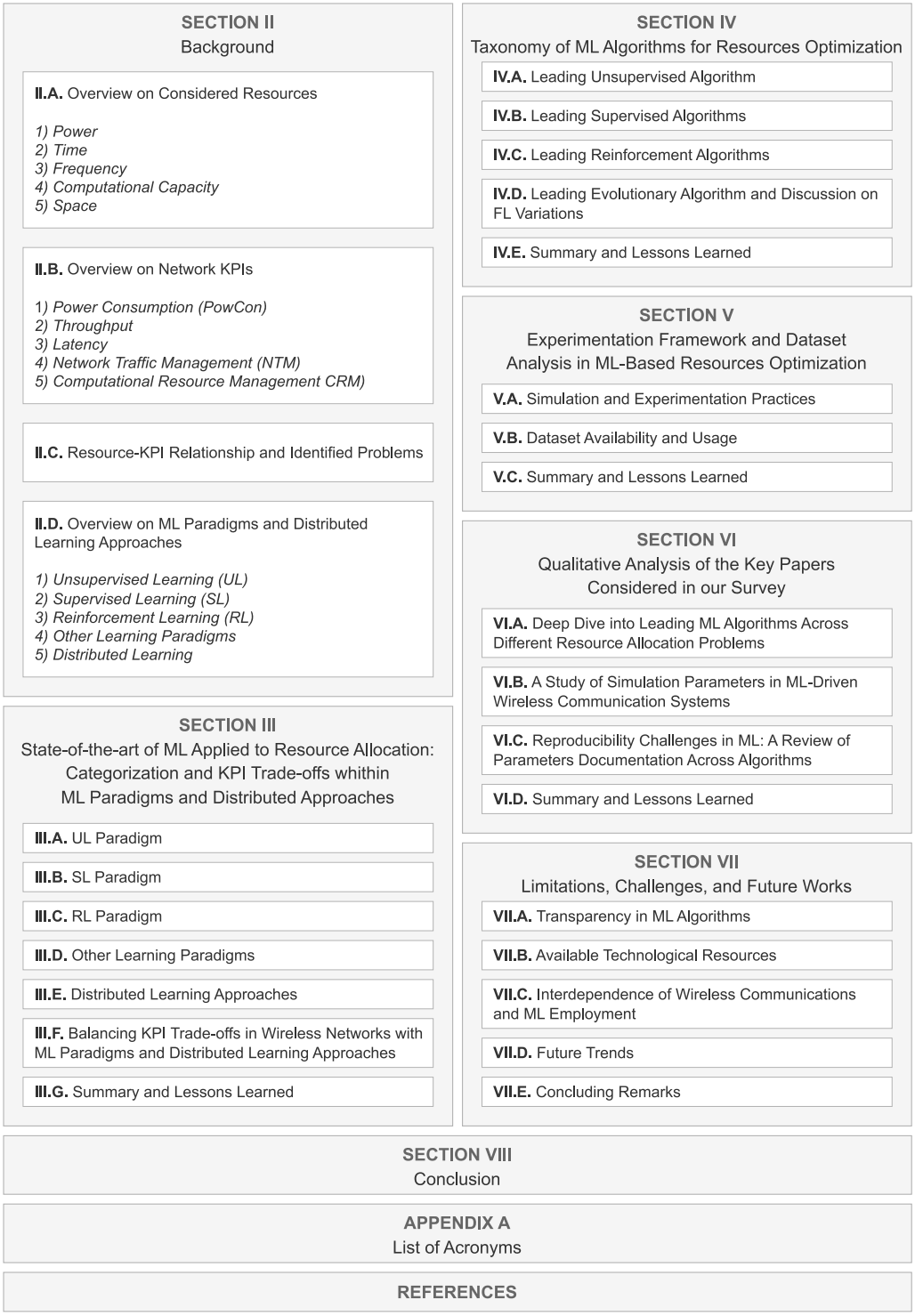


Fig. 1. The structure of this paper.

precision, real-time responsiveness, and seamless integration across devices and environments. To meet these demands, various network resources—such as power, bandwidth, processing capacity, and spatial coverage—can be managed and optimized, enabling the efficient and reliable delivery of these services. ML techniques play a critical role in this optimization, enabling dynamic resource allocation, predictive management, and adaptive tuning of network parameters to respond intelligently to fluctuating demands and environmental conditions, ultimately enhancing overall network performance.

For example, localization services are essential in 5G and 6G mobile networks to provide accurate information on the location and movement of devices and people, playing a key role in applications such as navigation, logistics, and

security [1]. Crucial resources for this service include location accuracy, response time, and signal availability. Accuracy is critical to ensure precise positioning, which is essential for real-time applications like vehicle tracking and safety systems. Response time is equally important, particularly for systems like autonomous vehicles, where rapid decision-making relies on the most current location data. Signal availability ensures reliable functioning of positioning services even in challenging environments, such as indoors or underground. ML techniques, including Convolutional Neural Networks (CNNs), can enhance positioning accuracy by combining signals from multiple sources, while predictive algorithms optimize signal coverage and continuity, adapting to the context to ensure higher reliability.

Additionally, sensing services involve collecting and analyzing environmental data, which is crucial for a wide range of applications, from environmental monitoring to health devices [1]. Key resources for this service include accuracy, sampling frequency, coverage, and energy. Sensor accuracy is essential for high-quality, reliable data, supporting precise monitoring and control systems decision-making. Appropriate sampling frequency captures environmental changes efficiently, while broad coverage maximizes the monitored area with minimal sensor redundancy. Energy is also a critical resource, as sensors in remote or hard-to-reach areas must operate for extended periods without maintenance. ML models can effectively optimize these resources by adjusting sampling frequency based on data patterns and prolonging sensor life by activating them only as needed, enabling more efficient and cost-effective data collection.

Furthermore, computing services in 5G and 6G networks encompass the network's capability to process and analyze data efficiently, often leveraging distributed learning techniques to allocate tasks dynamically between devices and the network edge [3]. This approach not only reduces latency but also enables the network to adapt swiftly to real-time demands. The increasing reliance on innovative wireless networks and Software-defined Radios (SDRs) further underscores the importance of properly managing computational resources, such as processing capacity, memory, storage, and bandwidth. Processing capacity supports the execution of complex ML-driven tasks, while memory ensures the retention of critical contextual information to enhance decision-making. Local storage reduces dependency on backhaul networks, improving system responsiveness, and efficient bandwidth utilization facilitates seamless communication between devices and edge servers, minimizing delays. By leveraging ML, these resources can be dynamically predicted, allocated, and optimized, ensuring that distributed computing applications operate with high precision and efficiency, ultimately supporting the diverse demands of modern wireless networks.

Lastly, communication services are fundamental to next-generation networks, serving as the backbone for positioning, sensing, and computing by enabling fast, reliable data transmission across devices [1]. Their effectiveness hinges on optimizing resources like power, time, frequency, spatial allocation, and computational capacity, all of which impact user experience and service quality. For instance, low-latency, high-bandwidth communication is essential for real-time positioning applications such as autonomous driving and emergency response. Similarly, communication efficiency supports sensing by ensuring the continuous flow of environmental data from distributed sensors, where power optimization helps reduce energy consumption, crucial for low-cost, battery-powered devices in environmental monitoring and healthcare. Effective optimization of communication resources thus minimizes interference, improves data throughput, and conserves energy, leading to an enhanced overall performance that benefits both the end-user experience and the delivery of critical services.

Given the significance of communication as the core enabler for all other network services, ML has emerged as a superior approach over traditional techniques for optimizing communication resources. ML algorithms are particularly effective in adapting dynamically to fluctuating network conditions, predicting resource demands, and managing interference, enabling more precise and efficient communication across diverse network scenarios. Similarly, computing services have become increasingly essential in modern networks, driven by the advancements in ML through distributed learning and the growing adoption of Software-Defined Networks (SDNs) and SDRs, which demand more efficient allocation of computational resources. These developments highlight the critical need to optimize both communication and computing resources. Based on this foundational role of communication and computing services and the demonstrated effectiveness of ML for resource optimization, our survey will focus on ML-driven solutions for enhancing communication and computing service resources in wireless network systems. Additionally, it will analyze KPIs such as latency, throughput, Power Consumption (PowCon), Network Traffic Management (NTM), and Computational Resource Management (CRM), demonstrating how ML-driven optimization improves network performance, user experience, and the network's capacity to meet diverse demands across various services.

To achieve this, Figure 2 illustrates the mapping of the theoretical foundation necessary for this technical survey, which consists of describing: (i) the allocation of resources to be considered for wireless communications; (ii) the selected KPIs for network analysis, their synergy with the allocated resources, and the main problems encountered in wireless systems based on these indicators; (iii) and the commonly used learning paradigms and distributed learning approaches to solve these problems. The overview of each of these concepts, as well as a broader discussion about Figure 2, will be presented in the following sections.

A. Overview on Considered Resources

Among the main critical resources in wireless communications, as depicted in Figure 2, we considered power, time, frequency, computational capacity, and space [15]. These allocation resources are described below, in addition to their relevance in the context of wireless communications, considering user experience and network cost savings.

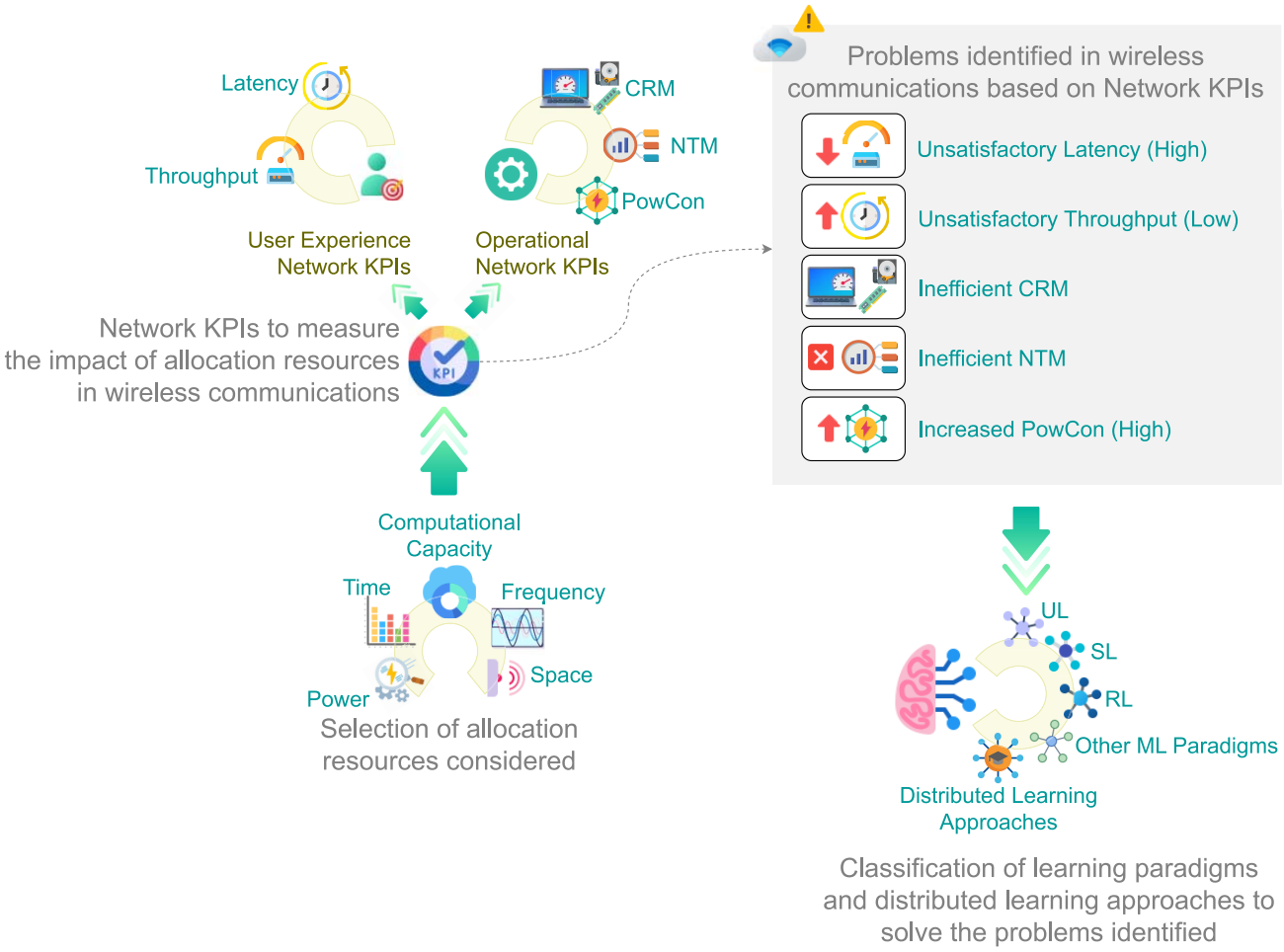


Fig. 2. Mapping of the theoretical foundation necessary for a better understanding of this technical survey: critical allocation resources, network KPIs, problems to be addressed in wireless communications based on mapped network KPIs, and finally learning paradigms and distributed learning approaches commonly used to solve these problems.

1) Power: Power in wireless communication systems refers to the amount of energy used to transmit signals over the air, and it is a critical resource that affects signal reach, obstacle penetration, and communication clarity. Managing power involves adjusting transmit power to ensure adequate signal quality while minimizing energy use [16]. Optimizing power allocation enhances user experience, reduces operational costs, and supports sustainability efforts by lowering energy consumption and the carbon footprint of wireless networks [17].

2) Time: Time in wireless communication refers to the allocation of transmission intervals, organizing when data packets are sent, which impacts system capacity and user Quality of Service (QoS). Effective time management is crucial for real-time applications like video streaming, ensuring smooth data flow without delays [18]. Optimizing time allocation improves user experience through reduced latency and higher data throughput while financially benefiting operators by enabling more users to be served with the same infrastructure [19]. Additionally, it enhances Energy Efficiency (EE), reducing the network's overall energy demand [20].

3) Frequency: Frequency refers to the bandwidth occupied by a signal, and effective allocation is essential to minimize interference and maximize network capacity and QoS [21]. Optimizing frequency allocation enhances user experience with reliable, high-quality communication while increasing network capacity and scalability, allowing for more users and services. The efficient use of frequency maximizes spectrum value and promotes environmental sustainability through more efficient energy use per unit of data transmitted [22].

4) Computational Capacity: Computational capacity refers to the processing power available to execute essential signal processing algorithms, encryption, and other critical telecommunications tasks, including AI-based algorithms [23]. This resource supports complex functions like error correction and modulation, ensuring high-quality communication services. Optimizing computational capacity enhances user experience by enabling advanced services, such as high-definition video streaming and secure communication [24]. Efficient allocation reduces operational costs by minimizing hardware upgrades and energy consumption, offering financial benefits and reducing environmental impact by optimizing existing resources and lowering electronic waste [25], [26].

5) Space: Space is a critical resource for system capacity in a given area, utilizing techniques like beamforming, which directs signal transmission using antenna arrays to enhance

signal quality, extend range, and reduce interference [27], [28]. This improves system efficiency by concentrating energy where needed [29]. Optimizing spatial allocation through beamforming strengthens connections, increases network capacity, and reduces interference. Financially, it allows for more efficient infrastructure use, potentially reducing the need for additional Base Stations (BSs). Environmentally, precise energy focus lowers power waste, contributing to the sustainability of wireless communication systems [30].

B. Overview on Network KPIs

Considering that the resource allocation optimization significantly impacts a wireless communication system, we can monitor the network through KPIs. Therefore, it is possible to establish a relationship between critical allocation resources and network KPIs, map the main problems commonly identified in wireless communication systems, and evaluate network performance to diagnose possible improvements regarding the optimization of these resources.

As illustrated in Figure 2, network KPIs are classified into (i) user experience and (ii) operational. User experience KPIs measure the satisfaction of a wireless system's end-user and are related to the QoS. Operational KPIs assess the network's internal performance and resource management, focusing on cost-effectiveness and productivity. In this context, user experience KPIs selected to be examined in our survey include throughput and latency, while operational KPIs involve PowCon, NTM, and CRM [31], [32], [33], [34], [35], which are critical for monitoring network performance and efficiency. These network KPIs are described in detail below.

1) Power Consumption: The PowCon KPI measures energy consumption in wireless systems, mainly the power allocated to the transceivers [36]. Monitoring PowCon is crucial for reducing operational costs, as energy usage affects the financial performance of network operators, and for promoting environmental sustainability by lowering the carbon footprint. Additionally, efficient power management improves the lifespan of battery-powered devices, enhancing user experience. Overall, tracking PowCon is vital for optimizing a network's EE and reducing energy use without compromising QoS [37].

2) Throughput: Throughput, measured in bits per second (bps), indicates the volume of data successfully transmitted over a network within a specific time frame, serving as a key metric for network performance [38]. High throughput is essential for supporting data-intensive applications such as video streaming, online gaming, and large file transfers. Monitoring throughput is critical for network management as it directly impacts user satisfaction and QoS. Additionally, it allows administrators to identify and address infrastructure bottlenecks, facilitating targeted improvements to meet increasing data demands and maintain a competitive edge [39].

3) Latency: Latency measures the delay between a user's action and the network's response, a critical metric for real-time applications like Voice over Internet Protocol (VoIP), online gaming, and interactive services [40]. Low latency ensures quick response times, which is essential for a seamless user experience. Monitoring latency is crucial for meeting the real-time demands of modern applications, as high latency can cause delays, communication issues, and a poor user experience, negatively impacting a service provider's reputation. By tracking latency, operators can optimize networks, reduce delays, prioritize traffic, and enhance overall QoS [41].

4) Network Traffic Management: NTM focuses on managing and distributing data traffic efficiently across a network to maintain optimal performance and avoid congestion, particularly during peak usage times [42]. Unlike throughput, which measures data transmission capacity, NTM assesses how data packets are managed, prioritized, and fairly distributed among users to enhance QoS, especially in time-sensitive scenarios. Monitoring NTM enables network operators to make informed decisions regarding capacity planning, network design, and congestion management, ensuring critical services receive necessary bandwidth while maintaining overall network efficiency and reliability [43].

5) Computational Resource Management: The CRM KPI evaluates the management of computational resources through task scheduling, offloading, and ML techniques to reduce complexity [44]. This involves efficiently allocating processing power across servers, routers, and network devices to balance load and prevent bottlenecks. Monitoring CRM is vital for maintaining network performance as dynamic resource allocation becomes essential in increasingly complex, data-intensive networks. It helps optimize resource use, ensuring scalability, supporting new services, and maintaining high-quality user experiences [45].

C. Resource-KPI Relationship and Identified Problems

The analysis of the previously presented KPIs helps identify which resources are optimized in the network and which could be better utilized to enhance communication system performance. These KPIs reveal resource allocation issues and their impact on wireless communications. Figure 3 presents a heat map depicting the relationship between critical resources and network KPIs.

According to Figure 3, power has the greatest impact on the PowCon and throughput KPIs. Optimizing power allows transceivers to reach users with minimal energy, reducing overall consumption. Improved EE increases Signal-to-Interference-plus-Noise Ratio (SINR), enhancing Modulation and Coding Scheme (MCS) and signal reliability, which boosts throughput by providing more stable connections. Power also moderately affects latency, NTM, and CRM. Better power allocation reduces Packet Error Rate (PER) and the demand for packet re-transmissions, lowering latency, while dynamic adjustments of transmission power based on traffic demand improve NTM. Additionally, enhanced EE can simplify multiple access techniques, stabilizing transmission and reducing computational load during data re-transmission or error correction, indirectly benefiting CRM [46].

Efficient time allocation within the transmission window significantly impacts latency and throughput by minimizing delays and improving data delivery. This reduces the need for retransmissions due to errors and increases transmission capacity, as more data can be sent in the available periods,

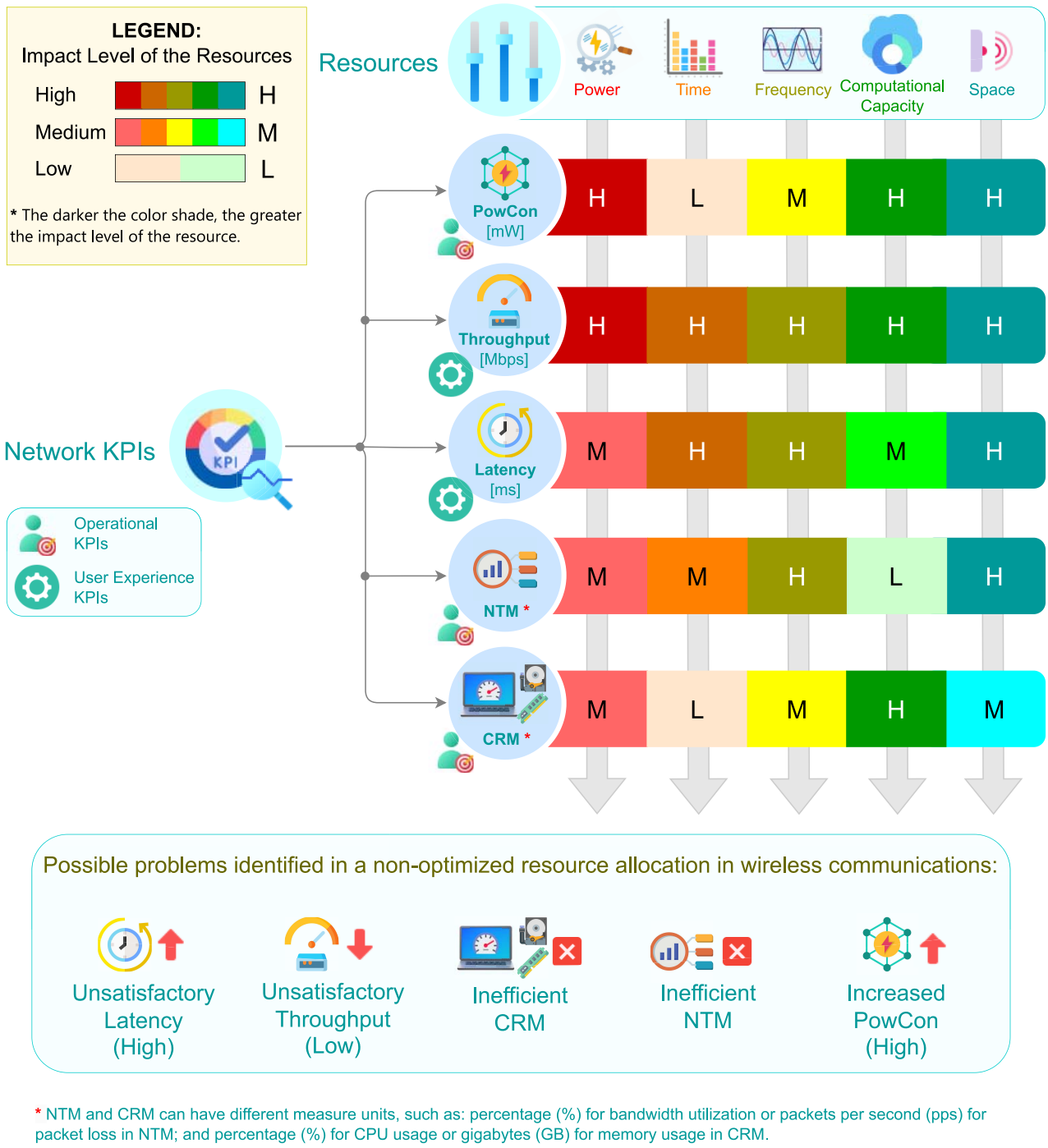


Fig. 3. The heat map representation of the relationship between resources and network KPIs.

as seen with Faster than Nyquist (FTN) signaling [47]. Time also moderately influences NTM, as dynamic time adaptation allows for better management of transmission parameters, such as bandwidth and priority, based on traffic demand. Although time has no direct impact on PowCon and CRM, precise time management can improve these KPIs by optimizing transceiver scheduling, reducing energy consumption, and supporting more complex detectors for efficient data processing [48].

Frequency allocation, which determines available bandwidth, strongly impacts throughput, latency, and NTM. Efficient frequency use maximizes throughput, prevents interference and congestion, and reduces delays. Larger bandwidths decrease frame duration and latency, while shorter time-domain signals require wider bands for low-latency transmission. Dynamic frequency prioritization improves traffic management, optimizing communication for multiple users and enhancing NTM. Frequency also moderately affects PowCon and CRM. From a PowCon perspective, the frequency used significantly influences the power required for transmission, impacting energy consumption. Regarding CRM, higher

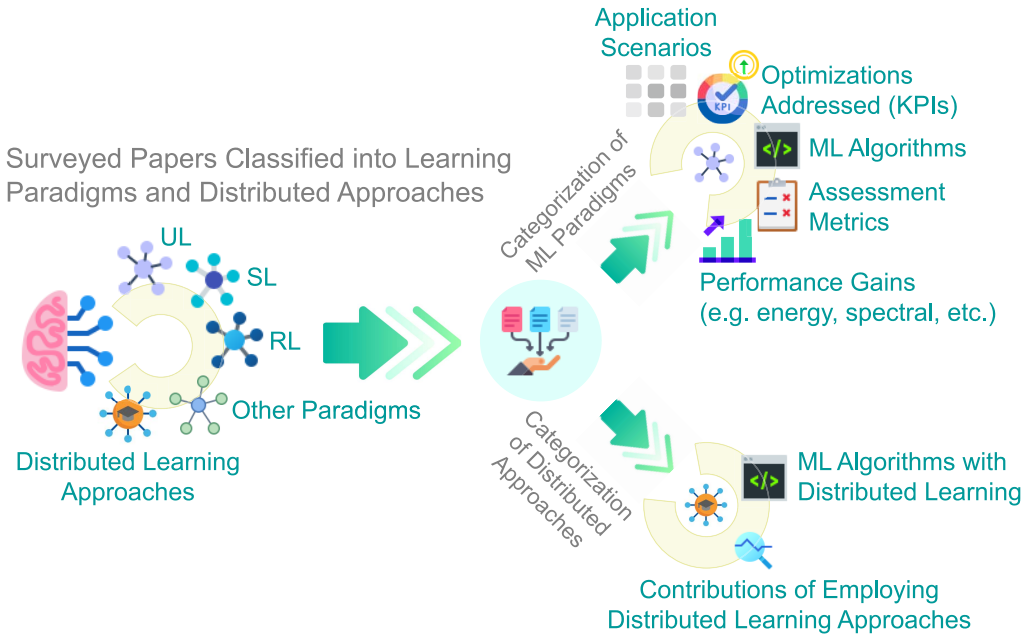


Fig. 4. Overview of information extracted from the selected papers to establish the state-of-the-art of our survey and enable data analyses.

bandwidth can increase system complexity, requiring more advanced converters and processors [48].

The allocation of computational capacity significantly affects CRM, PowCon, and throughput KPIs. Optimizing resources like memory, Graphics Processing Unit (GPU), and Central Processing Unit (CPU) prevents bottlenecks and enhances overall system performance, improving transmission efficiency and reducing energy consumption [49]. Greater computational availability allows for the use of complex algorithms to improve signal detection with lower Signal-to-Noise Ratio (SNR), enhancing PowCon [50]. Additionally, optimized algorithms can increase throughput by enhancing techniques like Multiple-Input Multiple-Output (MIMO). Efficient allocation of computational capacity moderately impacts latency by improving data processing efficiency, thus reducing delays. Indirectly, this resource affects NTM by enabling more complex algorithms for efficient traffic management and dynamic adjustment of transmission parameters, such as bandwidth and priority [51].

The space resource plays a crucial role in network KPIs such as PowCon, throughput, latency, and NTM. Spatial resources provide diversity and reduce transmission power while maintaining QoS, lowering energy consumption by efficiently targeting users with minimal power. Techniques like beam steering minimize signal dispersion and serve multiple users effectively. MIMO schemes leverage space diversity to boost system robustness and SE, significantly impacting throughput, latency, and NTM. Space resources also moderately affect CRM due to the higher complexity of algorithms to explore the spatial dimension demand, including MIMO and beamforming [52].

In the context of wireless communication networks, optimizing specific KPIs can have varying impacts on other KPIs, depending on the specific scenarios and the interconnected nature of these performance metrics. Considering PowCon, throughput, latency, NTM, and CRM, each KPI's optimization can influence the others positively, negatively, or in a context-dependent manner.

Focusing on PowCon, which aims to reduce transceivers' transmission power to lower energy consumption, its effect on throughput depends on the network environment [53]. In urban areas with dense small cells, reduced transmission power can prevent interference and enhance throughput. However, in rural areas, this may weaken the signal, decreasing throughput due to lower SNR. Latency is similarly affected, where power reduction can either lower retransmissions and latency or worsen signal quality, increasing delay. Optimizing PowCon can also improve or degrade NTM and CRM efficiency based on how resources and computational loads are managed.

Throughput optimization can significantly impact other KPIs [54]. In a well-optimized network, increasing throughput can shorten data transmission time, reducing energy consumption (PowCon). However, in cases where higher throughput demands increased transmission power or energy-intensive techniques, energy consumption may rise. The relationship with latency is similarly complex, since higher throughput can reduce latency by speeding up data transmission, but if achieved through greater processing complexity, it might increase latency. High throughput usually enhances NTM, allowing better traffic flow, but in congested networks, prioritizing throughput could worsen NTM. Regarding CRM, while higher throughput can reduce retransmissions and enhance resource efficiency, it may also heighten computational load if intensive processing is required.

Latency optimization can significantly enhance network performance, but with trade-offs [55]. Lower latency often improves PowCon by making transmissions more energy-efficient, though it may increase energy consumption if

higher transmission power is needed. Latency reduction typically boosts throughput by enabling faster data transmission, but certain techniques may prioritize speed over data volume, impacting throughput. In terms of NTM, lower latency helps manage traffic by allowing quicker responses, though excessive focus on latency could create traffic imbalances. Regarding CRM, latency reduction usually improves Computational Efficiency (CE) by speeding up task processing, but if it increases complexity, resource usage may rise, potentially affecting CRM.

NTM optimization plays a crucial role in network performance with varied effects [56]. Effective traffic management enhances PowCon by using transmission power efficiently, reducing energy consumption. However, in congested networks, more power may be required to maintain quality, increasing energy use. Optimizing NTM often boosts throughput by balancing network load, whereas inefficient management or overload can decrease throughput. Good traffic management generally reduces latency by preventing congestion, but excessive load balancing can raise latency due to additional processing. Efficient NTM also benefits CRM by avoiding bottlenecks, though it may strain resources if traffic management becomes too complex.

Finally, CRM is crucial in maintaining network performance. Optimized CRM often reduces PowCon by minimizing unnecessary processing and improving resource allocation, particularly in virtualized environments [57]. However, prioritizing energy savings over transmission power may harm EE if signal quality deteriorates. The effect on throughput is complex: efficient CRM can improve throughput by freeing computational capacity, yet in resource-limited settings, it might reduce this KPI. Effective CRM usually lowers latency by streamlining task processing, though higher processing complexity can increase latency. Regarding NTM, optimized CRM typically supports better traffic management by preventing computational overloads, but overly restrictive CRM could hinder traffic management.

In summary, the interdependence of KPIs in wireless networks requires careful consideration, as the effects can be context-dependent, and optimizing one can positively or negatively impact others. A holistic approach, accounting for all scenarios and network conditions, is essential for effective KPI management. Efficient resource allocation leads to performance gains related to, for example, EE, SE, Latency Efficiency (LE), Packet Delivery Efficiency (PDE), and CE [58], [59], [60], [61]. EE involves minimizing energy usage while maintaining target performance in terms of Bit Error Rate (BER) or PER. SE maximizes data transmission within a given frequency spectrum. LE reduces end-to-end response time. PDE ensures high successful data delivery rates, and CE optimizes the exploitation of available computational resources.

While the primary goal of optimizing computational capacity is to enhance CE, optimizing other resources—such as power, time, frequency, and space—can also contribute to improving CE indirectly. By reducing transmission PowCon, increasing throughput, minimizing latency, and managing traffic more efficiently, these resource allocation optimizations contribute to better CE. ML techniques can replace conventional methods to reduce computational complexity, thereby improving CE without significantly affecting the overall network performance in terms of resource allocation optimization [62]. Metrics such as accuracy, Mean Squared Error (MSE), Mean Absolute Error (MAE), and Cumulative Distribution Function (CDF) are used to compare the performance of ML algorithms to conventional methods [63]. While CE can also reduce PowCon and latency, its focus differs from EE and LE, which aim to minimize energy consumption and transmission time. Efficient radio resource management, such as through programmable SDR and Cognitive Radio (CR), reduces collisions, retransmissions, and power usage, thus optimizing PowCon [64] of network devices. In contrast, CE maximizes the computational capacity of CPUs and GPUs, improving access to memory and storage to enhance performance by reducing processing time, energy consumption, and latency [65].

Optimizing critical resources not only improves network KPIs but also enhances Quality of Experience (QoE). Increasing throughput boosts the network's data transmission capacity within a given time-frequency grid, supporting data-intensive applications like high-definition video streaming, online gaming, and 3D video by enabling faster data transfer. Reducing latency is vital for providing a responsive user experience, especially in real-time applications such as video calls, industrial processes, and augmented/virtual reality, where the information life cycle is short [66]. Additionally, effective NTM enhances QoE by prioritizing latency-sensitive applications, ensuring timely delivery and adequate throughput. Finally, efficient CRM further improves network performance by speeding up task processing [67].

Regarding cost reduction, power allocation optimization leads to significant energy savings and lower operational costs [68]. Enhancing SE reduces the need for additional spectrum, which can also result in significant cost savings [69]. Optimizing NTM, throughput, and latency makes the network more efficient, saves energy, and avoids resource wastage. Furthermore, by optimizing CRM and improving CE, it is possible to reduce the use of servers and Information Technology (IT) infrastructure, leading to savings in hardware investment and maintenance of these resources [70].

Based on the analysis of Figure 3, along with the critical resources and network KPIs previously outlined, several issues arising from non-optimized resource allocation in wireless communications can be identified. These include unsatisfactory latency (high), unsatisfactory throughput (low), inefficient CRM and NTM, and increased PowCon (high). Consequently, we conclude that a thorough analysis and optimization of the KPIs discussed in this paper, through more effective resource allocation, can address these challenges and significantly enhance the performance of wireless networks.

D. Overview on ML Paradigms and Distributed Learning Approaches

Given the significance of optimizing critical resource allocation in wireless communications, both for network operational aspects and user experience, as well as for meeting diverse application demands, various technologies have been explored,

and the utilization of AI to address these challenges has increasingly been contemplated.

AI is a field of computer science that focuses on creating intelligent systems capable of performing tasks that typically require human intelligence. Within AI, ML stands as a powerful subset, enabling computers to learn from data and improve performance without explicit programming [71]. As ML algorithms continue to advance, their applications have become increasingly diverse, with different learning paradigms and distributed learning approaches being used to meet various demands better.

ML can be defined as a class of algorithms that allow systems to automatically improve their performance in specific tasks through data or experience. A key characteristic of ML is that the algorithm adapts or adjusts its internal parameters based on feedback, thus enabling the system to make increasingly accurate decisions or predictions over time. From this perspective, ML algorithms share the core trait of learning from data or experience and improving their performance over time.

In contrast, other optimization solutions, such as meta-heuristic algorithms (e.g., Tabu Search), do not involve learning in the same sense. These algorithms explore the solution space using predefined strategies to avoid local optima but do not adapt or improve based on past experience or data. Meta-heuristics operate through fixed search procedures and rules rather than improving their model or solution dynamically through feedback.

In the realm of telecommunications, the integration of ML techniques and algorithms holds immense potential for resource allocation optimization. By harnessing AI capabilities, wireless networks can dynamically adapt to changing conditions and varying traffic demands, resulting in improved efficiency and optimized resource utilization. ML algorithms can, for example, predict network congestion, optimize spectrum allocation, and schedule resources efficiently, leading to better user experiences, reduced latency, and cost savings [72], [73], [74]. Additionally, ML-driven enhancements in security measures, anomaly detection, and fault management contribute to increased network reliability and overall performance, elevating the telecommunications industry to new heights of innovation and efficiency [75], [76].

To solve the problems related to the previously described KPIs analyzed in wireless communications networks, it is possible to develop algorithms related to different ML paradigms, which can be classified as: UL, SL, RL, among other ones, including evolutionary algorithms. In addition, it is also crucial to discuss distributed learning approaches, which enable the training process to be distributed across multiple nodes in the network, leveraging different ML paradigms to enhance scalability, reduce latency, and efficiently utilize distributed computational resources.

In this context, these learning paradigms share the core trait of learning from data or experience and improving their performance over time, which can be classified as an ML algorithm. For example, SL algorithms are considered ML because they learn from labeled datasets to map inputs to desired outputs. They adjust their models to minimize errors in future predictions. UL algorithms, like K-Means, learn from data without explicit labels by discovering underlying structures or patterns in the data. RL algorithms, such as Q-Learning, improve by interacting with an environment and optimizing their actions based on feedback in the form of rewards. Even evolutionary algorithms, such as genetic ones, fit under ML when they evolve solutions over time-based on fitness evaluations, simulating a learning process through the iterative improvement of populations.

The following sections provide an overview of the learning paradigms commonly used for the resource allocation optimization and consequently solving the main problems in wireless communication systems [77]. In addition, it also introduces the distributed learning approach and how it can contribute to addressing these problems.

1) Unsupervised Learning: UL is an ML algorithm used to identify patterns in data without any labels [78], [79]. This means there is no explicit feedback on the accuracy of the output, since the model learns to structure the data on its own through methods like clustering, dimensionality reduction, and association. The process involves analyzing datasets to find natural groupings, correlations, or structures without prior knowledge of the outcomes. Algorithms in this category work by exploring the data and attempting to find a hidden structure or intrinsic patterns. From this premise, UL is most commonly applied in areas where understanding the underlying structure of the data is crucial, but explicit labels are absent or hard to obtain [80].

Specifically, UL techniques like clustering are highly relevant for tasks such as user behavior analysis, resource management, and anomaly detection in wireless networks. Clustering can be applied to user grouping, where devices with similar communication patterns and user behaviors are clustered to optimize network traffic, enabling a more efficient spectrum sharing and allowing networks to dynamically allocate resources to different users based on their real-time needs [81]. In spectrum sensing, UL algorithms can identify spectrum usage patterns across various frequencies, detecting idle channels or underutilized bands to optimize spectrum allocation and minimize interference.

Additionally, reducing computational overhead in high-dimensional resource allocation problems is also essential due to the need for real-time decision-making across large datasets. Wireless systems must process complex information, including multiple frequencies and users, to allocate resources efficiently and avoid interference. Since high-dimensional data increases the computational burden, techniques like Principal Component Analysis (PCA) can reduce this complexity by focusing on the most relevant features, enabling faster decision-making and more efficient resource allocation without compromising network performance [82].

2) Supervised Learning: SL is an ML paradigm where an algorithm learns from labeled training data, understanding the relationship between the input variables and the output [83]. The training data consists of input-output pairs, where the correct answer is given, and the model makes predictions or

decisions based on new, unseen data. The learning algorithm incrementally improves its accuracy by minimizing the difference between the actual and predicted outputs. Common approaches include regression for continuous outputs and classification for discrete outputs. Based on this principle, this type of learning is widely applied in scenarios where historical data are useful to predict future events [84].

3) Reinforcement Learning: RL is a type of ML where an agent learns to make decisions by taking actions in an environment to achieve some goals. Unlike SL, the agent is not told which actions to take but instead must discover which actions yield the most reward by trying them. The learning process involves receiving feedback through rewards or penalties and using this feedback to learn the best strategy or policy to accomplish the task [85]. This iterative learning process of trial and error sharpens the agent's decision-making abilities over time. In accordance with this foundation, RL is extensively applied in areas requiring decision-making under uncertainty. In this case, this learning paradigm can be a powerful tool for solving complex optimization problems that involve sequential decision-making [85].

In wireless communications, the design of reward structures plays a critical role in guiding RL agents toward optimal solutions. These reward functions are often formulated to balance multiple, potentially competing objectives, such as reducing latency, maximizing throughput, and minimizing energy consumption. For example, a reward function might be structured as a weighted combination of these objectives, where the agent earns higher rewards for actions that improve overall network performance while simultaneously conserving energy. In multi-objective optimization scenarios, this approach ensures that the RL agent does not prioritize one metric at the expense of others, leading to more balanced resource management [86].

For example, optimizing both throughput and PowCon simultaneously is crucial for maintaining efficient network performance while conserving energy. Throughput, which measures the rate of successful data transmission, is essential for supporting high-demand applications such as video streaming or online gaming. However, focusing solely on maximizing throughput can lead to increased transmission power, which in turn raises energy consumption and may cause excessive interference in the network. Conversely, prioritizing PowCon reduction without considering throughput can result in poor data transmission rates, affecting the overall QoS for users. For instance, if power is reduced excessively in an attempt to conserve energy, the signal strength might weaken, leading to lower data rates and a degraded user experience [87].

To address this, a well-designed reward structure in RL for wireless systems often uses a weighted combination of both throughput and PowCon. In this context, a reward function might assign positive rewards for actions that increase throughput while simultaneously penalizing excessive power usage. By adjusting the weightings, the RL agent can learn to strike a balance between maintaining high throughput and minimizing PowCon ensuring that neither metric is optimized at the expense of the other. This balanced approach leads to more efficient resource management and optimized network performance, demonstrating the critical role that carefully crafted reward structures play in guiding RL agents toward optimal outcomes in time-varying systems, as exemplified by wireless communications [88].

4) Other Learning Paradigms: Since there are several ML algorithms that do not fit into the previously mentioned learning types, these approaches are classified in this survey in the group named as other learning paradigms. For example, FL fits best in this group as it is not exclusive to UL, SL, or RL, but rather a flexible approach for various learning paradigms, rooted in the concept of distributed learning, which will be discussed in the next section.

In addition, some types of algorithms, such as evolutionary ones, are also better categorized in "Others" as they adopt their own unique learning approaches, distinct from traditional paradigms previously described. Genetic Algorithm (GA), a type of evolutionary computation, is an ML algorithm that falls under a different paradigm of learning [89]. Inspired by the process of natural selection, it emulates the concept of survival of the fittest to optimize solutions. By mimicking biological evolution, GA iteratively evolves a population of candidate solutions through selection, crossover, and mutation operations to find the best fit for a given problem [90]. This approach allows it to explore a broader solution space, making it suitable for various optimization tasks where traditional learning paradigms may not be as effective [91].

In the context of solving complex optimization problems, similar to GAs, meta-heuristic algorithms can also be designed to explore vast and potentially non-linear solution spaces to find optimal or near-optimal solutions [92]. While effective in certain contexts, these algorithms, such as Simulated Annealing [93] and Tabu Search [94] are generally more focused on local search optimization. They are often used where a refined local search is more critical than global exploration, which may limit their applicability in scenarios requiring extensive search space exploration, such as those found in wireless communications with diverse conditions and variables. Given that meta-heuristic optimization algorithms, such as Simulated Annealing and Tabu Search, are not categorized as ML algorithms, our survey will not consider them.

5) Distributed Learning: Distributed Learning refers to a broad concept where data or computational processes are distributed across multiple nodes or devices to perform learning tasks. This approach commonly handles large-scale datasets or complex computations by dividing the workload among several entities, such as servers, edge devices, or user devices [95]. Examples include training ML models on clusters of servers to share the computational burden, FL where data remains localized on individual devices in a decentralized architecture and only model updates are shared to promote privacy and parallel processing systems that partition datasets for simultaneous analysis [96]. Transfer Learning (TL) can also facilitate distributed learning by enabling pre-trained models to be fine-tuned on localized data, reducing computational demands, training time, and improving convergence [97], [98]. The primary goal of distributed learning is to enhance efficiency, scalability, and sometimes privacy, depending on the implementation.

In addition, distributed learning can be applied to various learning frameworks, including SL, RL, and others. For example, in distributed SL, training data can be partitioned across multiple nodes, where each node independently trains a portion of the global model using local data, and updates are aggregated centrally or collaboratively. Similarly, in distributed RL, agents in a network can independently explore and share their experiences to collectively optimize a policy [99], [100].

In wireless communication systems, distributed learning can significantly improve resource allocation optimization by leveraging the decentralized and heterogeneous nature of these networks. For instance, BSs can act as distributed agents in a multi-cell wireless network, independently learning and optimizing resource allocation policies, such as power control, spectrum allocation, or user scheduling, based on local environmental conditions. These local decisions can be aggregated or shared among neighboring nodes to improve global performance while minimizing signaling overhead and latency. Distributed learning can achieve near-optimal resource allocation by enabling collaboration across network nodes while reducing computational bottlenecks and maintaining system robustness [95].

III. State-of-the-Art of ML Applied to Resource Allocation: Categorization and KPI Trade-Offs Within ML Paradigms and Distributed Approaches

This section presents a comprehensive review of the ML paradigms applied to resource allocation for solving the main problems identified in wireless communications and it is organized based on Figure 4. For this survey, we categorized all selected papers considering the aforementioned learning paradigms and distributed learning approaches, mapping the most commonly used ML algorithms and metrics, in addition to the optimized network KPIs, which can consequently monitor some challenges identified in wireless communications and are described in Section II-B.

Accordingly, the methodology for selecting papers for this survey adopted, as inclusion criteria, relevant articles published in different research databases in the field of AI, specifically within the context of ML techniques applied to resource allocation optimization in wireless communications systems, considering the critical resources, network KPIs, and learning paradigms previously defined. Conversely, exclusion criteria were considered: the exclusion of surveys and tutorials as part of the selected works for the technical analysis of this paper and the exclusion of papers published before 2019. Therefore, Table II summarizes the methodology used to compile the extracted data for our survey.

As indicated in Table II, this comprehensive and most up-to-date survey comprises a total of 130 research papers with publication periods ranging from 2019 to 2024. These papers were collected from multiple academic databases, including IEEE Xplore, Elsevier Science Direct, and SpringerLink. Therefore, through Table II, it is possible to observe a certain oscillation in the number of articles published over the period of publication year, which may have been influenced by various factors. One of these factors, for instance, could be related to the pandemic period of COVID-19, which affected the editorial processes of academic journals with potential delays in article reviews and publications. Additionally, the prioritization of urgent research and concentrated efforts on studies related to the disease, along with limitations in many parts of the world that could impact research conduct due to travel restrictions and access to laboratories and academic resources, are also determining factors for the potential impact of the pandemic on the number of published articles. Nonetheless, we can observe that research related to ML techniques applied to resource allocation optimization in wireless communications systems has been growing in recent years.

In this manner, our technical survey provides a comprehensive data extraction from the selected works based on the KPIs for the assessment of network issues and conditions. To systematically organize our state-of-the-art, the surveyed papers are categorized into subsections that address two primary dimensions: (i) the learning paradigms applied to resource allocation optimization, which include UL, SL, RL, and other methods such as evolutionary algorithms, discussed in Sections III-A through III-D, respectively; and (ii) the use of distributed learning approaches, which are essential for enhancing scalability, reliability, and performance in dynamic and resource-constrained wireless environments, detailed in Section III-E. Additionally, in Section III-F, we provide a more detailed analysis of balancing KPI trade-offs in wireless communications using advanced ML paradigms and distributed learning approaches, exploring how different algorithms manage multiple network resources to enhance performance across various scenarios.

For Sections III-A–III-D, the categorization of the selected works by ML paradigm is organized in accordance with Tables III to VI. These tables are structured to organize and synthesize the most relevant data from the analyzed papers, with the details being described below.

The first column assigns a unique reference number to each paper, with an asterisk (*) or double asterisks (**) indicating works using two or three distinct learning paradigms, respectively. These papers appear in multiple tables, and their performance data should be analyzed holistically across paradigms. The second column provides a brief reference description. The third column summarizes the implementation or test scenario, detailing the application and/or network features.

The fourth column highlights the network KPIs addressed, including PowCon, throughput (Thrput), latency, NTM, and CRM, which reflect improvements tied to the critical resource allocation problems in wireless communications previously discussed in Section II-B. A "✓" symbol identifies the KPIs optimized by each paper.

The fifth column lists the implemented algorithms, while the sixth one describes the evaluation metrics used to assess performance. It includes, for example, supervised model metrics, cumulative rewards in RL, graphs illustrating the statistical probability of specific parameters, such as outage for non-scheduled traffic or data packet delivery, experimental tests conducted through simulations, or even real-world scenarios, among others.

TABLE II
SUMMARY OF PAPERS SELECTED FOR THE STATE-OF-THE-ART OF OUR TECHNICAL SURVEY

Database Platform	Publication Year						Total per Database Platform
	2019	2020	2021	2022	2023	2024	
IEEE Xplore	5	14	9	5	5	9	47
Science Direct	0	3	1	4	5	6	19
SpringerLink	7	8	11	14	18	6	64
Total per Year	12	25	21	23	28	21	**130**

TABLE III
TABLE OF DATA EXTRACTION FROM PAPERS CONSIDERING UL PARADIGM

#	Reference	Application Scenario	Optimizations Addressed (KPIs Analysis)					Unsupervised Learning Algorithm	Assessment Metric	Performance Gain				
			PowCon	Thrput	Latency	NTM	CRM			EE	SE	LE	PDE	CE
1*	[101]	C-RAN					✓	K-means	Accuracy					✓
2*	[102]	D2D multicast networks	✓	✓				SOM	EE-throughput graphs	✓	✓			
3*	[103]	Networks with flying BSs		✓				K-means++	MSE, avg throughput, accuracy		✓			✓
4	[104]	Non-orthogonal SATCOM	✓					DNN	CDF of sum-rate graph					✓
5*	[105]	UDN	✓	✓		✓		K-means	EE graph, throughput graph, QoS, convergence speed	✓	✓		✓	
6	[106]	Cognitive radio system	✓					DNN, Ensemble DL	Sum SE and comp. time graphs		✓			✓
7*	[107]	Wireless sensor networks				✓		K-means	Accuracy, precision, recall, false positive rate, F1-score			✓	✓	✓
8*	[108]	Net. slicing for C-RAN				✓	✓	GAN	SE, delay graphs		✓	✓	✓	✓
9*	[109]	FMCW radar systems					✓	DCGAN	mAP, RQS					✓
10	[110]	RAN design in ultra-dense urban scenario		✓		✓		Constrained K-means, SOM	Area SE, and average throughput per user graphs		✓			
11*	[111]	Mobile CC environment					✓	K-means	Delay, machine use graphs					✓
12**	[112]	IRS-Aided MISO-NOMA	✓	✓				K-GMM	Sum rate vs IRS, power graphs	✓	✓			
13**	[113]	MISO-NOMA net. w/ IRS	✓	✓				K-GMM	SE vs power, IRS graphs	✓	✓			
14*	[114]	Fiber wireless access networks			✓	✓		K-means	PDR, resource use, blocking prob., traffic offloading ratio and delay graphs			✓	✓	
15	[115]	MIMO-NOMA	✓	✓	✓	✓		SVD with DL	Cost function, SNR, max. power, and no. of user graphs based on BER, SE, throughput, MSE, BLER, NMSE, and effective/sum rates	✓	✓	✓	✓	

Finally, the last column presents the results, including performance gains classified as EE, SE, LE, PDE, and CE, with a "✓" symbol indicating improvements explicitly reported in the papers.

The aggregated information presented in Sections III-A–III-E will serve as a foundation for subsequent qualitative and quantitative analyses, enabling the derivation of insights into current research trends and the identification of potential areas for future advancement. All acronyms and terminologies presented in Tables III to VII, related to application scenarios, ML algorithms, and evaluation metrics, are comprehensively defined in the Appendix to ensure clarity and consistency throughout the discussion.

To consolidate key findings, provide highlight insights, and outline directions for future research, a summary subsection of the observed state-of-the-art is also included at the conclusion of this section. It aims to investigate and identify patterns from the collected data and derive lessons learned, including, for instance: (i) limitations and challenges in developing ML solutions for optimizing multiple KPIs within a single system; (ii) performance analysis of the selected works concerning efficiency gains; (iii) motivations behind the broader use of specific ML algorithm types for certain scenarios compared to others; (iv) and key synergies and trade-offs among the different KPIs analyzed.

Subsequently, the following subsections are presented, respectively: (i) the categorization of the papers based on the different ML paradigms studied (UL, SL, RL, and others); (ii) the classification and analysis of the selected papers employing distributed learning approaches; (iii) balancing KPI trade-offs in wireless networks with advanced ML paradigms and distributed learning; and (iv) a summary and lessons learned.

A. UL Paradigm

In this section, we provide the state-of-the-art considering the selected works related to the use of UL paradigm applied to resource allocation in wireless communications. Table III summarizes the data gathered from 15 papers addressing UL-related algorithms. From Table III, we can observe a variety of scenarios applied to the UL paradigm, such as Cloud-Radio Access Network (C-RAN), Wireless Sensor Network (WSN), and Ultra-Dense Networks (UDNs). In terms of problems commonly discussed in wireless communications and solved using UL algorithms, we can highlight solutions aimed at reducing energy consumption and increasing throughput, resulting, consequently, in a higher number of works focused on improving network performance from the standpoint of spectral and computational efficiencies.

We can identify the most used metrics graphs of energy and spectral efficiency analysis and the average throughput for network performance evaluation, followed by metrics for assessing the applied model, such as loss functions (e.g., MSE) or accuracy.

B. SL Paradigm

Table IV summarizes the data gathered from the surveyed papers related to using the SL paradigm. As indicated, 64 studies using SL approaches were surveyed, comprising 36 distinct supervised algorithms. Some examples of wireless networks explored include UDNs, Data Center Network (DCN), Fog-computing-based Radio Access Network (F-RAN) WSN, C-RAN, and Cognitive Radio Network (CRN). Indeed, we can notice a wide applicability of SL algorithms in 5G and Beyond fifth Generation of Wireless Communication

Technology (B5G) networks, focused on industrial and IoT scenarios. Overall, the works involving the use of supervised algorithms address the network KPIs considered in this survey in a well-distributed manner, offering solutions to a wide range of problems in wireless communications. We can highlight a significant use of SL approaches to reduce energy consumption and improve efficiency in managing the network traffic, achieving performance gains in all types of efficiency explored.

Furthermore, Table IV shows that different assessment metrics were addressed, such as graphs related to efficiency gain (e.g., SE and EE), error rate (e.g., Symbol Error Rate (SER) and BER), and computational resources (e.g., computational time and processing delay) to evaluate network performance. Regarding the evaluation of the model trained, different loss functions were considered (i.e., Root Mean Square Error (RMSE) and MSE), in addition to diverse metrics to assess the quality and behavior of the predictions of the SL algorithms addressed (i.e., accuracy, precision, recall, F1-Score).

C. RL Paradigm

The works centered in RL approaches are reviewed in Table V. Here, we have a total of 61 surveyed papers with ML proposals regarding the RL paradigm, presenting a wide variety of solutions through 21 distinct reinforcement algorithms. Similar to the state-of-the-art for SL works, it can be observed that RL proposals also seek to solve a diverse range of issues discussed here for wireless systems, including solutions for reducing energy consumption and improving traffic management. These studies also achieved performance gain results in all the efficiency perspectives analyzed.

As indicated in Table V, the scenarios that explore the use of RL algorithms are also diverse, where we can highlight, as with the different types of networks already mentioned in previous paradigms, MEC-based networks and 5G scenarios focused primarily on reliable and low-latency communications. As assessment metrics, a variety of network evaluation means can be noticed, such as error and loss graphs (e.g., BER and Packet Loss Ratio (PLR)), computational analyses of time and processing delay, and analysis graphs of network KPIs, such as throughput and energy. However, as expected, the works focused on RL present as a focus for evaluating the trained model, the approach of reward based on training, and the algorithm's behavior over time.

D. Other Learning Paradigms

From Table VI, we can observe 18 selected articles addressing learning paradigms not previously mentioned for resource allocation optimization. It includes evolutionary methods, specific distributed learning approaches, and custom or unique ML algorithms. The application scenarios involving the use of these algorithms include CRN, WSN, C-RAN, Wireless Body Area Network (WBAN), and SDN, among others. With a total of 10 distinct algorithms developed, the majority of the papers analyzed implement evolutionary algorithms, with a major focus on solving problems in wireless communications aimed at reducing energy consumption.

Consequently, the results obtained favor energy and SE as performance gains in the network. For this purpose, different metrics were addressed in the studies to assess network performance, such as graphs of the discussed efficiencies (e.g., EE and SE), graphs for analyzing energy consumption and computational processing of the network, and graphs for evaluating data error and loss (e.g., PER, and blocking and dropping probability graphs).

E. Distributed Learning Approaches

Table VII provides a comprehensive analysis of 23 of the 130 papers surveyed that adopt distributed learning approaches to address resource allocation optimization challenges in wireless communication systems. The first column identifies the paper reference. The second column details the ML algorithms employed in combination with distributed learning approaches for each paper. Lastly, the third column provides a description of the contributions of distributed approaches combined with the ML algorithms used in these works to enhance the performance and outcomes of the proposed solutions.

According to the second column of Table VII, it is evident that various ML paradigms, including UL, SL, and RL, have been employed. This highlights the versatility of distributed learning, demonstrating its adaptability across different ML techniques. These paradigms are often tailored to the unique requirements of distributed setups, such as preserving privacy, reducing latency, or enhancing CE, showcasing how distributed learning can bridge algorithmic diversity with network optimization needs.

Moreover, the third column provides valuable insights into how distributed learning contributes to resource allocation optimization. The articles span various approaches, such as employing RL to enable local decision-making by network agents, using FL to maintain privacy while training models, and integrating technologies like blockchain and edge computing to enhance robustness and scalability. A recurring theme is using distributed agents that independently learn and optimize tasks while collaborating to refine global policies, such as in hierarchical RL frameworks or asynchronous FL setups. This illustrates a consistent effort to leverage distributed learning for scalability, efficiency, and adaptability in dynamic environments.

From the analysis, a few patterns and gaps emerge. RL appears as a dominant paradigm for distributed decision-making, often paired with hierarchical or multi-agent structures to manage complex environments. FL is frequently used to address privacy and data locality concerns, particularly in IoT and edge network scenarios. However, some challenges are presented, such as effectively handling communication overhead among distributed agents and ensuring convergence in heterogeneous network settings. Additionally, while many studies focus on EE and latency reduction, there is a limited exploration of the integration between distributed learning and other advanced paradigms and hybrid ML techniques. Future research could address these gaps, focusing on cross-paradigm integration, unreliable network robustness, and improved resource-constrained environment methods.

TABLE IV
TABLE OF DATA EXTRACTION FROM PAPERS CONSIDERING SL PARADIGM

#	Reference	Application Scenario	Optimizations Addressed (KPIs Analysis)					Supervised Learning Algorithm	Assessment Metric	Performance Gain				
			PowCon	Thrput	Latency	NTM	CRM			EE	SE	LE	PDE	CE
1	[116]	Hybrid Intra-DCN				✓		NARNN	MAE and MSE metrics				✓	
2*	[117]	C-RAN	✓	✓				LSTM	PowCon consumption graph	✓	✓			
3	[118]	AP selection strategy in wireless networks		✓				Random Forest	Throughput via iperf			✓		
4	[119]	UDN		✓				ANN	RMSE metric		✓			✓
5	[120]	Wireless-UDN	✓					ANN, SVM	Resource consumption graph	✓				
6	[121]	IoT scenario	✓		✓			ANN (SGD algorithm)	MSD metric	✓		✓		✓
7*	[122]	Streaming over F-RAN		✓		✓		LSTM	Bit rate, rebuff. ratio		✓			
8	[123]	Wireless comm.				✓		SVM	Accuracy metric				✓	
9	[124]	Multi-cell OFDM net.	✓		✓	✓		DNN	SE curve, CPU time, accuracy		✓	✓		✓
10	[125]	VNF			✓		✓	Linear, Ridge Regressions, Random Forest, Gradient Boosting, SVR, MLP	RMSE, CPU usage, total delay graph			✓		✓
11	[126]	Network slicing in vehicular networks			✓			ConvLSTM	Network slicing delay graph			✓		
12*	[101]	C-RAN					✓	SVR, Linear Regression, ANN	Accuracy					✓
13	[127]	Vehicular comm. net.	✓		✓			Logistic Regression	Latency, BER, power graphs	✓	✓			
14**	[112]	IRS-Aided MISO-NOMA	✓	✓				LSTM	Sum rate vs IRS, power graphs	✓	✓			
15	[128]	IoV					✓	RNN	Accuracy					✓
16	[129]	Wireless net. virtualization				✓	✓	LSTM	Delay, energy, utility graphs		✓			✓
17	[130]	Mobility Multi-cell OFDM	✓		✓	✓		DNN	SE curve, CPU time, accuracy		✓	✓		✓
18*	[103]	Networks with flying BSs		✓				DNN	MSE, avg throughput, accuracy		✓			✓
19	[131]	Wireless network edge					✓	LMS, MLP, SVM	Accuracy, NMSE, MSD metrics					✓
20	[132]	Multi-access edge comp.					✓	FNN	MSE, histogram, CDF graphs					✓
21**	[113]	MISO-NOMA net. w/ IRS	✓	✓				LSTM	SE vs power, IRS graphs	✓	✓			
22	[133]	Downlink MC-NOMA net.			✓			ANN (SGD-based)	MSE metric			✓		✓
23	[134]	Wireless sensor networks				✓		CART	accuracy, F1-score, recall heat, confusion matrices; delay				✓	✓
24	[135]	Intelligent IoT app.					✓	CNNs	graphs; inference accuracy					✓
25*	[136]	LoRaWAN for IoT net.	✓			✓		Random Forest, SVM, Linear Regression, LDA, Gaussian Neive Bayes, K-NN	Accuracy, packet reception ratio, energy/packet	✓			✓	
26	[137]	LoRa-enabled IoT app.				✓		LSTM, GRU, SVM, DNN	Accuracy, energy consumption, packet success ratio graphs	✓			✓	
27	[138]	Manufacturing systems	✓					LSTM	Energy consumption table	✓				
28	[139]	NFV net. architectures				✓	✓	LSTM	RMSE, AMAE metrics					✓
29	[140]	CC using IoT					✓	Random Forest, SVM, ANN, Decision Tree, GBR	Accuracy					✓
30*	[107]	Wireless sensor net.				✓		Linear Regression, ID3, ANN, CART	Accuracy, precision, recall, false positive rate, F1-score			✓	✓	✓
31	[141]	5G network slicing				✓		Random Forest, SVM, MLP, GBR	MAE, RMSE metrics	✓				
32*	[142]	NOMA framework	✓		✓		✓	CNN (for FEEL)	Energy consumption graphs	✓		✓		✓
33	[143]	Beyond 5G RAN slicing					✓	ARIMA, LSTM, DNN Gradient Boosting	MAPE Cost and CPU time graphs					✓
34	[144]	Wireless sensor networks	✓					Ensemble ML	RMSE, accuracy	✓				
35	[145]	C-RAN	✓	✓			✓	SVM	Power consumption, computing resources, and data rate graphs	✓	✓			✓
36	[146]	Wireless sensor network					✓	ANN	RMSE, MPE, MAPE, R, T-test, p-value					✓
37	[147]	Massive MIMO in 5G	✓					RNN	EE graph	✓				
38*	[109]	FMCW radar systems					✓	F-RCNN	mAP, RQS					✓
39	[148]	SS in Cognitive Radio				✓		ANN	Accuracy, detection probability		✓			
40	[149]	Cloud-assisted IoT comm.					✓	Logistic Regression	Delay, response time, BW graphs					✓
41	[150]	OFDM comm. systems			✓			ANN, RNN, LSTM	SER and BER graphs			✓	✓	
42	[151]	AP selection in heterogeneous fixed wireless networks	✓					K-NN, ANN	Confusion matrix, accuracy, CDFs (received signal power, SNR, etc.)	✓				
43	[152]	Malicious user identification for cognitive radio networks				✓		SVM	Graphs of probability and energy of detection for malicious user id.		✓			
44*	[153]	Cognitive radio networks				✓		BPNN, ANFIS	Accuracy, spectrum decision graph, blocking and dropping prob. graphs		✓			
45	[154]	Constrained C-RAN	✓		✓			CNN	EE, computing time, delay graphs	✓		✓		✓
46	[155]	5G heterogeneous networks	✓			✓		DL based Encoder with NN	Throughput, PLR graphs accuracy, precision, recall, F1-score	✓			✓	
47	[156]	Next generation comm. net.	✓	✓				ANN	BER, SER, throughput, SNR info.	✓	✓		✓	
48	[157]	GERAN/EUTRAN		✓		✓		NARNN	Error predict., throughput graphs		✓		✓	
49	[158]	Cognitive radio networks					✓	DNN, SVM	Secrecy rate, outage prob. graphs					✓
50*	[159]	SDN-based AP selection in next-generation wireless networks				✓		MLP	CDF of loss graph, Q-Score calculation, accuracy				✓	
51	[160]	Video streaming over wireless networks			✓	✓		LASSO, SVR, Ridge Regression, ANN	APSNR, SSIM, MSAD, VQM, and MSE graphs; MOS			✓	✓	
52	[161]	MIMO wireless systems				✓		ANN	Confusion matrix, MSE, MAE, F1-score, accuracy				✓	
53*	[162]	CR-based IoT				✓		FRST-SVM	PER, channel utilization graphs		✓		✓	
54	[163]	OFDM/NOMA-assisted B5G comm. systems	✓	✓				SVM	Sum capacity, power consumption graphs, AMC success rate	✓	✓			
55	[164]	Antenna selection in MIMO systems	✓	✓			✓	CNN	EE, capacity and accuracy graphs	✓	✓			✓
56*	[165]	uRLLC in industrial wireless networks		✓	✓			DNN	BER, latency, PLR, throughput, timing complexity graphs			✓	✓	✓
57	[166]	NOMA networks	✓	✓			✓	Ensemble ML (combining CNN, FNN and Random Forest)	Sum rate by no. of sub-channels and users, power predictions, R-Squared, MSE, RMSE, accuracy, time delay	✓	✓			✓
58*	[167]	UAV communications	✓	✓		✓		ESN (RNN)	Throughput and UAV trajectories graphs; reward; RMSE; comp. time	✓	✓	✓		✓
59	[168]	5G networks		✓	✓	✓		LSTM	Accuracy, throughput, handover latency, success rate graphs		✓	✓	✓	
60	[169]	B5G/6G networks	✓					ENN-CAWOA	MAE, MSE, MAPE, RMSE, energy demand, prediction of resources-based load graphs	✓				✓
61	[170]	UAVs for 5G and beyond nets.	✓	✓		✓		ANN	PLR variation, throughput CDF graphs	✓	✓		✓	
62	[171]	5G networks		✓				DNN, SVM, Random Forest, B-KNN	Train./exec. times graph, throughput, accuracy, precision, recall, F1-score		✓			
63	[172]	IoT networks			✓	✓	✓	RNN	Accuracy, loss, transaction time, and energy consumption graphs			✓	✓	✓
64*	[173]	6G IoT networks		✓	✓		✓	CNN	Computational, response, and aggregation times; accuracy; throughput; latency; delay; and delivery ratio			✓	✓	✓

F. Balancing KPI Trade-Offs in Wireless Networks With ML Paradigms and Distributed Learning Approaches

In wireless communication systems, optimizing KPIs such as throughput, PowCon, latency, NTM, and CRM requires addressing complex trade-offs. For instance, increasing throughput often entails higher PowCon or latency due to the need for more robust signaling or additional computational resources to handle the increased traffic demand. These

TABLE V
TABLE OF DATA EXTRACTION FROM PAPERS CONSIDERING RL PARADIGM

#	Reference	Application Scenario	Optimizations Addressed (KPIs Analysis)					Reinforcement Learning Algorithm	Assessment Metric	Performance Gain				
			PowCon	Thrput	Latency	NTM	CRM			EE	SE	LE	PDE	CE
1	[174]	uRLLC application	✓			✓		Q-Learning	Traffic outage probability	✓	✓		✓	
2	[175]	UDN	✓					Event-triggered Q-Learning	Cumulative reward	✓				
3	[176]	IoT networks	✓		✓			Q-Learning, DQN	Power-latency cost reward	✓		✓		
4*	[122]	Streaming over F-RAN		✓		✓		DRL	reward, bit rate, rebuff. ratio		✓			
5	[177]	Multi-UAV comm.	✓					Q-learning, DQN	reward, system utility graph	✓				
6**	[112]	IRS-Aided MISO-NOMA	✓	✓				DQN	Sum rate vs IRS, power graphs, reward	✓	✓			
7	[178]	MEC-based networks					✓	Q-Learning, Multistack RL	Reward, max. delay graph					✓
8*	[102]	D2D multicast networks	✓	✓				Q-Learning	EE-throughput graphs, reward	✓	✓			
9	[179]	Remote education		✓		✓		MARL	Mean and STD, QoS, reward			✓	✓	
10	[180]	Satellite net. for IoRT	✓					ACRL, Q-Learning	Reward, data vs battery graph	✓				
11**	[113]	MISO-NOMA net. w/ IRS	✓	✓				DQN	SE vs power, IRS graphs, reward	✓	✓			
12	[181]	uRLLC vehicular networks				✓		DQN	Reward, data transfer reliability				✓	
13*	[105]	UDN	✓	✓		✓		Q-Learning (MARL)	Reward, EE graph, throughput graph, QoS, convergence speed	✓	✓		✓	
14*	[136]	LoRaWAN for IoT networks	✓			✓		EXP4	Accuracy, reward, packet reception ratio, energy/packet	✓			✓	
15	[182]	Cognitive M2M networks	✓					Q-Learning, DRL	Graphs of devices EE, reward	✓				
16	[183]	5G-oriented distrib. CC		✓		✓		Q-Learning	Reward, user throughput graph		✓		✓	
17	[184]	MEC 5G IoT networks					✓	ACDRL, DQN	Energy, latency graphs; reward					✓
18*	[108]	Net. slicing for C-RAN				✓	✓	DRL	SE, delay graphs, reward		✓	✓	✓	✓
19	[185]	Digital twins in 6G net.			✓	✓		SACRL, DDPG, CRL	Reward; flow rate, delay graphs			✓	✓	
20	[186]	Indoor SCN				✓		Q-Learning	Reward, avg. utility CDF				✓	
21	[187]	LAA/WLAN coexist. net.				✓		Q-Learning	Reward, throughput graphs		✓			
22	[188]	Wireless sensor networks	✓				✓	Nash Q-Learning	Convergence time and average power consumption graphs, reward	✓				✓
23	[189]	Cooperative relay networks		✓		✓		RL algorithm (unspecified)	Reward, avg. throughput graph	✓			✓	
24*	[111]	Mobile CC environment					✓	DRL	Delay, machine use graphs, reward					✓
25	[190]	Security-aware MEC					✓	DRL	Delay, energy consumption graphs, reward					✓
26	[191]	Heterogeneous edge computing					✓	A3C, DQN, DDQN , Prioritized Relay DQN	Number of tasks and consumption graphs, reward					✓
27	[192]	Real-time applications in edge-cloud computing					✓	DQN	Cumulative reward; app. response time, energy consumption graphs					✓
28	[193]	Power line, wireless comm.	✓					Q-Learning	PSD, outage prob. graphs, reward	✓	✓			
29	[194]	MEC systems					✓	Q-Learning, DQN	Sum cost of tasks graphs, reward					✓
30	[195]	Wireless sensor networks	✓			✓		Nash Q-Learning	Reward, mean lifetime of nodes and engaged sensor device graphs	✓			✓	
31	[196]	Dense cloud NOMA SCN	✓			✓		MARL	Net. MOS and throughput graphs		✓			
32	[197]	Vehicular networks		✓	✓	✓		Q-Learning	Throughput, avg. delay and packet delivery ratio graphs, reward		✓	✓	✓	
33	[198]	DCNs		✓	✓	✓		DQN (DRL)	Throughput, RTT, and CDFs of link BW utilization graphs; reward		✓	✓	✓	
34	[199]	B5G-ultra-dense wireless cellular networks	✓					DQN (DRL)	Reward, MSE, EE graphs	✓				
35	[200]	WBAN	✓	✓				RL with Fuzzy-Logic	EE, throughput, PLR graphs	✓			✓	
36	[201]	D2D underlay networks	✓			✓		DDQN (MARL)	Reward, sum data rate graphs	✓	✓			✓
37	[202]	Next generation multi operator cellular networks		✓	✓	✓		DQN	Reward; PDR; and throughput, delay, resource utilization graphs	✓	✓	✓	✓	
38	[203]	D2D mobile edge networks					✓	DDPG, DQN	Offloading ratio, acceleration ratio, and hit ratio graphs, reward					✓
39	[204]	MEC networks					✓	DQN, ACDRL	latency and energy graphs; reward					✓
40	[205]	IRS-aided covert wireless communication systems	✓					DDPG	Reward, covert rate graphs	✓				
41	[206]	MEC					✓	Q-Learning	Energy consumption, processing time delay graphs; reward					✓
42	[207]	D2D communications	✓	✓				Distributed RL, SARSA	Throughput, SE, EE, fairness, and sum of rate graphs, reward	✓	✓			
43	[208]	MEC					✓	DQN	Energy consumption and latency graphs; reward					✓
44	[209]	CSMA/CA in wireless comm. (WLANs, MANETs)			✓			Q-Learning	Throughput, PLR, speed, and delay graphs; reward			✓	✓	
45	[210]	Wireless sensor networks				✓		Q-Learning (MARL)	Packet delivery ratio, latency, energy consumption graphs; reward	✓		✓	✓	
46	[211]	MTC random access in LTE-Advanced				✓		Q-Learning	Success prob., no. of retransmissions, service time graphs; reward				✓	
47*	[159]	SDN-based AP selection in next-generation wireless networks				✓		DDQN , DQN, Q-Learning	CDF of Q-score, loss, and reward graphs, accuracy, MSE				✓	
48	[212]	SDVN architecture			✓	✓		DQN	delay, packet loss graphs; reward			✓	✓	
49	[213]	UAV-assisted 5G mMTC network slicing	✓					DDQN, DQN, Q-Learning	EE graphs, reward	✓				
50*	[165]	uRLLC in industrial wireless networks		✓	✓			DQN	BER, latency, PLR, throughput, timing complexity graphs; reward			✓	✓	✓
51	[214]	Dynamic video streaming				✓		DQN, DQNReg	QoE parameters graphs, reward				✓	
52	[215]	RAN slicing (uRLLC, eMBB)		✓	✓			DQN	Reward, avg. throughput and delay		✓	✓		
53	[216]	LPWAN	✓	✓	✓		✓	DQN	Throughput, delay, exec. time, EE, energy consumption, SINR, tx. rate, remaining resource graphs; reward	✓	✓	✓	✓	✓
54*	[217]	UDNs	✓	✓		✓	✓	DQN	SE, EE, and fairness graphs; reward	✓	✓		✓	✓
55*	[218]	O-RAN (uRLLC, eMBB)	✓	✓	✓	✓	✓	DRL	SE, fairness, data rate, and tx. error probability graphs; reward		✓	✓	✓	✓
56*	[219]	SD-IoUAVs	✓		✓		✓	A3C	Accuracy, system cost, latency and energy consumption graphs; reward	✓		✓		✓
57*	[167]	UAV communications	✓	✓		✓		NAF-based DRL	Throughput and UAV trajectories graphs; reward; RMSE; comp. time	✓	✓	✓		✓
58*	[220]	Wireless networks					✓	Model-based RL	Loss time, energy consumption, and completion time of FL graphs for BW and CPU frequency of users					✓
59	[221]	Slicing in 5G wireless communications		✓	✓	✓	✓	STAC	Prob. of slides acceptance, time complexity, and overall utility vs. slice requests graphs; reward		✓	✓	✓	✓
60*	[222]	Networks with SWIPT and MC-NOMA	✓		✓		✓	DDPG	Reward, LOS, NLOS, data size, energy consumption, no. of UEs graphs	✓		✓		✓
61	[223]	Next-generation cloud-enabled non-terrestrial networks	✓	✓	✓	✓		DDQN (DRL)	EE vs. PowCon by nodes and throughput vs. max. transmit power graphs	✓	✓	✓		

challenges can be addressed differently depending on the ML paradigm applied, making it crucial to understand how each approach can contribute to balancing these KPIs.

UL algorithms excel at identifying patterns in unlabeled data, enabling adaptive approaches to KPI optimization in wireless networks. For example, in C-RANs, the K-Means

TABLE VI
TABLE OF DATA EXTRACTION FROM PAPERS CONSIDERING OTHER ML PARADIGMS

#	Reference	Application Scenario	Optimizations Addressed (KPIs Analysis)					Other ML Algorithm	Assessment Metric	Performance Gain				
			PowCon	Thrput	Latency	NTM	CRM			EE	SE	LE	PDE	CE
1*	[117]	C-RAN	✓	✓				GA	Power consumption graph	✓	✓			
2*	[142]	NOMA framework	✓		✓		✓	FedAvg (FEEL)	Energy consumption graphs	✓		✓		✓
3	[224]	SDN-enabled fog comp.					✓	Collaborative ML	Delay, power graphs					✓
4	[225]	WBAN	✓					Distributed learning	Transmission power cost graph	✓				
5*	[153]	Cognitive radio networks				✓		GA	spectrum decision graph, blocking and dropping probability graphs		✓			
6	[226]	UAV-assisted WSN	✓					GA	Harvested energy and SE graphs	✓	✓			
7	[227]	Wireless sensor networks	✓					GA, MA	Energy consumption-related graphs	✓				
8	[228]	mmWave Massive MIMO	✓				✓	EDA, GA	EE, SE graphs	✓				✓
9*	[162]	CR-based IoT				✓		GA	PER, channel utilization graphs		✓		✓	
10	[229]	Wireless FL networks	✓		✓		✓	FL	Max. energy consumption, local dataset size, max. comput. frequency, latency	✓		✓		✓
11	[230]	Unreliable and resource-limited wireless networks	✓		✓	✓	✓	FL-GR	Accuracy; impacts of energy and delay in wireless parameters (e.g., no. of RBs)	✓		✓		✓
12*	[217]	UDNs	✓	✓		✓	✓	NSGA-II	SE, EE, and fairness graphs; reward	✓	✓		✓	✓
13*	[218]	O-RAN (uRLLC, eMBB)	✓	✓	✓	✓	✓	Distributed Learning	SE, fairness, data rate, and tx. error probability graphs; reward		✓	✓	✓	✓
14*	[219]	SD-IoUAVs	✓		✓		✓	A3FN	Accuracy, system cost, latency and energy consumption graphs; reward	✓		✓		✓
15*	[220]	Wireless networks					✓	FL	Loss time, energy consumption, and completion time of FL graphs for BW and CPU frequency of users					✓
16*	[114]	Fiber wireless access net.			✓	✓		GA	PDR, resource use, blocking prob., traffic offloading ratio and delay graphs			✓	✓	
17*	[222]	Networks with SWIPT and MC-NOMA	✓		✓		✓	FL	Reward, LOS, NLOS, data size, energy consumption, no. of UEs graphs	✓		✓		✓
18*	[173]	6G IoT networks		✓	✓		✓	FL	Computational, response, and aggregation times; accuracy; throughput; latency; delay; and delivery ratio			✓	✓	✓

algorithm can cluster devices based on similar traffic or energy consumption patterns. This clustering allows for efficient resource allocation, reducing PowCon in low-priority nodes without significantly compromising overall network throughput. Additionally, algorithms such as Generative Adversarial Networks (GANs) can simulate complex network environments, enabling the prediction of future congestion or identifying high-demand areas. This facilitates proactive adjustments to allocated resources, balancing PowCon and throughput effectively.

In SL, algorithms utilize historical data with multiple KPIs already optimized and balanced as a reference to train models capable of predicting configurations that maximize performance. For instance, Deep Neural Networks (DNNs) can learn patterns of user mobility and traffic demands to recommend optimal resource allocation strategies that simultaneously optimize throughput and PowCon. However, the success of these approaches is heavily dependent on the quality and robustness of the datasets used for training. In dynamic network environments, the need for frequent retraining to adapt the model to changes can pose a significant challenge, limiting the effectiveness of SL under rapidly evolving conditions.

To address dynamic and continuously changing environments, RL offers a powerful approach to managing trade-offs in wireless networks. RL algorithms leverage reward structures to guide the learning process toward solutions that optimize multiple KPIs simultaneously. For example, in a C-RAN scenario, an RL agent can be rewarded based on a weighted combination of throughput maximization and PowCon minimization, allowing the network to dynamically adapt to fluctuating traffic conditions and energy demands. Through techniques such as Deep Q-Networks (DQNs), RL agents can learn policies that directly account for network variability, continuously refining resource allocation strategies. By exploring and exploiting the environment, RL ensures that trade-offs are not static but evolve dynamically with the network, ensuring optimal KPI balance under changing conditions.

Finally, hybrid approaches and distributed learning methods provide significant opportunities to enhance KPI trade-offs in wireless networks. Combining paradigms, such as using GA to explore large solution spaces and RL to refine these solutions, can lead to more effective optimization strategies. For instance, in the context of throughput and PowCon, GAs can identify promising initial configurations, which are then dynamically adjusted by RL agents to account for real-time conditions. Moreover, distributed learning frameworks, where multiple agents or nodes collaborate to optimize KPIs, enable scalable solutions for large-scale networks. Distributed RL, for example, can allow multiple RL agents operating at different network layers—such as edge devices and central controllers—to share insights, reducing latency and improving resource allocation efficiency. Similarly, distributed supervised or unsupervised learning approaches can aggregate data across the network to refine models while minimizing computational overhead at individual nodes. These distributed methods enhance robustness and scalability, ensuring that network-wide trade-offs are balanced across diverse and dynamic environments.

In summary, each ML paradigm offers unique strengths in addressing the challenges of KPI optimization and trade-offs in wireless networks. UL algorithms uncover latent patterns, SL models provide predictive capabilities based on historical data, RL adapts to dynamic environments, and hybrid approaches leverage the strengths of multiple paradigms. By incorporating distributed learning strategies, these approaches can be further enhanced, enabling efficient and scalable optimization solutions that balance trade-offs across complex and dynamic wireless communication scenarios.

TABLE VII
TABLE OF DATA EXTRACTION FROM SURVEYED PAPERS CONSIDERING DISTRIBUTED LEARNING APPROACHES

Ref.	Algorithm(s) used	Description of the distributed learning contribution for the proposed solution
[105]	K-Means and Q-Learning	The proposed strategy employs distributed MARL, where each FBS acts as an agent using a cooperative Q-learning algorithm, leveraging cloud-stored historical data for collaborative decision-making, improving EE, and reducing computational complexity in UDNs.
[108]	GAN and DRL	The article proposes a Twin-GAN-based DRL framework for joint communication and computational resource allocation in hybrid C-RAN and O-RAN, with distributed DRL agents using GANs to manage bandwidth and computational resources independently while collaborating to optimize 5G network performance in real-time through decentralized decision-making.
[138]	LSTM	The proposed architecture integrates big data with distributed ML for real-time processing and optimization in large-scale manufacturing systems, using cloud-based platforms to aggregate and analyze data collected at the factory floor, applying Long Short-Term Memorys (LSTMs) NNs for energy consumption forecasting and anomaly detection.
[142]	FedAvg with CNN)	The proposed framework leverages FEEL enabled by NOMA, where devices perform local training and share model updates with an edge server, ensuring data privacy while optimizing energy consumption and communication resource allocation in a distributed learning setup.
[173]	FL with CNN	The article proposes a distributed learning framework in 6G, combining FL and blockchain, where IoT devices train local models while preserving privacy and security, and use distributed hash tables for intelligent resource allocation, network slicing, and optimized QoS in 6G IoT networks.
[179]	MARL	The proposed Hierarchical MARL framework improves video resource allocation in remote educational networks by dividing decision-making between a master controller prioritizing video classes and slave controllers independently, optimizing scheduling rules, enabling distributed, hierarchical learning to enhance service quality for multiple users.
[183]	Q-Learning	The proposed system employs cooperative Q-learning, where small-cell BS act as distributed agents exchanging local state information to independently optimize resource allocation while periodically sharing updates with a centralized controller for global policy refinement.
[184]	ACDRL and DQN	The model uses distributed natural Actor-Critic-based DRL, with edge devices collaborating to manage computational tasks and communication resources locally, while a central server aggregates their updates for global optimization in real-time.
[188]	Nash Q-Learning	The proposed algorithm applies distributed RL in wireless sensor networks, where each node learns and decides locally on actions like activating or hibernating to optimize coverage and connectivity without relying on a central controller.
[196]	MARL	The proposed multi-agent learning framework for dense small networks with NOMA enables distributed agents (small BSs) to learn and execute resource allocation decisions locally, while the cloud aggregates updates to refine global policies and improve convergence.
[199]	DQN (DRL)	The proposed distributed DRL model for UDN power control trains policies centrally, then enables small base stations to adjust power levels autonomously based on their local states and neighbor interactions, optimizing EE in dense environments.
[201]	DDQN (MARL)	The proposed scheme uses distributed DDQN-based DRL for D2D networks, where each D2D pair acts as an independent agent making local decisions on sub-channel and power allocation, while a central controller refines policies during training for distributed execution.
[207]	Distributed RL with SARSA	The proposed SARSA-based reinforcement learning technique enables distributed D2D devices to allocate spectrum and power independently using local observations, while sharing value functions with neighbors to optimize network-wide efficiency.
[210]	Q-Learning (MARL)	The article proposes a distributed RL approach for energy-efficient routing in WSNs, where multiple agents operate collaboratively to learn the network's current state, select optimal paths based on Q-values, and share future rewards, improving EE and network lifetime through decentralized decision-making.
[212]	DQN	The proposed architecture consists of an intelligent and resizable control plane for SDVN that uses DRL to dynamically deploy distributed controllers in vehicular networks, where the learning agent optimizes controller placement by balancing load based on local traffic conditions in a distributed manner.
[218]	Distributed Learning with DRL	The article proposes a distributed learning framework leveraging DRL within the O-RAN architecture, where distributed agents at the network edge handle eMBB-uRLLC multiplexing in real-time, while a central training unit optimizes global resource allocation policies.
[219]	A3FN with A3C	The article proposes an asynchronous FL framework for the IoUAVs, where distributed learning is employed to train local models on UAV and IoT nodes, transmitting parameters to a mobile-edge computing server for global aggregation while optimizing resource allocation and reducing latency.
[220]	FL with a model-based RL	The article proposes a model-based RL approach for FL resource allocation, where distributed learning is utilized to optimize communication resources and federated model parameters dynamically, improving convergence speed and EE in heterogeneous network environments.
[222]	FL with DDPG	The article proposes a DDPG-based FL framework, leveraging distributed learning with SWIPT, in addition to MC-NOMA, to optimize power and CPU resource allocation dynamically, minimizing energy consumption under latency constraints in energy-efficient FL systems.
[224]	Collaborative ML	The proposed Collaborative ML framework trains local models on fog nodes with private data, sharing only model parameters for central aggregation, preserving data privacy while enabling distributed decision-making across the network.
[225]	Distributed Learning algorithm	The proposed transmission power oriented AI based on distributed learning algorithm for WBANs uses distributed learning, where body sensors autonomously select transmission parameters based on local conditions and neighbor states, achieving energy-efficient power control without central coordination.
[229]	FL	The article proposes a novel TDMA-based scheme for wireless FL networks, where distributed learning is used to allow users to train local models and transmit them sequentially while optimizing latency by addressing resource heterogeneity through parallel local training and model transmission for different users.
[230]	FL-GR	The article proposes a FL framework with Gradient Recycling, which utilizes distributed learning by recycling historical gradients of unscheduled and transmission-failure devices to enhance the robustness and efficiency of training in resource-limited and unreliable wireless networks.

G. Summary and Lessons Learned

Table VIII presents an overview of the synthesized data from the surveyed studies, facilitating a comprehensive analysis of key findings and lessons learned. It categorizes works based on NOA (Number of Optimizations per Article), highlighting the learning paradigms employed, the inclusion of distributed learning approaches, and the performance gains achieved. The table also introduces PPO (Papers Per Optimization) and PKS (Papers by KPI Simultaneity), which quantify studies optimizing specific KPI sets and those addressing multiple KPIs simultaneously. The analysis reveals an inverse relationship between NOA and PKS, indicating that as the number of simultaneously optimized KPIs increases, fewer studies are found. Most works (53.08%) optimize

a single KPI, while only 5.38% address four or more, primarily due to hardware limitations, exponential computational complexity, and conflicting trade-offs between KPIs, such as PowCon and latency. To overcome these challenges, advanced ML-based strategies have been explored, including: hierarchical Multi-Agent Reinforcement Learning (MARL) for problem decomposition and dynamic learning from the environment; FL for decentralized training across edge devices to improve privacy and latency; TL for improved adaptability by reusing knowledge from previous optimizations to reduce computational overhead; and Reconfigurable Intelligent Surface (RIS)-aided ML to dynamically manipulate wireless propagation and enhance SE and EE. These approaches enable scalable, real-time multi-resource optimization, addressing the constraints inherent in wireless communication environments.

Table VIII also highlights that SL and RL are the predominant paradigms in ML-based resource allocation optimization, representing 49.23% (64 of 130) and 46.92% (61 of 130) of the surveyed studies, respectively. SL is widely used due to the availability of labeled telecommunications data, enabling precise optimization of well-defined objectives by leveraging historical correlations between resource allocation and network performance metrics. Conversely, RL is particularly effective in dynamic environments, as it learns through interaction, allowing adaptation to variations in network load, user demand, and channel conditions [231]. Other approaches complement these paradigms, such as UL, which excels in exploratory data analysis and clustering user behaviors to improve QoS [232], and evolutionary algorithms, which provide robust solutions for non-linear and multidimensional optimization problems when labeled data is scarce [233]. Additionally, hybrid techniques combining SL for direct optimization and UL for uncovering insights from unlabeled data enhance performance [234]. Distributed learning, employed in 23 of 130 papers, has also emerged as a key solution, enabling parallel processing across multiple nodes, reducing latency, enhancing scalability, and improving system resilience to failures while efficiently managing vast and dynamic wireless network data. These findings underscore that SL excels in structured scenarios with abundant labeled data, RL is crucial for adaptive optimization in dynamic systems, while hybrid and distributed learning approaches bridge gaps between individual paradigms. This highlights the necessity of tailored, scenario-specific ML strategies to fully exploit resource optimization in modern wireless networks.

With respect to performance gains achieved through resource allocation optimization in wireless communications, Table VIII indicates a general balance among the efficiencies evaluated, except for LE, which was comparatively less prioritized. This is largely due to the complexity of optimizing latency, as it often involves trade-offs with other factors. Lowering latency typically requires high-performance, cost-prohibitive hardware [235], and many wireless networks and devices have inherent limitations that make substantial reductions impractical. In contrast, EE and CE, addressed in 56 and 59 of 130 papers, respectively, are prioritized due to environmental concerns, mobile and IoT device constraints, data processing needs, and economic incentives. EE is particularly critical for reducing carbon emissions and operational costs in telecommunications [236], [237], as well as for extending battery life in mobile and IoT devices [238], [239]. Additionally, CE plays a crucial role in reducing hardware costs [240] and efficiently handling large-scale data processing, ensuring seamless and fast service delivery. These findings suggest that while LE optimization faces greater practical and economic challenges, EE and CE improvements are more feasible and align with both technological advancements and sustainability goals, making them key focus areas in resource allocation strategies.

Finally, Figure 5 summarizes the surveyed studies based on the optimized KPIs, considering different ML paradigms and distributed learning approaches. PowCon was the most researched KPI (57 papers, 43.85%), reflecting its impact on EE and operational costs. Single-KPI optimization relied mainly on SL and RL, while multi-KPI optimization revealed synergies between PowCon and throughput (19 papers), requiring trade-offs between EE and data rates. RL was predominant in optimizing PowCon and NTM, demonstrating its ability to manage dynamic environments. Distributed learning was mainly explored in CRM-related PowCon optimization, enhancing energy-efficient resource allocation. Throughput, studied in 37 papers (28.46%), is crucial for high-data-rate applications but was rarely optimized alone (only five papers, with five of them using SL). Instead, it often appeared in multi-KPI optimization, particularly with PowCon, NTM, and latency, where RL proved effective, especially in throughput-NTM optimization (13 of 18 papers). The limited exploration of distributed approaches for throughput suggests that centralized optimization remains more efficient in maximizing data rates without the added complexity of distributed architectures.

Latency, although critical for delay-sensitive applications, was the least studied KPI (33 papers, 25.38%) due to its complex trade-offs and network constraints [241], [242]. SL was common in single-KPI latency optimization, while RL dominated multi-KPI scenarios, particularly in latency-NTM and latency-throughput optimization. Distributed learning approaches appeared moderately, possibly due to their relevance in large-scale systems where latency reductions require decentralization. NTM, examined in 53 papers (40.77%), is essential for QoS and QoE, with 18 papers using SL for its standalone optimization. Strong synergies with throughput (18 papers) and PowCon (16 papers) highlight its role in network efficiency, while RL excelled in multi-KPI NTM optimization. The limited use of distributed learning in NTM suggests future research opportunities in large-scale traffic management. Finally, CRM, studied in 44 papers (33.85%), had the highest proportion of single-KPI optimization (24 papers) and relied heavily on distributed learning, emphasizing the need for decentralized computation. The frequent use of evolutionary methods and FL also reflects the growing demand for adaptive ML solutions in CRM.

In summary, the emphasis on PowCon as the most studied KPI highlights the critical need for EE in wireless systems, driven by sustainability and cost reduction goals. In contrast, the lower focus on latency stems from the challenges in modeling and optimizing time-sensitive parameters despite its

TABLE VIII
TABLE OF DATA SYNTHESIS OVERVIEW RELATED TO RESOURCE ALLOCATION OPTIMIZATION USING ML TECHNIQUES APPROACHED IN THE SURVEYED PAPERS

		Learning Paradigm				Distributed	Performance Gain						
NOA	**Network KPIs Optimized**	UL	SL	RL	Other	**Approach**	EE	SE	LE	PDE	CE	**PPO**	**PKS**
1	PowCon	2	6	8	3	3	17	3	0	0	3	19	69
	Throughput	1	4	0	0	0	0	3	1	0	2	4	
	Latency	0	3	1	0	0	0	0	4	2	1	4	
	NTM	1	12	7	2	1	3	5	2	13	2	18	
	CRM	3	11	12	2	3	0	0	0	0	24	24	
2	PowCon and throughput	3	5	5	1	1	8	7	0	2	0	8	36
	PowCon and latency	0	3	1	0	0	4	1	3	0	2	4	
	PowCon and NTM	0	2	5	0	2	5	3	0	4	1	6	
	PowCon and CRM	0	0	1	1	1	2	0	0	0	2	2	
	Throughput and Latency	0	1	2	0	0	0	1	2	1	1	2	
	Throughput and NTM	1	2	4	0	2	1	4	1	4	0	6	
	Latency and NTM	1	1	2	1	1	0	0	4	4	0	4	
	Latency and CRM	0	1	0	0	0	0	0	1	0	1	1	
	NTM and CRM	1	2	1	0	1	0	2	1	1	3	3	
3	PowCon, throughput and NTM	1	2	2	0	1	3	3	1	2	1	3	18
	PowCon, throughput and CRM	0	3	0	0	0	3	3	0	0	3	3	
	PowCon, latency and NTM	0	2	0	0	0	0	2	2	0	2	2	
	PowCon, latency and CRM	0	1	2	4	3	4	0	4	0	4	4	
	Throughput, latency and NTM	0	1	3	0	1	1	4	4	4	0	4	
	Throughput, latency and CRM	0	1	0	1	1	0	0	1	1	1	1	
	Latency, NTM and CRM	0	1	0	0	0	0	0	1	1	1	1	
4	PowCon, throughput, latency and NTM	1	0	1	0	0	2	2	2	1	0	2	6
	PowCon, throughput, latency and CRM	0	0	1	0	0	1	1	1	1	1	1	
	PowCon, throughput, NTM and CRM	0	0	1	1	0	1	1	0	1	1	1	
	PowCon, latency, NTM and CRM	0	0	0	1	1	1	0	1	0	1	1	
	Throughput, latency, NTM and CRM	0	0	1	0	0	0	1	1	1	1	1	
5	PowCon, throughput, latency, NTM and CRM	0	0	1	1	1	0	1	1	1	1	1	1
Total number of papers:		15	64	61	18	23	56	47	38	44	59	130	130

relevance for delay-sensitive applications. SL is preferred for single-KPI optimization due to its simplicity and efficiency in structured scenarios where labeled datasets can train models to achieve specific objectives, while RL excels in multi-KPI optimization by learning trade-offs dynamically, making it particularly effective in complex, multi-agent environments. For example, RL can balance PowCon and throughput by adjusting system parameters in real-time, reducing energy consumption during low demand and optimizing transmission power under high data traffic to enhance performance while maintaining EE. Additionally, the limited adoption of distributed learning, except in CRM, reveals an opportunity to improve scalability and decentralization in modern networks. The success of ML-based optimization depends not only on selecting the appropriate paradigm but also on tailoring algorithms to exploit specific KPIs synergies and trade-offs while considering constraints such as computational capacity, energy consumption, and scalability. Future research should explore hybrid approaches that combine SL's predictive capabilities with RL's adaptability, alongside distributed frameworks, to manage the growing complexity of wireless networks and enhance multi-KPI optimization.

IV. TAXONOMY OF ML ALGORITHMS FOR RESOURCES OPTIMIZATION

To perform an in-depth performance and algorithm use analysis for optimizing critical resources in wireless communications, this section provides a detailed taxonomy aimed at identifying the algorithms that prevail over others for this task and consequently understanding the primary reasons and advantages in their implementation for this purpose, considering their suitability according to the applications demand. Figure 6 presents the taxonomy of the learning paradigms and, subsequently, ML algorithms approached in the surveyed papers.

According to the Figure 6 previously provided, a wide variety of ML algorithms can be observed for the task of optimizing resources in wireless communications, totaling 77 different ones. It is essential to clarify that the purpose of Figure 6 is to classify the taxonomy of algorithms according to the learning paradigms analyzed in this paper (UL, SL, RL, and others). This clarification is particularly relevant since solutions incorporating distributed learning approaches are addressed across various paradigms and algorithms, making it inappropriate to confine them to a single category.

In addition, some algorithms fit into more than one learning paradigm, such as DNN, given their flexibility to adapt depending on the application or demand to be met. Therefore, while DNNs are more commonly used in SL tasks, they can be adapted and applied in different learning paradigms. In this context, DNNs can be used as SL, for example, to classify network traffic into different categories such as video streaming, file transfers, and more [243]. Prioritizing critical applications like voice calls over less critical ones like video streaming can improve service quality and optimize NTM.

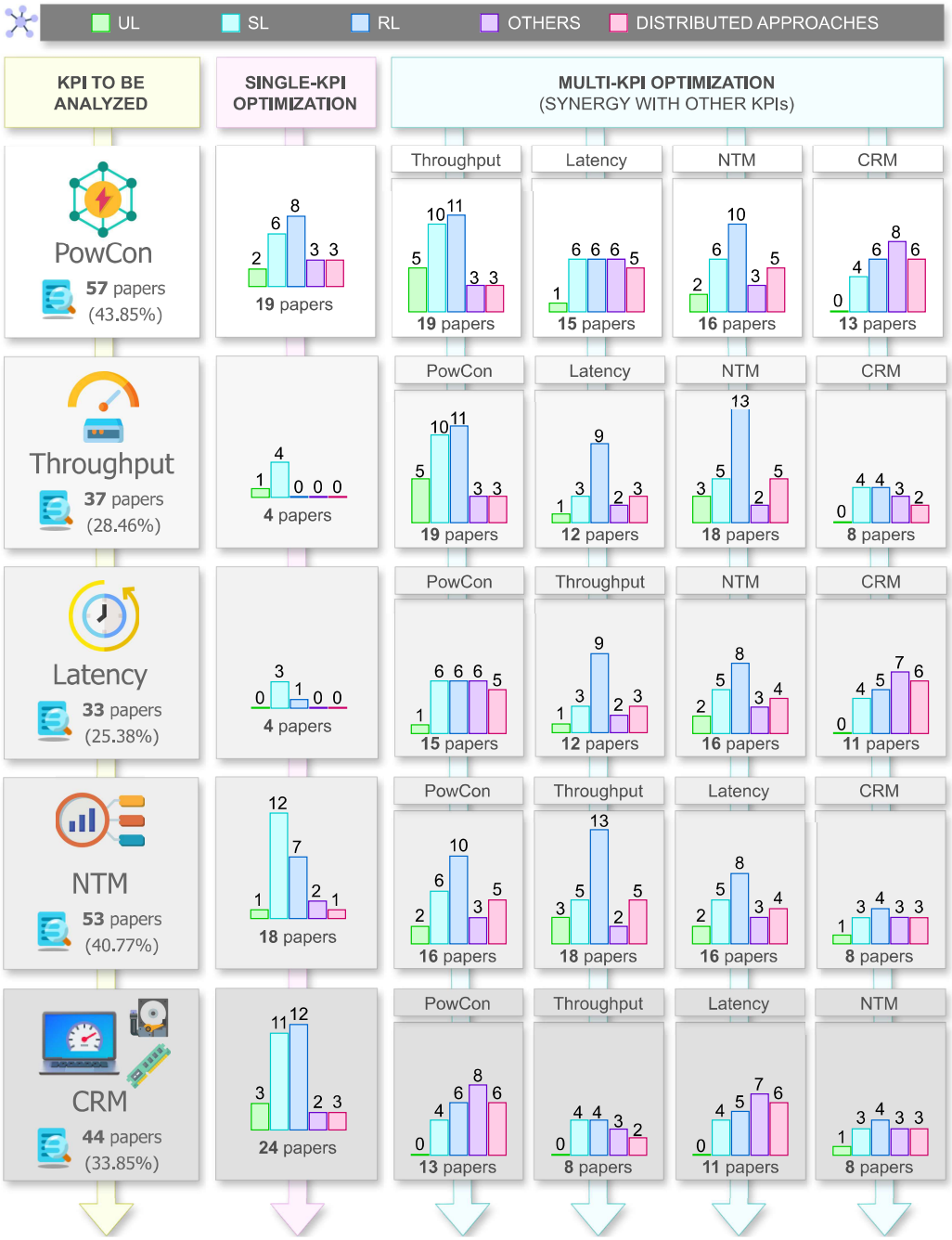


Fig. 5. Summary of the KPIs analysis in ML-based solutions for wireless communications in the selected papers of this survey.

Moreover, considering the UL paradigm, DNNs can be utilized to cluster telecommunications customers based on their usage patterns. For instance, the network can segment customers into groups with similar behaviors, such as heavy mobile data users, frequent callers, etc [244]. These groups can be used to optimize pricing plans or allocate network resources more efficiently.

Analyzing the behavior present in the ML algorithm taxonomy provided in Figure 6, we can conclude that for each learning paradigm, there is a range of specific algorithms, highlighted in purple, commonly selected that seem to have greater relevance and efficiency for use in the task of optimizing resources in wireless communications. The following subsections discuss the leading ML algorithms classified into different learning paradigms and pointed out in Figure 6, followed by a summary and lessons learned overview that consolidates key insights from this discussion.

A. Leading Unsupervised Algorithm

Regarding UL, we can emphasize K-Means, highlighted in purple in Figure 6, and found in 5 out of 15 instances of algorithms within this paradigm for resource allocation optimization, along with appearances in its variations such

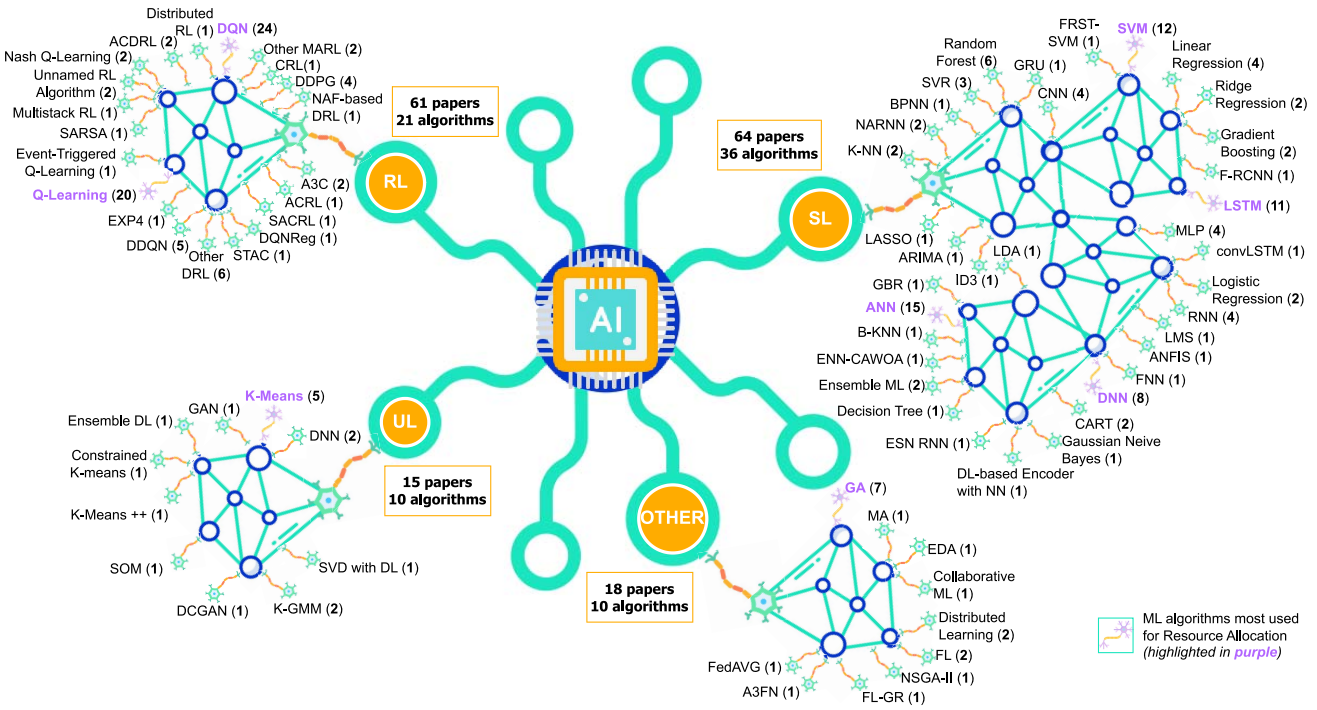


Fig. 6. Taxonomy of ML algorithms approached in the surveyed papers.

as K-means++ and Constrained K-means. The K-Means algorithm is an UL technique used to cluster data into different groups. Initially, the user defines the number of clusters (K) and selects random points from the data as the initial centroids. Each data point is then assigned to the cluster whose centroid is nearest, usually using the Euclidean distance as the distance metric. After this initial assignment, the position of each centroid is recalculated as the mean of the points assigned to it. This process of assignment and update repeats until the centroids no longer change significantly or until a maximum number of iterations is reached [245].

The main parameters of the K-Means algorithm include the number of clusters K, the convergence criteria (which can be the maximum number of iterations or the minimum change in centroid positions between iterations), and the distance metric, which is usually the Euclidean distance but can be other metrics depending on the problem. The choice of initial centroids can influence the final result, and techniques like K-Means++ improve this selection for better convergence [245].

In the context of optimizing resource allocation in wireless communication systems, K-Means can be used to group users based on their traffic characteristics or geographic location to optimize resource allocation, determine the best locations for cell towers by clustering high-demand areas, and manage interference zones more effectively. However, the algorithm has some limitations and difficulties, such as determining the optimal number of clusters, which can be challenging and usually requires heuristic methods. Additionally, K-Means is sensitive to the initial choice of centroids, which can lead to convergence to a local minimum. Another limitation is that the algorithm assumes clusters are spherical and of similar size, which may not always be true. In high-dimensional data, the distance metric may lose meaning, complicating clustering [105].

Due to its simplicity and efficiency, K-Means is computationally less intensive than other clustering algorithms, making it suitable for large-scale datasets and real-time applications, which are common in telecommunications. K-Means is versatile and can be applied to various domains and data types. In telecommunications, where diverse types of data are generated, K-Means can be used for clustering and pattern discovery in data related to power allocation, traffic management, or other resource allocation optimization tasks [105], [246]. Moreover, this algorithm is well-established, widely used, and has a robust ecosystem of libraries and tools for implementation [247]. It can quickly provide insights into resource allocation and distribution without the need for complex model selection or training procedures, which is particularly valuable in real-time decision-making scenarios. Furthermore, K-Means is well-suited for identifying distinct resource clusters, helping to categorize and manage resources more effectively. Its simplicity, efficiency, and adaptability make K-Means a preferred choice for resource allocation optimization in wireless communications, enabling quick and practical solutions to this specific task.

In addition to K-Means, the K-Means Gaussian Mixture Model (K-GMM) algorithm is also well-explored. Unlike K-Means, which assumes that clusters are spherical and uses only the mean as the centroid, assigning each point to the nearest centroid based on Euclidean distance, K-GMM models each cluster as a Gaussian distribution with mean, covariance matrix, and a mixing coefficient, allowing for soft assignments where each point has a probability of belonging to multiple clusters. While K-Means is simpler and faster, ideal for

well-separated spherical clusters, K-GMM is more flexible for varied shapes but is more complex and computationally intensive.

K-GMM is known for its capability of capturing non-linear relationships in data, making it suitable for various resource allocation optimization scenarios where the relationships between parameters may not be linear. This non-linearity can be prevalent in telecommunications systems due to the intricate interplay of factors like signal propagation, traffic variations, and network configurations. K-GMM's capacity to model complex data patterns enhances its effectiveness in this domain. It can be particularly useful when dealing with use cases where the specific relationships between resource parameters are unknown a priori. In resource allocation optimization, it's often challenging to define these relationships accurately, making an algorithm like K-GMM, which can adapt and discover such relationships from data, quite valuable. In summary, these advantages give it an edge over other UL algorithms in this specific context.

B. Leading Supervised Algorithms

Figure 6 shows that SL algorithms were the most commonly used when compared to other learning paradigms. Furthermore, it can be observed through the figure a much wider variety of SL algorithms in comparison to other learning types, which can be explained by the substantial availability of frameworks and tools to support these approaches. Given that supervised algorithms are commonly used, not only for the purpose of resource allocation optimization in wireless communications but in general, there is a wealth of learning materials, as well as numerous open-source libraries and tools widely available for implementing these algorithms, making the implementation easier and more accessible.

An interesting aspect to observe in Figure 6 is that among the primary SL algorithms implemented for resource allocation optimization, highlighted in purple, one of them is the Artificial Neural Network (ANN), mentioned in 15 different surveyed papers, and it may refer to a more generic and broad term for ML algorithms. This implies that some of the researched works did not provide further specifications regarding the ML algorithm used to perform the optimization, affecting their understanding and reproducibility.

ANNs are computational models inspired by the human brain's functioning, consisting of nodes (neurons) organized in layers: an input layer, one or more hidden layers, and an output layer. Each neuron in a layer receives weighted inputs, applies an activation function (such as Rectified Linear Unit (ReLU), sigmoid, or hyperbolic tangent), and transmits the output to the neurons in the next layer. The learning process in ANNs involves initializing the weights of the connections between neurons randomly, followed by optimizing these weights using algorithms like back-propagation, which minimizes an error between the predicted and actual outputs [248].

In the context of wireless communication systems, ANNs can be used for tasks like dynamic resource allocation, traffic demand prediction, and network parameter optimization. However, ANNs have limitations, such as the need for large amounts of training data and the difficulty in interpreting what the network is learning. Additionally, they may struggle with problems where the temporal sequence of data is important, as they lack internal memory to handle temporal dependencies [119], [119], [146], [148]. In addition to ANN, we can highlight DNNs, Support Vector Machines (SVMs), and Long Short-Term Memory (LSTM) networks as the most commonly used supervised learning algorithms for resource allocation optimization.

DNNs are an extension of ANNs, characterized by having a large number of hidden layers. The greater the number of layers, the more complex the representations the network can learn, allowing DNNs to be applied to highly complex problems such as image recognition and natural language processing. The initialization of DNNs follows the same principle as ANNs, with weights initially assigned randomly and optimized through back-propagation. However, due to the network's depth, methods like regularization and batch normalization are often used to avoid problems like overfitting and to stabilize the training process [249].

In wireless communications scenarios, DNNs can be employed to optimize spectrum allocation, enhance channel coding efficiency, and predict communication channel quality. DNNs, however, face challenges such as high computational cost, the need for large amounts of labeled data, and complexity in interpreting results. Additionally, training DNNs can be prone to issues with vanishing or exploding gradients, especially in very deep networks. Present in 8 different papers, a possible justification for the widespread use of DNNs in the studies is their ability to handle complex data, as they can learn complex relationships within datasets, which is crucial for optimizing resources in wireless communications. Consequently, DNNs can model non-linear patterns and extract relevant data features, making them suitable for tasks such as transmission power adjustment and network traffic management [165].

SVMs are SL algorithms used for both classification and regression. The primary parameters of an SVM algorithm include the penalty parameter and kernel function. The penalty parameter works by finding the hyperplane that best separates the data into different classes and controls the trade-off between maximizing the margin and minimizing classification error. The kernel function, such as linear, polynomial, or Radial Basis Function (RBF), transforms the data into a higher-dimensional space to separate the classes better. In this context, initializing SVM involves selecting the appropriate kernel and configuring the penalty parameter, along with any additional parameters related to the chosen kernel. These settings are crucial for optimizing the algorithm's performance in classification tasks [250].

In the domain of wireless networks, SVM can be used for tasks such as channel classification, anomaly detection in network traffic, and channel condition prediction. The limitations of this algorithm include being less efficient on large datasets and the challenge of selecting the ideal kernel and parameters. Additionally, SVM is not naturally extendable to multi-class problems, requiring approaches like "one-vs-all" to handle multiple classes [145], [152].

Approached in 12 articles of our survey, a likely reason for the extensive use of the SVM may be associated with its flexibility, as it is known for its generalization capability and its ability to handle high-dimensional datasets effectively. Therefore, SVMs are effective in identifying separation hyperplanes, which is useful in classification problems, such as identifying types of traffic or setting power limits.

LSTMs are a variant of Recurrent Neural Networks (RNNs), designed to handle problems where temporal dependencies are important. Unlike traditional RNNs, LSTMs have a special cell structure with gates that regulate the flow of information, allowing the network to retain relevant information over long periods and avoid the vanishing gradient problem. This makes LSTMs ideal for tasks like time series forecasting, speech recognition, and machine translation [251].

The key parameters of an LSTM network include the number of units in each LSTM cell, which determines the memory capacity, the input and output gate weights, which control the flow of information into and out of the cell, and the forget gate, which determines what information should be discarded over time. These parameters help the LSTM network manage long-term dependencies in sequential data [251].

Regarding wireless communication frameworks, LSTMs can be applied to predict traffic demand based on historical patterns, optimize resource allocation according to temporal changes, and improve interference management. Despite their advantages, they are more complex and require more computational power for training compared to traditional RNNs. Additionally, LSTMs training can be slow due to the need to process long sequences of data [138].

Designed in 11 different researched works, LSTM can be considered a standout algorithm due to its characteristics related to temporal sequences and sequential data. In telecommunications scenarios, where data often exhibit sequential patterns and temporal dependencies, LSTMs are highly valuable. They are a form of RNN capable of capturing contextual information over time, making them applicable to tasks such as flow prediction, QoS control, and computational resource management in network servers.

In addition to the aforementioned advantages, scalability, performance on large datasets, and continuous training and learning can also set these algorithms apart from others. DNNs and LSTMs can be scaled to handle large volumes of data and high complexity. These algorithms are also adaptable and can learn from constantly changing data, which is particularly valuable in wireless communications, where network conditions may vary over time. Furthermore, SVM, DNN, and LSTM are recognized for their ability to handle a substantial amount of information, which is crucial in telecommunications scenarios, such as in the case of achieving more efficient resource allocation optimization.

However, it is important to note that the choice of the algorithm depends on the specific characteristics of the problem and the available data. In some cases, simpler SL algorithms, such as linear regression or decision trees, may suffice. The selection of the ideal algorithm for optimizing resources in wireless communications should be based on the nature of the problem, the available data, and the desired outcomes. Additionally, the combination of multiple SL algorithms, such as ensemble ML, can also be beneficial in certain scenarios.

C. Leading Reinforcement Algorithms

Alongside SL algorithms, RL algorithms were the most commonly used ones for resource allocation optimization in wireless communications systems. As illustrated in Figure 6, in this learning paradigm, Q-Learning and DQN stood out as the most addressed among the researched works, being present in 20 and 24 papers of the total number of articles, respectively.

Q-learning aims to find the optimal decision-making policy in an environment to maximize cumulative reward. It works by associating a Q-value (quality) with each state-action pair, representing the expected reward of taking a particular action in a given state and subsequently following the optimal policy. The main equation used in Q-Learning is the Bellman Equation, which updates the Q-values based on the immediate reward received and the Q-value of the next state [252].

The key parameters of Q-learning include the learning rate, which determines how much new Q-values replace the old ones, and the discount factor, which weighs the importance of future rewards compared to immediate rewards. Initializing the algorithm involves defining a Q-table with initial values (usually zero) for all state-action pairs and setting up the initial policy, which may be exploratory (such as the ϵ-greedy policy) to ensure the agent explores the environment [252].

In the context of wireless communication systems, Q-Learning can be applied to, for example, spectrum allocation optimization, where the agent learns to allocate resources efficiently based on environmental feedback, such as signal quality or interference. However, Q-Learning faces challenges in environments with large state spaces, where the Q-table can become extremely large and difficult to manage. Additionally, Q-Learning can be slow to converge in complex environments, especially when the action space is also large [187], [187], [193].

In contrast, DQN is an extension of Q-Learning that uses DNNs to approximate the Q-value function, enabling the algorithm to handle continuous or very large state spaces where the traditional Q-table approach would be infeasible. In DQN, the Neural Network (NN) takes the current state as input and outputs Q-values for all possible actions, trained using the temporal difference method with mini-batches of experiences stored in an experience replay buffer [253].

The main parameters of DQN include the architecture of the neural network (such as the number of layers and neurons), the learning rate for optimizing the network, the size of the experience replay buffer, and the target update rate, which controls how often the target network is updated to reduce correlation in the training data. Initializing DQN involves setting up the neural network, the exploration policy, and defining parameters like the buffer size and update rate [253].

DQN can be applied to dynamic resource allocation problems in wireless communications, where the environment's state changes constantly, and decisions need to be made in real-time. It is particularly useful in scenarios where the state space is very large or continuous, such as networks with many

users or frequencies. However, DQN faces limitations such as the need for significant computational power to train the NN and sensitivity to hyper-parameters, which can lead to instabilities during training [181], [192], [212].

As a result, we can observe through the statistics presented in our survey several reasons that justify why these algorithms are selected more frequently to address this type of task. One possible reason is that both algorithms have effective mechanisms for balancing the exploration of new actions and the exploitation of known actions in the environment. This is crucial in telecommunications, as finding the balance between allocating available resources and trying new approaches is essential. Additionally, since Q-Learning and DQN are capable of modeling the environment by building a Q-function that estimates the expected reward for each action in each state, it can be valuable for predicting how different actions will affect resource allocation and then optimizing decisions based on this estimation.

Furthermore, it is important to highlight that the Q-Value method approach, which aims to learn the Q-function upon which these two algorithms are based, is conceptually simpler to understand and implement. The e-greedy strategy typically used to balance exploration and exploitation is also easy to implement and effective for efficiently exploring the environment. In this context, these algorithms usually choose actions based on the estimated Q-value but occasionally make random choices to explore new actions. This approach to Q-value updates can also provide greater training simplicity, as other RL algorithms, such as policy-based methods, can be more complex in terms of calculations and fine-tuning.

Moreover, Q-Learning and DQN can be used in continuously evolving environments, capable of adapting their policies as the environment evolves. This is quite common in wireless communications, where network conditions and traffic change constantly. Not less important, the fact that these algorithms are flexible regarding offline and online training enables them to be trained before being applied to the actual telecommunications environment, allowing the optimization of policies based on historical data and ideal configurations.

Given the paramount importance of CE in telecommunications, Q-learning and DQN also prove to be highly effective in terms of computational resources. While some other RL algorithms aim to learn stochastic policies, such as Deep Deterministic Policy Gradient (DDPG), Q-Learning and DQN tend to learn deterministic policies, where the chosen action is the one that maximizes the estimated Q-value for a specific state. Consequently, this simplifies the implementation, as there's no need to deal with randomness in actions, making exploration more effective and predictable.

When analyzing the degree of complexity proposed by the Q-Learning and DQN algorithms, it should be considered that both work well when the action space is discrete, which simplifies action selection compared to environments with continuous action spaces. Various approaches can be undertaken in scenarios of resource allocation optimization in wireless communications with discrete action spaces. Adjusting transmission power in wireless communication systems, such as cellular networks, the action space can be discrete, involving choosing discrete power levels, such as low, medium, or high. In telecommunications networks, bandwidth allocation or time slot assignment can also be done discretely. For example, in the allocation of time slots in Time Division Multiple Access (TDMA) networks, the action space consists of allocating or not allocating a time slot to a specific user. Regarding NTM analysis, in traffic routing in networks, the choice of paths for forwarding data packets can be a discrete process, where routers can choose from a discrete set of output ports to direct packets. Furthermore, the selection of channels and frequencies available for wireless communications can also be discrete so that devices choose from a set of available channels not in use by other devices.

For the DQN approach in particular, other aspects may be relevant in choosing this algorithm, such as experience replay and DL techniques. DQN incorporates a replay memory that stores past experiences, allowing the agent to train efficiently in batches of training data and avoiding learning instability, leading to improved training efficiency. Furthermore, to approximate the Q-values, DQN uses DNNs, enabling the algorithm to handle more complex state and action spaces and providing a richer representation of the environment, which can lead to more effective exploration and better generalization.

Although there are several advantages and reasons for the more extensive use of Q-Learning and DQN when compared to other RL algorithms, the choice of the ideal algorithm will depend on the application's requirements, the specific problem's characteristics, and the complexity of the environment. Q-Learning is a solid choice for problems with a discrete action space and a small to moderate state space. DQN, on the other hand, is more suitable for problems with continuous action spaces and complex state spaces because it uses neural networks to estimate the Q-functions.

In summary, while there are various RL approaches, including DDPG, Double Deep Q-Network (DDQN), State-Action-Reward-State-Action (SARSA), and Actor-Critic Reinforcement Learning (ACRL), among others, Q-Learning and DQN often excel in resource allocation optimization scenarios for wireless communications due to their overall effectiveness and more straightforward implementation. However, the algorithm selection should be based on the nature of the problem and the available computational resources.

D. Leading Evolutionary Algorithm and Discussion on FL Variations

Although "Distributed Learning" and "Collaborative ML" can be applied to algorithms across different paradigms, such as UL, RL, and SL, and therefore do not fit exclusively into a single category, some papers presented only these terms as definitions for the proposed algorithms in their respective solutions. Consequently, the lack of further details about the types of algorithms used motivated the classification of these papers under the "Others" category.

Furthermore, as shown in Figure 6, four of the ten algorithms classified as "Others" are variations of FL (classic FL, Federated Averaging (FedAvg), Asynchronous

Federated Framework Network (A3FN), Federated Learning with Gradient Recycling (FL-GR)), highlighting the growing interest in this approach for optimizing resource allocation in wireless communications. FL allows multiple devices to collaborate in training a global model without directly sharing local data, thus preserving privacy and reducing the need for data transfers over congested networks. For instance, in a 6G network, FL can be utilized to optimize bandwidth allocation, where edge devices locally train models to predict resource demand, sharing only parameter updates (such as gradients) with a central server. This approach enhances network efficiency while safeguarding users' sensitive data.

Among the algorithms not classified as UL, SL, and RL, the GA, highlighted in purple in Figure 6, can be regarded as the most influential one for performing resource allocation optimization in wireless communications systems, being implemented in 7 different works.

GA is an optimization technique inspired by natural selection and biological evolution processes. It seeks optimal or near-optimal solutions to complex problems by simulating the evolutionary process, where a population of potential solutions (chromosomes) evolves over several generations. Each solution is evaluated by a fitness function that measures its quality concerning the problem at hand. The main operators used in GA are selection, crossover, and mutation. Selection chooses the fittest solutions for reproduction, crossover combines parts of two or more parents to generate offspring, and mutation introduces random variations to maintain genetic diversity in the population [254].

The key parameters of this algorithm include population size, crossover rate, mutation rate, and the number of generations. Initializing GA involves creating an initial population of randomly generated solutions, which are then evaluated by the fitness function. Over generations, the solutions evolve, with the best solutions (those with higher fitness) having a greater chance of being selected for reproduction until a stopping criterion is reached, such as the maximum number of generations or population convergence [254].

In the realm of wireless communication environments, GA can be applied to complex optimization problems, such as spectrum allocation, network parameter optimization, and network planning. It is particularly useful in scenarios where the search space is large and non-linear, allowing the algorithm to explore different areas of the solution space efficiently. However, GA has limitations, such as dependency on the quality of the fitness function and the need for a large number of fitness evaluations, which can be computationally expensive. Additionally, GA may converge to sub-optimal solutions if the population diversity decreases too quickly [117].

As shown through the statistical analysis presented in this paper, GA is an evolutionary algorithm commonly used for resource allocation optimization in wireless communication systems, which can be attributed to its adaptability and ability to explore a wide solution space. GA is advantageous in this context because it can effectively handle complex, non-linear, and non-convex optimization problems, which are prevalent in telecommunications. It can simultaneously optimize multiple objectives, making it suitable for optimizing various resources like power, time, frequency, computational capacity, and space. Furthermore, GAs can adapt to dynamic environments and handle constraints effectively.

However, GAs also have some limitations. They require careful tuning of parameters, and the search process may be computationally expensive. GAs tend to be slower than some other optimization techniques, which could be a disadvantage when real-time or low-latency decisions are essential. Additionally, GAs might not always guarantee the global optimal solution, and convergence is not guaranteed, so multiple runs or variations of the algorithm may be required. Despite these drawbacks, the flexibility and robustness of GAs make them a popular choice for resource allocation optimization in wireless communications.

E. Summary and Lessons Learned

A comprehensive taxonomy of ML algorithms for resource allocation optimization in wireless communications was performed in our survey for categorizing these techniques into key learning paradigms: UL, SL, RL, and evolutionary approaches. Among the most prevalent algorithms, K-Means stands out in the UL category for clustering and identifying patterns in network data. In the SL paradigm, ANN, DNN, LSTM, SVM are highlighted for their adaptability and effectiveness in a wide range of optimization tasks. Meanwhile, Q-Learning and DQN dominate the RL landscape, and GA is identified as a leading approach among evolutionary techniques.

The findings reveal that the choice of ML algorithm strongly depends on the characteristics of the task and the learning paradigm. For instance, K-Means, a prominent UL algorithm, demonstrates efficiency in clustering users or resources, which can optimize network layouts and traffic management. However, its assumptions regarding cluster shapes and initialization can limit its performance with non-spherical or high-dimensional data. In contrast, K-GMM offers enhanced flexibility, making them suitable for non-linear and complex data distributions.

Supervised algorithms, such as ANNs and DNNs, are well-suited for modeling complex relationships in tasks such as traffic prediction and resource allocation. In addition, LSTMs excel in scenarios requiring temporal sequence modeling, making them effective for dynamic resource allocation and interference management. SVMs are versatile for both classification and regression tasks, performing well in high-dimensional spaces but requiring careful parameter selection. Despite their strengths, supervised approaches face challenges such as high computational demands and the need for substantial labeled datasets, making efficient implementation crucial in wireless environments.

RL approaches, particularly Q-Learning and DQN, have proven effective in dynamic and evolving wireless communication environments. Their ability to balance exploration and exploitation makes them ideal for spectrum and power allocation tasks. DQNs, in particular, extend the applicability of traditional Q-Learning by leveraging NNs to approximate Q-values, enabling solutions in large or continuous state

spaces. Despite these strengths, RL methods often require significant computational resources and careful parameter tuning to ensure stability during training. Evolutionary algorithms, such as GA, excel in solving non-linear, multi-objective optimization problems. Their adaptability to large-scale scenarios makes them valuable for optimizing complex systems like wireless communications, although their computational cost and slower convergence may pose challenges in scenarios demanding low latency.

This taxonomy also underscores the versatility of certain algorithms, such as DNNs, which bridge multiple paradigms due to their adaptability. Their use in hybrid approaches, for instance, combining SL and RL techniques, showcases their potential to address multifaceted optimization challenges. The growing interest in distributed methods, like FL, highlights the importance of privacy-preserving approaches in modern wireless systems, particularly as the field advances toward 6G networks.

The analysis emphasizes that while no single algorithm universally outperforms others, combining different approaches or tailoring algorithms to specific tasks can significantly improve performance and efficiency. Future research should explore integrating emerging technologies, such as quantum-enhanced ML algorithms, to address the computational constraints and scalability issues that remain limitations in current methods. These advancements could unlock new possibilities for real-time and efficient resource allocation in increasingly complex wireless communication systems.

V. Experimentation Framework and Dataset Analysis in ML-Based Resources Optimization

This section aims to survey the relevant information provided by the authors of the selected works that contribute to the development and even reproduction of experimentation framework scenarios commonly used for resource allocation optimization approaches using ML techniques, especially in wireless communication systems. For this purpose, two distinct approaches were considered for analysis: (i) information necessary to build the simulation or experimentation environments used, including, for example, its type, software components, frameworks, Integrated Development Environments (IDEs), simulators, emulators, programming languages, libraries, and additional tools; (ii) and a survey of works that provide access to databases and/or datasets used, as well as a detailed description of the dataset contents. These two approaches are detailed below, along with a discussion of the lessons learned from the results obtained in these analyses.

A. Simulation and Experimentation Practices

In our comprehensive analysis of 130 articles focused on resource allocation optimization for wireless communication systems utilizing ML algorithms, we aimed to assess the reproducibility of the validation scenarios and environments used for testing these ML-based proposals. Our investigation centered around five key aspects: (i) general information of the paper, such as reference and publication type, classified as Book, Conference Paper (CP), and Journal or Magazine (J/M); (ii) whether the articles conducted tests using simulation (S), field testing or experimentation (E), or both; (iii) the disclosure of Softwares (SWs), Frameworks (FWs), and/or IDEs used to aid in building simulation and experimentation scenarios, such as for generating synthetic data or essential parameters for simulating a wireless communication environment; (iv) the mention of programming languages, libraries, and additional tools that offer further insights into the environment and testing processes; and (v) the utilization of other specific simulators, emulators, and/or Simulation Databases (SimDBs) critical for building the simulation environment. Table IX summarizes this relevant information, which aids in the process of building the simulation and experimentation scenarios used in the surveyed papers.

Based on Table IX, we can note that from the 130 articles analyzed, a substantial majority (128 articles) employed a simulation environment, with only five exploring experimental scenarios involving additional equipment. Notably, 40 out of 130 articles provided details on the software, frameworks, and IDEs employed, highlighting Matrix Laboratory (MATLAB) as the most prevalent tool used in 28 studies, followed by PyCharm, referenced in four approaches. MATLAB's widespread adoption can be attributed to its complete and comprehensive tools and powerful mathematical and graphical capabilities, making it an ideal choice for modeling and simulating complex systems in wireless communications. PyCharm, while less common, offers a robust environment for Python development, facilitating seamless integration with ML and other libraries found in Python's vast ecosystem.

Experimental approaches frequently incorporated software such as Laboratory Virtual Instrument Engineering Workbench (LabVIEW) and GNU Radio, primarily to support tests involving SDR equipment. Examples include the use of Universal Software Radio Peripheral (USRP) 2954-R and USRP N210, as noted in [105] and [148]. These tools are preferred for their ability to interact with hardware components, such as Field-Programmable Gate Array (FPGA)-based ones, and their versatility in real-world signal processing applications.

Regarding programming languages, considering that a significant portion of the surveyed papers employ the MATLAB tool, this environment is consequently widely adopted in the proposed works, as it is also a proprietary multi-paradigm programming language and numerical computing platform developed by MathWorks. In addition, Python emerged as the dominant choice, mentioned in 32 articles, while C++ was employed in one study. Python's popularity can be attributed to its extensive range of ML libraries, ease of use, and strong community support. C++, although less frequently mentioned, provides high performance and control over system resources, which can be beneficial for specific simulation tasks.

In terms of libraries and additional tools, 30 articles provided relevant information, with TensorFlow (TF) and Scikit-learn (Sklearn) being the most prominent, featured in 19 and four articles, respectively. TensorFlow's comprehensive ecosystem and scalability make it suitable for complex ML models, while Sklearn's simplicity and efficiency cater to basic ML tasks. Additionally, network performance testing tools

TABLE IX
TABLE OF RELEVANT INFORMATION FOR BUILDING SIMULATION AND EXPERIMENTATION SCENARIOS USED IN THE SURVEYED PAPERS

Ref.	Pub. Type	Information to Build Simulation Environment				
		S	E	SW, FW, IDE *	Languages, Libs, Tools	Sim/Emu, SimDB
[101]	CP	✓		MATLAB	Python, Sklearn, TF	
[102]	J/M	✓				
[103]	J/M	✓				
[104]	CP	✓		MATLAB (Opt. Toolbox)	Python 3.7, TF	
[105]	J/M	✓	✓	MATLAB, LabVIEW		
[106]	J/M	✓			TF	
[107]	J/M	✓		PyCharm	Python, TF	
[108]	J/M	✓			5GLENA Module	NS-3
[109]	J/M		✓		TF	
[110]	J/M	✓				
[111]	J/M	✓				OpenCellID Database (Sprint MNO)
[112]	CP	✓				
[113]	J/M	✓				
[114]	J/M	✓		MATLAB 2023a		
[115]	J/M	✓		MATLAB 2020a		
[116]	CP	✓				
[117]	J/M	✓		Anaconda	Python, TF	
[118]	CP	✓				Mininet Wi-Fi
[119]	J/M	✓				LTE-Sim
[120]	Book	✓			Python, GUROBI solver, Sklearn	
[121]	CP	✓				
[122]	J/M	✓			Python	OpenAirInterface
[123]	J/M	✓				
[124]	CP	✓				
[125]	CP	✓				
[126]	CP	✓				
[127]	CP	✓				
[128]	CP	✓				
[129]	CP	✓		MATLAB (DL Toolbox)		
[130]	CP	✓				
[131]	J/M	✓		MATLAB R R2019b		
[132]	CP	✓		MATLAB (Std. non-linear opt. solver)		
[133]	CP	✓				
[134]	CP	✓			Python	
[135]	J/M	✓		MATLAB		
[136]	J/M	✓			Python	LoRaSim
[137]	J/M	✓			Sklearn	NS-3
[138]	J/M	✓			Python	
[139]	J/M	✓				
[140]	J/M	✓			Python, Matplotlib	
[141]	J/M	✓		Jupyter notebook	Python API	Mininet Wi-Fi
[142]	J/M	✓				
[143]	J/M	✓			Python, KPIC-Lite, TF, Resource library	SUMO
[144]	J/M	✓		MATLAB		
[145]	J/M	✓		MATLAB R2016b		
[146]	J/M	✓		MATLAB (NN toolbox)		
[147]	J/M	✓				
[148]	J/M	✓	✓	GNU Radio, MATLAB		
[149]	J/M	✓				
[150]	J/M	✓		MATLAB		
[151]	J/M	✓			Python, TF, Sklearn	
[152]	J/M	✓		MATLAB		
[153]	J/M	✓				
[154]	J/M	✓				
[155]	J/M	✓				NS-3
[156]	J/M	✓				
[157]	J/M	✓				
[158]	J/M	✓				
[159]	J/M	✓			TF, DL library, Iperf 2.0.8	Mininet Wi-Fi
[160]	J/M		✓	MATLAB		
[161]	J/M	✓				
[162]	J/M	✓		MATLAB		
[163]	J/M	✓				
[164]	J/M	✓		MATLAB		
[165]	J/M	✓			Python	
[166]	J/M	✓				
[167]	J/M	✓				
[168]	J/M	✓		MATLAB		
[169]	J/M	✓				
[170]	J/M	✓		MATLAB 2020b		
[171]	J/M	✓		RapidMinerTM		WinProp
[172]	J/M	✓				IoTSim-Osmosis Simulator
[173]	J/M	✓		Pytorch 1.7.1., Docker, IntelliJ IDEA 2020.1.1 c	Python 3.8.5, Azul JDK 11.0.12, Kotlin 1.2.71, CUDA 10.2	
[174]	J/M	✓				
[175]	J/M	✓				
[176]	J/M	✓				
[177]	CP	✓				
[178]	J/M	✓				
[179]	J/M	✓				5G-Scheduler C/C++ Simulator (LTE-Sim)
[180]	J/M	✓		MATLAB, STK	Python	
[181]	J/M	✓			Python 3.7.6, Hmmlearn 0.2.7, TF 1.13	
[182]	J/M	✓			TF 1.0	
[183]	J/M	✓				
[184]	J/M	✓				
[185]	J/M	✓		PyCharm	Python 3.6, TF 1.14 with CUDA	Internet topology zoo Database
[186]	J/M	✓				
[187]	J/M	✓		MATLAB		
[188]	J/M	✓			Python	OMNeT++
[189]	J/M	✓				
[190]	J/M	✓		PyCharm	Python, TF 1.10.0	
[191]	J/M	✓			Python, TF 1.10.0	
[192]	J/M	✓			Tool to build DAGs	
[193]	J/M	✓			Python	Python Simulator
[194]	J/M	✓			NumPy (Python), TF	Python Simulator
[195]	J/M	✓			Python	NS-2
[196]	J/M	✓				
[197]	J/M	✓		MATLAB 2019b		
[198]	J/M	✓		Ryu Controller	Iperf, Ping	Mininet Wi-Fi
[199]	J/M	✓				
[200]	J/M	✓				Castalia Platform
[201]	J/M	✓			Python 3.7.4, TF 1.14.0	
[202]	J/M	✓				
[203]	J/M	✓			Python 3.9.9	
[204]	J/M	✓			Python Env.	
[205]	J/M	✓				
[206]	J/M	✓				
[207]	J/M	✓		MATLAB 2021a		
[208]	J/M	✓		PyCharm 2021	Python 3.8.5	
[209]	J/M	✓		ns3gym RL FW		NS-3
[210]	J/M	✓				NS-2
[211]	J/M	✓		MATLAB		
[212]	J/M	✓		Docker, ODL SDN Controller	Python 2.7.17, TF 1.15.0, Keras 2.3.1	SUMO, Mininet Wi-Fi
[213]	J/M	✓			Python, NumPy, Conda	
[214]	J/M	✓				
[215]	J/M	✓		MATLAB	RL Library	
[216]	J/M	✓		Python Platform	Python	
[217]	J/M	✓		MATLAB (DL toolbox)		
[218]	J/M	✓				
[219]	J/M	✓			TF 2.1.0	
[220]	J/M	✓				
[221]	J/M	✓			Python 3, TF	
[222]	J/M	✓				
[223]	J/M	✓				
[224]	J/M	✓	✓		CloudSim Library	iFogSim
[225]	J/M	✓		MATLAB		
[226]	J/M	✓				
[227]	J/M	✓			C++	
[228]	J/M	✓				
[229]	J/M	✓				
[230]	J/M	✓				

* We have already considered that for the studies using the MATLAB tool, its respective high-level programming language, specialized in numerical computations, data analysis, and algorithm development, is also being employed.

such as Iperf and Ping were also highlighted, demonstrating their relevance in evaluating network capacity and latency.

Furthermore, 25 articles specified the use of simulators, emulators, or simulation databases, with Network Simulator (NS) versions 2 and 3 being the most cited, appearing in six articles. These simulators are favored for their ability to model various network protocols and environments accurately, including 4G and 5G stacks. Mininet Wi-Fi Emulator, present in five studies, is noted for its effectiveness in simulating wireless networks, making it a valuable tool for research in this domain.

Despite these insights, our analysis revealed that 52 out of the 130 articles (40%) lacked sufficient information regarding the aforementioned aspects. This indicates a significant gap in the documentation of validation environments, which poses challenges to the reproducibility and validation of research findings.

In conclusion, while the prevalent use of certain software, libraries, and tools reflects their effectiveness and community preference, there is a need for improved transparency and detailed reporting in academic publications. By providing comprehensive information on the tools and methodologies employed, researchers can enhance the reproducibility and credibility of their work, facilitating further advancements in the field of ML-driven wireless communication systems. Future efforts should focus on establishing standardized guidelines for reporting simulation and experimentation setups, ensuring that research findings are both verifiable and extendable by the broader scientific community.

B. Dataset Availability and Usage

The primary objective of this section is to evaluate the availability and transparency of dataset sources used in studies that apply ML algorithms to optimize resource allocation in wireless communication systems. Given the importance of reproducibility in scientific research, understanding the extent to which datasets are accessible is crucial for validating and extending the findings of these studies.

Our analysis of 130 articles focused on identifying the extent to which authors provide sufficient information about the data sources used to train and evaluate their ML algorithms. This includes references, authors, platforms, repositories, and other essential details needed for data access. The results of this analysis are depicted in Figure 7. We can conclude only 18 articles (13.85%) provided relevant information regarding the sources or databases of their data samples.[1]

These articles detailed the datasets by offering references, identifying authors, or listing platforms and repositories, thereby facilitating access and replication of the study's data. Notably, 13 articles made their data and/or code available upon request from the authors or demand, highlighting a willingness to share but lacking immediate public accessibility.[2]

In contrast, a significant majority of 102 articles (78.46%) did not provide clear or sufficient information about their data sources. This lack of transparency poses a considerable challenge to the scientific community, as it hinders the ability to reproduce experiments and validate the results reported in these studies. The absence of detailed dataset information limits the potential for other researchers to verify findings, test the robustness of the algorithms used, or apply these algorithms to similar problems in the field.

The scarcity of publicly available datasets underscores the need for improved data-sharing practices within the wireless communications research community. By adopting standardized methods for documenting and sharing dataset information, researchers can enhance the reproducibility and impact of their work. Open access to datasets not only supports verification and validation efforts but also encourages innovation by enabling other researchers to explore new avenues of research using existing data.

Furthermore, the analysis reveals a significant gap in the provision of dataset source information within the examined literature. To advance the field and foster a more collaborative research environment, it is imperative that future publications prioritize transparency and accessibility of datasets, aligning with broader movements towards open science and data sharing [255].

Moreover, we can conclude that, although most articles present flowcharts and methodologies of the algorithms implemented to generate the dataset used in the proposed solution [166], or even snapshots of the developed dataset [173], numerous parameters are still required to reproduce the complete sample values faithfully. This is due to factors such as initialization or even randomness that may be involved in this equation. Therefore, it is reasonable to conclude that information regarding the dataset's source and availability is essential for the reproduction process of the solutions implemented in the studied articles to achieve the expected results.

Following this analysis, we present Table X (Part 1 and Part 2), detailing the datasets used in the surveyed papers, specifically highlighting the 18 papers for which access source references were provided by the authors. This table includes the name of the dataset, information regarding the source provided by the authors, a description of the data sample contents, and the specific role of each dataset in executing the task aimed at the ML algorithm referenced in the work for optimizing resource allocation. Table X (Part 1 and Part 2) is presented below, aiming to offer a comprehensive overview of the datasets and to facilitate further exploration and validation by the research community.

In the context of wireless communication research, datasets serve a vital role in addressing various tasks associated with resource allocation optimization using ML, and the analysis of the selected articles in Table X (Part 1 and Part 2) reveals several common data types that these datasets embody, each contributing to different aspects of the problems being tackled.

A significant portion of the datasets focus on network traffic data, which is crucial for tasks such as traffic prediction and QoS enhancement. For example, the Milan Cellular Traffic Dataset [126] and the Abilene Network Dataset [185] provide detailed records of network usage, including Short Message Service (SMS), phone calls, and Web traffic, as well as real-world traffic traces from backbone networks. These datasets enable researchers to predict future traffic patterns and optimize resource allocation strategies, ensuring that bandwidth and computational resources are efficiently distributed to meet demand. Similarly, the Deutsche Telekom Network Traffic Dataset [139] and the Wireless Local Area Network (WLAN) Throughput Dataset [214] offer insights into network performance metrics such as throughput and latency, which are essential for evaluating QoS and developing adaptive rate adaptation frameworks in dynamic network environments.

Another common data type observed is related to user and device preferences, which aids in modeling user behavior and optimizing content delivery within networks. Datasets like the MovieLens Dataset [102] and the MovieLens 1M Dataset [203] contain user ratings of movies, allowing researchers to predict content demand and optimize caching strategies. By understanding user preferences, ML models can anticipate which content is likely to be requested, thereby reducing latency and improving data delivery efficiency. The Moore Dataset [155], which includes traffic traces for classifying IoT devices based on traffic patterns, supports tasks related to device profiling and resource management. Additionally, the Liver Patient Dataset [224], consisting of medical data for classifying patients based on health metrics, demonstrates the applicability of ML in resource management

[1]The 18 papers that provide some information of sources to access the datasets used are summarized in Table X.

[2]The 13 papers that provide data and/or code available upon request from the authors or demand are presented in [140] [142], [143], [152], [156], [162], [168], [185], [191], [197], [198], [209], and [213].

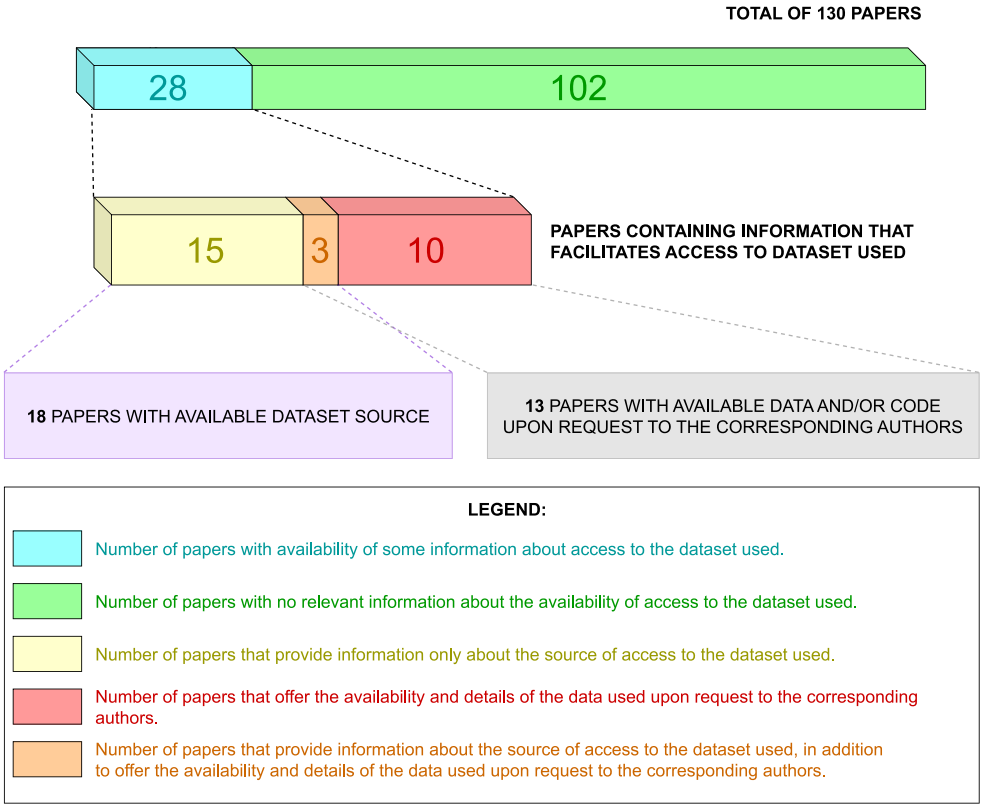


Fig. 7. Provision of information on the dataset source used in the selected papers of our survey.

within fog computing environments, where processing tasks are distributed across multiple nodes.

In addition, datasets like Mixed National Institute of Standards and Technology (MNIST), Canadian Institute for Advanced Research (CIFAR)-10 and CIFAR-100 [123], [142], [230], [230], and [131], while originally designed for image classification tasks, can be leveraged in wireless communication resource allocation by serving as proxies for testing ML models under realistic constraints. For example, these datasets can be used to simulate resource demand patterns by mapping images or their features to communication tasks, such as user requests or signal patterns. This enables the evaluation of model performance in allocating resources like bandwidth or power efficiently, ensuring minimal interference and optimal throughput, similar to how image features are categorized in classification problems.

Energy and resource utilization metrics form another key data type, providing insights into power consumption and resource management. The GWAT-13 Materna Dataset [128], which includes data on Virtual Machine (VM) resource utilization, is used to optimize resource management strategies in cloud and edge networks, minimizing over-provisioning and enhancing energy efficiency. Similarly, the Squid Cache and Nginx Proxy Datasets [125] offer performance measurements under various resource configurations, enabling predictions of Virtual Network Function (VNF) resource requirements and improving energy efficiency in virtualized environments. In addition, Energy Usage and Weather Datasets [169] are combined to create a unified dataset for modeling and forecasting hourly energy demand to train and validate predictive models, evaluated importance in forecasting energy consumption.

Lastly, datasets focusing on mobility patterns and positioning data play a critical role in optimizing resource allocation in dynamic mobile networks. The Simulation of Urban MObility (SUMO)-Based Dataset [143], synthesized using the SUMO framework, provides mobility traces from urban simulations and network traffic types, which are used to optimize ML models for 5G radio access network RAN slicing and resource allocation. This dataset helps simulate real-world traffic conditions and radio signal variations, offering valuable insights into the impact of mobility on network performance. Additionally, the Hydraulic System Monitoring (HSM) Dataset [131] involves sensor data for monitoring hydraulic systems, providing insights into mobility and sensor positioning for ML tasks focused on energy-delay-accuracy trade-offs. Furthermore, China Mobile User Demographics Dataset [170] can be used to identify the location of active users at given timestamps, divided into zones and sub-zones, enabling the prediction of user density and activity distribution across the area.

In summary, the examination of datasets used in ML-driven resource allocation research reveals key patterns and challenges in the field. While this section focuses on analyzing the common data types and their roles within the studied works, a deeper exploration of insights, potential benchmarks for standardization, and future trends will be discussed in Section VII-A. These insights aim to enhance transparency and reproducibility in the literature, providing a pathway

TABLE X
TABLE OF SURVEYED PAPERS PROVIDING INFORMATION REGARDING THE SOURCE OF THE DATASET USED IN THEIR RESPECTIVE WORKS. IT INCLUDES THE AUTHORS' NAMES, REFERENCES, PLATFORMS, REPOSITORIES, DATABASES, OR ESSENTIAL INFORMATION TO ACCESS THE DATASET USED. (PART 1)

Ref.	Dataset Name	Source Information	Description	Role in the Paper
[102]	MovieLens dataset	GroupLens Research Group	Collection of movie ratings, containing detailed information on user preferences and ratings for various movies.	To determine user preferences, which helps in forming D2D multicast clusters. These clusters improve resource allocation and EE by grouping users with similar preferences for more efficient data distribution.
[123]	MNIST dataset	NIST	Large database of handwritten digits used for training image processing systems. It consists of 60,000 training images and 10,000 testing images. Each image is a 28x28 pixel gray-scale image of a single digit from 0 to 9.	To train and test how a SVM classifier based on a data-importance aware radio resource allocation scheme allocates more radio resources to data that are deemed more important for the ML task. In the context of the MNIST dataset, "important" data could be those samples that are closer to the decision boundary of the classifier and, hence, more critical for accurate model training.
[125]	Squid cache and Nginx proxy datasets	SNDZoo project	Each dataset contains about 4.6 million data points with performance measurements under varying resource configurations. They capture the relationship between maximum sustainable traffic load and CPU requirements for VNFs.	To train ML models that predict VNF resource requirements based on performance measurements, aiming to improve the accuracy of dynamic resource allocation, thereby enhancing QoS and minimizing over or under-allocation of resources.
[126]	Milan cellular traffic dataset	Part of a multi-source dataset of urban life in Milan and the Province of Trentino, compiled by Barlacchi et al.	Records of cellular network usage in Milan, Italy. It includes data on SMS, phone calls, and web traffic, recorded at an hourly interval across a grid overlay of 100100 squares.	To train a ConvLSTM NN for predicting future service traffic in vehicular networks, which is used to optimize the allocation of network resources, ensuring efficient and timely distribution based on anticipated demand.
[128]	GWAT-13 Materna dataset	Open-source directory associated with a German cloud provider	Data on VM resource utilization, including CPU, RAM, and storage utilization metrics collected over a three-month period from a German cloud provider, covering 850 VMs on average.	To train a ML model to predict future resource utilization of VMs. This allows for proactive resource management in edge and cloud networks, optimizing resource allocation, minimizing over-provisioning, and enhancing EE.
[131]	MNIST, HSM, Synthetic and real, and Gas sensor array datasets	MNIST is from NIST, while HSM and Gas sensor array are provided by UCI ML Repository. Synthetic and real datasets are generated through specific experiments conducted by the researchers	HSM involves physical measurements from sensors to monitor hydraulic system conditions. MNIST consists of images of handwritten digits. The synthetic and real datasets are used for various modeling purposes, while the gas sensor array dataset contains sensor data for gas mixtures [123].	To evaluate the performance of ML models in resource allocation tasks within wireless edge ML systems. HSM is employed to analyze the energy-delay-accuracy trade-offs in a NN-based classification task. Similarly, MNIST is used to study the classification accuracy under different constraints of energy and delay in order to optimize resource allocation while ensuring performance metrics.
[139]	Deutsche Telekom network traffic dataset	SNDLib	Network traffic data, which includes traffic volumes, flow data, and network topologies.	To simulate network traffic performance under various conditions and patterns, aiding in optimizing the allocation of network resources, such as bandwidth and computing power, and the effectiveness of different allocation strategies.
[142]	MNIST dataset	NIST	Similar to [123] and [131], it is represented by a large database of handwritten digits used for training image processing systems, containing 60,000 training images and 10,000 testing images, whose key feature is a 28x28 pixel gray-scale image of a single digit from 0 to 9.	To train a CNN model in a FEEL framework, simulating real-world scenarios in a distributed environment, where multiple devices perform local training and share model updates with a central server. It helps to optimize resource allocation (such as CPU frequencies and transmission powers) and reduce energy consumption and training delays.
[143]	SUMO-based dataset	Zenodo platform and previous work in MobiWac'21	Data synthetically generated using the SUMO framework. It includes individual traces derived from urban mobility simulations applied to real maps and empirical radio propagation models. The dataset also simulates various types of network traffic, such as web, voice, and video.	To simulate real-world traffic conditions and radio signal variations to optimize ML models for resource allocation in 5G RAN slicing. The dataset helps to train the KPIC-Lite model by providing various realistic traffic scenarios, and it is used to evaluate the performance of their resource allocation strategies under different conditions.
[155]	Moore dataset	Work of Andrew Moore. The repository or platform used was not explicitly mentioned.	Real-time traffic traces with 12 continuous features, containing 324,277 sample data flows for different traffic classes.	To train a DL-ENN model, which classifies IoT devices based on their traffic patterns and resource requirements, optimizing allocation resources, such as CPU and memory.
[169]	Bruce County hourly energy usage and weather data	Energy usage dataset is from IESO, while weather data is from an open-source environmental and natural resources data provided by the Canadian government.	Hourly energy usage dataset contains hourly energy consumption data for Bruce County, Ontario, recording electricity usage patterns over a decade (from 2010 to 2019). Weather data includes various weather attributes collected in Bruce County, including heat and air velocity.	The energy usage dataset enables the model to capture temporal patterns and variations in demand. The weather dataset contributes key environmental variables—heat and air velocity—identified as significant predictors of energy demand. These datasets validate the ML framework, which aims to predict hourly electricity load accurately while considering climatic influences and optimizing resource allocation.
[170]	China mobile user demographics dataset	Kaggle Platform	Information on mobile user activity patterns, timestamps, and geographical locations within a defined region. The data is tailored to identify the location of active users at given timestamps and divided into zones and sub-zones.	It serves as the input for training the NN used for the prediction of active D2D users and their precise locations within zones and sub-zones, forming the foundation for UAV trajectory planning and resource allocation. By leveraging this dataset, the ML model ensures optimal UAV movement and efficient resource sharing to maximize coverage and throughput while minimizing interference.

Continue on the next page

for advancing resource allocation optimization in wireless communications.

C. Summary and Lessons Learned

This section explored experimentation frameworks and dataset usage in ML-based resource allocation optimization, offering insights into simulation practices and dataset availability. The analysis revealed that a vast majority of the surveyed articles relied on simulations, with MATLAB and Python emerging as the most frequently employed tools due to their mathematical modeling capabilities and rich ML libraries, respectively. Despite the dominance of simulation-based studies, experimental works leveraging tools such as

TABLE X
TABLE OF SURVEYED PAPERS PROVIDING INFORMATION REGARDING THE SOURCE OF THE DATASET USED IN THEIR RESPECTIVE WORKS. IT INCLUDES THE AUTHORS' NAMES, REFERENCES, PLATFORMS, REPOSITORIES, DATABASES, OR ESSENTIAL INFORMATION TO ACCESS THE DATASET USED. (PART 2)

Ref.	Dataset Name	Source Information	Description	Role in the Paper
[185]	Abilene network dataset	SNDLib 1.0	Real-world traffic traces, collected by recording network traffic data from the Abilene backbone network, which was an Internet2 project. This dataset is used to analyze and simulate network traffic and routing.	To provide real-world traffic patterns, which are critical for training and validating models that predict network traffic and optimize resource allocation. The dataset simulates realistic network conditions and traffic loads to train and test ML algorithms designed to enhance the efficiency and accuracy of network resource allocation.
[203]	MovieLens 1M dataset	GroupLens Research Group	Real-world data on user ratings of movies. It includes 1,000,209 ratings for 3,952 movies by 6,040 users. Each rating ranges from 1 (worst) to 5 (best), and the dataset includes timestamps for each rating.	To simulate content requests within a network. The ratings provided by users are treated as requests for movie content, helping to model the demand for content within the network. This allows for the evaluation of different caching strategies aimed at optimizing resource allocation by predicting content popularity and user demand.
[214]	WLAN throughput and Big Buck Bunny video datasets	WLAN throughput dataset is available on the data.world platform, while Big Buck Bunny Video is provided by Blender Foundation.	WLAN throughput provides real-world network throughput traces, while Big Buck Bunny Video includes multiple profiles encoded in UHD using AVC and HEVC 4K formats, used for DASH.	WLAN throughput simulates realistic network conditions to train the RL-based rate adaptation framework, dynamically adapting video streaming quality. Big Buck Bunny video is a test content to evaluate ABR algorithm performance under video encoding profiles and network conditions.
[219]	MNIST dataset	NIST	As previously described in [123], [131], and [142] in this table, the dataset consists of handwritten digits used for training image processing systems.	To evaluate the performance of the proposed A3FN for IoT-enabled UAVs. MNIST serves as a benchmark to assess classification accuracy, training convergence, and resource optimization, demonstrating the framework's effectiveness in handling federated ML tasks under resource constraints.
[224]	Liver patient dataset	UCI ML Repository	It consists of 11 important variables, including age, gender, different enzymes and proteins, and a class attribute indicating the presence or absence of liver disease.	To train and validate ML models designed for predictive analytics within a fog computing environment, using real-world medical data to demonstrate how fog computing can handle intensive data processing tasks locally while minimizing the need for constant cloud communication.
[230]	MNIST, CIFAR-10, and CIFAR-100 datasets	MNIST if from NIST, while CIFAR-10 and CIFAR-100 are from CIFAR, Univerity of Toronto	MNIST, as presented in this table ([123], [131], [142], and [219]), is database of handwritten digits. CIFAR-10 consists of 60,000 32x32 color images divided into 10 classes (e.g., airplanes, cars, cats), with 6,000 images per class. CIFAR-100 is more complex, containing 60,000 images across 100 classes, organized into 20 superclasses, each with 5 subclasses. Both datasets are widely used for evaluating machine learning models in image classification tasks.	To assess the FL-GR framework's ability to handle classification tasks with varying levels of complexity and heterogeneity in data distribution. MNIST is used for simple digit classification, while CIFAR-10 and CIFAR-100 provide progressively more complex image classification challenges to demonstrate the robustness and scalability of FL-GR under different conditions.

GNU Radio and LabVIEW with hardware components like USRP were notably less common, underscoring the limited integration of real-world validation in the literature.

Simulation practices varied widely, with some articles detailing environments that included specific software frameworks, programming languages, and emulators, while others lacked sufficient documentation, making reproduction of results challenging. MATLAB was preferred for its robust mathematical toolkits, while Python's extensive libraries like TF and Sklearn offered flexibility for ML integration. Network simulators like NS-3 and Mininet Wi-Fi were also prevalent, reflecting their utility in modeling wireless communication systems, although transparency in configuration details was often lacking, impeding reproducibility.

Regarding datasets, three major types emerged: network traffic data, user behavior data, and energy/resource utilization data. Datasets such as the Milan Cellular Traffic Dataset and Abilene Network Dataset were instrumental for traffic prediction and QoS optimization, while others, like the GWAT-13 Materna dataset, focused on energy consumption and VM resource utilization in cloud environments. Behavioral datasets, such as MovieLens and Moore datasets, supported the optimization of content delivery and IoT resource management. However, only 13.85% of the articles provided sufficient dataset source information, with many datasets either proprietary or inaccessible, highlighting a significant gap in reproducibility and transparency.

VI. QUALITATIVE ANALYSIS OF THE KEY PAPERS CONSIDERED IN OUR SURVEY

While previous sections provided a comprehensive overview of the current literature by analyzing an extensive set of 130 papers, encompassing a broad spectrum of algorithms and proposals, and enabling the collection of data for a richer and more detailed statistical analysis, this section focuses on a more qualitative and technical examination. It delves into 27 key papers selected from the state-of-the-art through a rigorous filtering process, offering a detailed review of the ML proposals presented in these works. This approach allows for a deeper exploration of the methodologies and their applications, fostering a nuanced understanding of the contributions and advancements in the field.

Therefore, the criteria used to select the 27 key papers from the initial pool of 130 articles were: (i) exclusion of

conference and proceedings articles, retaining only more consistent and relevant works published in magazines, journals, and books, in addition to the exclusion of works that do not provide any information regarding the possible tools used for their development (information extracted from Table IX in Section V-A; and (ii) selection of works that propose only leading algorithms, as they are the most used according to the survey conducted in this paper to solve resource allocation problems in wireless communication systems (information extracted from the statistical analysis presented in Section IV and summarized in Figure 6).

From this perspective, we provide a detailed qualitative analysis of these papers, aiming to extract potential insights and future directions for the field of resource allocation optimization in wireless communication systems using the most used ML techniques to overcome these challenges. This qualitative analysis will be divided into three distinct approaches: (i) an in-depth analysis of leading ML solutions considering different resource allocation problems; (ii) a study of simulation parameters in ML-driven wireless communication systems; (iii) and a review of parameter documentation across algorithms to analyze reproducibility challenges in ML.

The first approach proposes a detailed summary of the proposals presented in the studies, along with the specific role of each algorithm within its respective context, highlighting their benefits, addressing their implementation limitations, and evaluating the processes and datasets used to train and guide these algorithms in executing the proposed tasks. The second and third approaches aim to perform a more thorough analysis of the simulation parameters and the documentation of key algorithm parameters presented in each of the 27 studies. We intend to guide future publications to align testing and experimenting scenarios with real-world conditions and challenges, thereby achieving more reliable and relevant results. Moreover, we analyze, for each leading algorithm, the key parameters necessary to reconstruct the proposed solution in the study, allowing for application and even fair comparison across diverse scenarios and methods. Finally, we conclude this section with a summary and lessons learned from the qualitative analysis conducted on the key papers using the three previously described approaches.

A. Deep Dive Into Leading ML Algorithms Across Different Resource Allocation Problems

This subsection presents a comprehensive qualitative analysis of 27 key papers, which were selected by applying the criteria previously established in Section VI to filter the surveyed papers from the state-of-the-art outlined in Section III. The analysis delves into the specific challenges each paper seeks to tackle, examining the processes defined by the proposed algorithms, which are tailored to the unique datasets and requirements of the respective studies, and evaluating how these methods strive to overcome the identified obstacles. Furthermore, the discussion highlights the limitations encountered in implementing these algorithms, providing a nuanced understanding of their practical constraints. By identifying patterns across various algorithms and challenges, this analysis aims to uncover the primary issues associated with their application and propose insights to guide the selection of suitable algorithms for distinct contexts, thereby enhancing their effectiveness in addressing resource allocation problems in wireless communications. Therefore, Table XI (Parts 1 to 4) meticulously compiles key information from the reviewed papers, detailing the problems addressed, the leading algorithms employed, the processes designed for solving these challenges, and the limitations identified. The detailed analysis below will further elaborate the findings and limitations, offering valuable perspectives on the application of ML algorithms in this domain.

In [105], a solution for severe interference, high computational complexity, and EE challenges in UDNs combines an improved dynamic K-Means clustering algorithm with a decentralized multi-agent Q-learning algorithm. This approach enhances EE and minimizes interference, significantly improving system performance. However, challenges remain with high computational overhead, limited scalability with increasing Femto-cell Base Stations (FBSs), and instability in dynamic conditions. The study in [114] presents a hybrid ML approach for resource management in Fiber Wireless (FiWi) networks, combining k-means clustering for network component optimization and Mountain Gazelle Optimization (MGO) with GA for traffic offloading. It improves resource utilization, packet delivery ratios, and offloading performance while reducing blocking probability and delays, though scalability and computational complexity remain challenges as network size grows. In [117], an LSTM model predicts network traffic with 90% accuracy, enabling a GA to optimize resource allocation in C-RAN. This reduces power consumption and improves resource utilization, though the computational complexity of LSTM and GA, along with adaptation difficulties in dynamic environments, highlight trade-offs between accurate prediction, efficient management, and low-latency demands.

The authors in [119] use an ANN to predict SINR values, enabling the construction of a conflict graph to model inter-user interference and improve resource allocation in UDNs, achieving a 186.68% increase in network throughput. However, the approach struggles in highly dynamic environments where interference changes rapidly, reducing prediction accuracy and resource allocation efficiency. The computational complexity of training and running the ANN, coupled with the need for continuous updates to the interference graph, poses challenges for real-time implementation. In [120], ANN and SVM are employed for resource allocation, with ANN predicting optimal decisions based on channel conditions and cell load, and SVM classifying resources using historical data. While these methods enhance allocation efficiency over traditional techniques, they face limitations in prediction accuracy and computational demands, which affect real-time performance. Additionally, the need for frequent retraining or adaptation to varying network scenarios highlights challenges in balancing efficiency, generalization, and adaptability.

In the study presented in [138], LSTM networks are employed to predict energy consumption in real-time for large-scale manufacturing systems, enabling dynamic rescheduling and

TABLE XI
IN-DEPTH ANALYSIS OF KEY PAPERS SURVEYED CONSIDERING SPECIFIC PROBLEMS TO BE SOLVED, LEADING ML ALGORITHMS USED, THEIR OBJECTIVES AND PROCESSES TO BE PERFORMED FOR SOLVING THESE PROBLEMS, AND LIMITATIONS FOUND TO OVERCOME THE CHALLENGES PROPOSED. (PART 1)

Ref.	Problems to be Solved	Algorithm(s)	Objective	Process	Limitations
[105]	• Severe interference; • High computational complexity; • EE.	K-Means	Grouping FBSs based on geographical location and traffic load.	• Location and traffic load consideration; • Dynamic clustering; • Cluster head selection based on distribution density of FBSs.	• Computational overhead; • Scalability; • Dynamic environment adaptation; • Resource allocation fairness.
		Q-Learning	Algorithm focuses on the efficient allocation of RBs among FUEs to maximize EE and minimize interference.	• **State:** each agent CH-FBS includes user association relationships, SINR, application priority, and delay parameters; • **Action:** select RB allocation and transmit power; • **Reward:** maximize EE while satisfying QoS constraints such as data rate and delay.	
[114]	• Inefficient resource utilization in FiWi networks; • Traffic congestion and imbalance in ONUs; • Difficulty in optimal placement of nodes and traffic offloading.	K-Means	To cluster wireless routers and optimize node placement in the FiWi network.	• Randomly initializes centroids; • Assigns wireless routers to nearest centroid; • Iteratively recalculates centroids until convergence; • Outputs clusters and node positions.	• Computational complexity with larger network sizes; • Limited by static conditions in clustering and optimization; • Potential network degradation under dynamic load variations.
		GA	To select the optimal routers for traffic offloading and minimize distance for efficient traffic redistribution in FiWi networks.	• **Initialization:** generate an initial population of router selections for offloading; • **Selection:** identify the best solutions based on fitness (distance min.) between nodes and routers; • **Crossover:** Combine selected solutions to create new ones, ensuring diversity and exploration of potential routes; • **Mutation:** introduce random changes to router selections, avoiding premature convergence and exploring new possibilities for traffic redistribution.	
[117]	• Dynamic traffic demands; • PowCon; • Resource allocation.	LSTM	To predict dynamic network traffic to enable proactive resource allocation.	• **Feature extraction:** use of features such as throughput, no. of UEs, no. of applications, and device types; • **Traffic prediction:** predict future traffic patterns, allowing the network to anticipate high and low demand periods.	• Prediction accuracy; • Computational complexity; • Real-time adaptation.
		GA	It is focused on optimizing resource allocation based on the predicted traffic from LSTM.	• **Initialization:** generate initial solutions (chromosomes) representing potential allocations of RRHs to BBUs; • **Selection:** select the best solutions based on a fitness function that minimizes power consumption and ensures QoS; • **Crossover:** combine selected solutions to create new ones, promoting the exchange of beneficial traits; • **Mutation:** introduce random changes to solutions to explore new possibilities and prevent from converging prematurely to sub-optimal resource configurations.	
[119]	• Severe co-channel interference; • Complex network optimization; • Scalability and practicality.	ANN	To construct an interference graph to represent inter-user interference relationships based on data rather than fixed thresholds or distances.	• **Data collection:** collect uplink SINR data and RB allocation information; • **Training:** train with this data to predict the SINR values, capturing the interference intensity from each potential interfering user; • **Graph construction:** use predicted SINR to construct a weighted conflict graph, where edges represent the interference intensity between users.	• Prediction accuracy; • Computational complexity; • Real-time adaptation.
[120]	• Severe interference; • High computational complexity; • Decentralized decision-making; • Resource allocation efficiency; • Real-time operation.	ANN	To predict optimal resource allocation decisions based on local network information.	• **Feature extraction:** extraction of local attributes like channel conditions, cell load levels, and network topologies; • **Training:** training using labeled data obtained from solving offline optimization problems; • **Decision-making:** make decentralized decisions during network operation, optimizing resource allocation based on real-time data.	• Prediction accuracy; • Computational complexity; • Generalization.
		SVM	To classify and allocate resources effectively by learning from past data.	• **Feature extraction:** similar to ANN, SVM uses local network features; • **Training:** training with the same labeled data used for ANN, focusing on separating different allocation scenarios; • **Decision-making:** to predict the best allocation strategies during network operation, ensuring optimal resource use.	

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optimized resource usage to enhance throughput and reduce energy consumption. However, integrating LSTM models requires significant infrastructure modifications and incurs high computational demands, especially in large-scale operations. The accuracy of input data is critical, as inaccuracies can lead to inefficient rescheduling, while the need for constant updates challenges real-time processing capabilities. In [145], a resource scheduling cost model is proposed for efficient computational resource allocation in Baseband Unit (BBU) pools, reducing power consumption in C-RAN. By leveraging SVM to estimate interference and optimize transmission power, the model achieves an 18% reduction in computing resource waste and a 10% decrease in power consumption. However, its computational complexity limits scalability and reliance on heuristics, and kernel function selection in SVM may result in sub-optimal performance, underscoring trade-offs between resource efficiency and computational demands.

TABLE XI
(Continued) IN-DEPTH ANALYSIS OF KEY PAPERS SURVEYED CONSIDERING SPECIFIC PROBLEMS TO BE SOLVED, LEADING ML ALGORITHMS USED, THEIR OBJECTIVES AND PROCESSES TO BE PERFORMED FOR SOLVING THESE PROBLEMS, AND LIMITATIONS FOUND TO OVERCOME THE CHALLENGES PROPOSED. (PART 2)

Ref.	Problems to be Solved	Algorithm(s)	Objective	Process	Limitations
[138]	• Inefficient resource allocation in manufacturing systems; • High energy consumption; • Bottlenecks in throughput.	LSTM	To predict energy consumption patterns in real-time, enabling dynamic rescheduling and resource reallocation for optimized resource usage.	• **Data collection:** gather large-scale, real-time energy consumption data from manufacturing systems; • **Feature extraction:** process the data to identify relevant features for time-series prediction; • **Training:** train using historical energy consumption data to learn long-term dependencies and patterns; • **Prediction:** forecast future energy consumption in real-time; • **Rescheduling:** dynamically adjust schedules and reallocate resources based on LSTM predictions to reduce energy consumption.	• Integration complexity; • Computational resource demands; • Dependency on data accuracy; • Real-time processing.
[145]	• Efficient allocation of computational resources in BBU pools; • Reduction of power consumption in C-RAN; • Minimization of computational complexity in resource scheduling.	SVM	Optimize resource allocation and minimize power consumption in C-RAN.	• **Feature extraction:** collect data of computing resource cost of the BBU pool and data transmission power and extract features such as interference levels, transmit power, and computing resource requirements; • **Training:** train model using extracted features and corresponding output samples, which represent probabilistic interference constraints; • **Decision-making:** estimate interference levels and optimize transmission power.	• Computational complexity; • Approximation and heuristics; • Dependence on Kernel function.
[146]	• Inefficient prediction of network lifetime, transmission power level, and inter-node distance in WSNs; • High computational complexity in MIP; • Energy dissipation.	ANN	To optimize network lifetime, transmission power level, and inter-node distance predictions with lower computational cost.	• **Feature extraction:** collect data on network lifetime, transmission power levels, and inter-node distances and extract features such as path loss, energy consumption, and network topology; • **Training:** train model using data from MIP model and implement MLP with back-propagation algorithm; • **Decision-making:** predict network lifetime, transmit power, and inter-node distances in real-time.	• Training data dependency; • Model complexity; • Scalability issues; • Integration challenges.
[148]	• SS; • Signal classification; • Optimization of detection probability.	ANN	To classify the state of the PU channel (busy or idle) based on features extracted from received signal data, thereby facilitating efficient SS.	• **Feature extraction:** capture signal data and extract features such as energy values and Zhang statistics from current and previous sensing events; • **Training:** train model with extracted features using labeled data with known channel states (idle or busy); • **Decision-making:** classify signal data, predicting whether the PU channel is idle or busy based on the learned mapping from input features to channel states	• Overfitting; • Hyper-parameter sensitivity; • Computational complexity; • Generalization.
[152]	• Malicious user identification; • Inefficient spectrum utilization; • Network security.	SVM	To classify legitimate SUs and MUs based on their energy detection reports, enhancing spectrum management in CRNs.	• **Feature extraction:** collect energy detection reports from SUs and MUs and extract features such as the energy levels reported by the users during SS; • **Training:** train using labeled data to distinguish between legitimate and MUs based on their energy reports.; • **Decision-making:** use a hyperplane to separate legitimate users from malicious ones, ensuring accurate detection and minimizing false positives	• Training data dependency; • Scalability issues; • Computational complexity; • Real-world variability.
[165]	• Severe interference; • High computational complexity; • EE.	DNN	To decode modulated SCMAs signals distorted by noise and interference, optimize the BERs and and improve reliability.	• **Feature extraction:** gather signal data, including SCMAs codewords and channel conditions, and extract relevant features such as energy values and interference levels. • **Training:** train using labeled data where the correct decoding of SCMAs signals is known; • **Decision-making:** decode new SCMAs signals under different channel conditions, reconstructing the original transmitted data with high accuracy.	• Training data dependency; • Scalability issues; • Computational complexity; • Real-world adaptation.
		DQN	To dynamically construct and decode SCMAs codebooks to maximize throughput, minimize latency, and adapt to varying network conditions.	• **State:** it includes information related to CSIs, SINRs, and the allocation of SCMAs codebooks used for encoding and decoding signals; • **Action:** it involves selecting or modifying SCMAs codebooks, allocating resources such as power and bandwidth to different users and channels, and determining the scheduling of data transmissions; • **Reward:** throughput improvement, latency reduction, error rate reduction, and EEs.	
[168]	• Inefficient mobility management in 5G; • Challenges in accurate user mobility prediction and seamless handovers; • Resource allocation issues in dynamic and dense network conditions.	LSTM	To predict user mobility patterns and locations using a deep residual LSTM model for proactive handover decisions and efficient resource allocation in 5G networks.	• **Feature extraction:** input historical user mobility data (e.g., locations, timestamps) to learn temporal patterns; • **Mobile prediction:** use residual connections in LSTM layers to capture short- and long-term dependencies for predicting future user locations; • **Handover optimization:** leverage predictions for proactive handover decisions and resource allocation to minimize latency and improve user experience.	• Computational complexity; • Sensitivity to the quality and volume of training data; • Potential difficulty in adapting to irregular and sudden changes in mobility patterns.

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The proposed work in [146] uses an ANN to predict network lifetime, transmission power levels, and inter-node distances in WSNs, achieving high accuracy with errors within 3% for a lifetime, 6% for power levels, and 3 meters for distance while reducing computation time compared to the Mixed Integer Programming (MIP) model. However, performance depends on training data quality, and the model faces scalability challenges, high computational demands, and integration difficulties across diverse WSN topologies. In [148], ANN enhances Spectral Sensing (SS) by classifying primary user Primary User (PU)

TABLE XI
(Continued) IN-DEPTH ANALYSIS OF KEY PAPERS SURVEYED CONSIDERING SPECIFIC PROBLEMS TO BE SOLVED, LEADING ML ALGORITHMS USED, THEIR OBJECTIVES AND PROCESSES TO BE PERFORMED FOR SOLVING THESE PROBLEMS, AND LIMITATIONS FOUND TO OVERCOME THE CHALLENGES PROPOSED. (PART 3)

Ref.	Problems to be Solved	Algorithm(s)	Objective	Process	Limitations
[170]	• Complex resource sharing and interference in D2D communication; • Difficulty in trajectory prediction for dynamic UAV-assisted relays; • Need for simultaneous uplink and downlink resource allocation in multi-hop D2D nets.	ANN	To predict UAV trajectories and identify active D2D users dynamically for optimal resource allocation and interference reduction, ensuring increased throughput and improved network efficiency.	• **Data extraction:** gather timestamps and user location data to train the ANN; • **Training:** use input features (day, hour, min, sec) to train the ANN for user location and activity prediction; • **Prediction:** determine zones/sub-zones with active users for UAV trajectory planning; • **Decision-making:** allocate resources based on predicted trajectories.	• Depends on precise training data for accuracy; • Increased computational demand for real-time trajectory and resource predictions; • Sensitivity to dynamic environmental changes.
[181]	• Lack of CSI for resource allocation; • High latency in decision-making for dynamic networks; • Efficient resource allocation in proactive vehicular networks.	DQN	To optimize the downlink data transmission success rate by efficiently allocating radio resources.	• **State:** it is defined by the RUs occupation matrix, which indicates whether each RU is occupied; • **Action:** it involves selecting which RBs to utilize for data transmission. The action space is managed to avoid unreasonable actions, reducing complexity; • **Reward:** based on successful data transmissions, penalizing failed attempts and incentivizing efficient resource use.	• Complexity and scalability; • Real-time adaptation; • Dependency on accurate reward signals.
[187]	• Coexistence of LAA and WLAN Systems; • Channel access mechanisms; • System throughput and fairness.	Q-Learning	To optimize channel access probabilities for LAA and WLAN systems, ensuring efficient and fair coexistence.	• **State:** it represents the current access probabilities of LAA and WLAN systems in the channel, defined by the probabilities that an LTE or Wi-Fi node transmits data in a given time slot; • **Action:** It involves adjusting the channel access probabilities for LTE and Wi-Fi nodes; • **Reward:** based on a proportional fair utility function that balances throughput and fairness of LTE and Wi-Fi systems.	• Scalability issues; • State space size; • Exploration vs. exploitation; • Dependence on accurate parameters.
[192]	• Efficient CRM in MEC environments; • Handling latency-sensitive applications; • Reducing unnecessary energy consumption and data transmission.	DQN	To optimize the offloading strategy for computation tasks in a vehicular edge computing environment.	• **State:** based on the topological sequence of the sub-tasks being scheduled, which includes the dependency relationships between sub-tasks, their execution status and resource availability; • **Action:** it represents the offloading decisions for each sub-task, such as: execute locally, offload to an edge server, or offload to the cloud server; • **Reward:** based on the reduction of energy consumption and latency, whose function normalizes these metrics to maintain a balance between them.	• State and action space complexity; • Convergence issues; • Generalization.
[193]	• PowCon in dual-media communication; • Adapting to mixed fading and impulse noise environments; • Strategies to maximize mutual information.	Q-Learning	To maximize mutual information in dual-media cooperative communication by optimizing transmission power and selecting the best relay algorithm.	• **State:** power levels and relay algorithms; • **Action:** choices of transmission power and relay algorithms, representing potential adjustments in the transmission strategy; • **Reward:** based on mutual information achieved, the function quantifies the effectiveness of the chosen power and relay strategy.	• Scalability issues; • Exploration vs. exploitation; • Dependence on accurate reward signals.
[194]	• Efficiently offloading computation tasks in dynamic vehicular networks; • Managing limited computational resources; • Minimizing latency while maintaining high performance.	DQN	To optimize the offloading decisions by approximating the Q-value function through a NN.	• **State:** it includes current task demands, available resources at edge servers, and network conditions; • **Action:** it involves selecting whether to offload tasks to edge servers or process them locally, determined by a NN; • **Reward:** based on task completion success, latency reduction, and resource utilization efficiency.	• Scalability; • Convergence time; • Generalization.
		Q-Learning	It focuses on iteratively improving the policy for task offloading and resource allocation by updating Q-values based on immediate rewards and future expected rewards	• **State:** similar to DQN, the state space covers offloading and resource allocation decisions; • **Action:** actions are selected based on a Q-table that maps state-action pairs to their respective Q-values, updated iteratively.; • **Reward:** based on immediate task completion success, latency, and resource usage, with Q-values updated to reflect both immediate and future expected rewards.	
[197]	• Minimizing communication latency; • Increasing throughput; • Efficiently managing network traffic; • Ensuring high packet delivery ratios; • Reducing PDR.	Q-Learning	To optimize routing protocols for VANETs to improve network performance metrics.	• **State:** road segment state (e.g., connectivity distribution, distance to destination, and road segment metrics); and vehicle states, such as speed, link reliability, the direction of movement, and buffer size. • **Action:** choosing the optimal road segment based on Q-values; and selecting the best intermediate vehicle for packet forwarding. • **Reward:** positive reward for successful packet delivery, minimized latency, optimized throughput, and efficient traffic management.	• Initial sub-optimal routing decisions during the exploration phase; • Computational overhead; • Scalability.

Continue on the next page

channels as busy or idle using features like energy values and Zhang statistics, improving detection probability by 63% over traditional methods in low SNR environments. Yet, the method struggles with overfitting, hyper-parameter tuning variations, high computational requirements, and inconsistent performance across different environments, reflecting challenges in balancing accuracy, generalization, and real-time deployment.

TABLE XI
(Continued) IN-DEPTH ANALYSIS OF KEY PAPERS SURVEYED CONSIDERING SPECIFIC PROBLEMS TO BE SOLVED, LEADING ML ALGORITHMS USED, THEIR OBJECTIVES AND PROCESSES TO BE PERFORMED FOR SOLVING THESE PROBLEMS, AND LIMITATIONS FOUND TO OVERCOME THE CHALLENGES PROPOSED. (PART 4)

Ref.	Problems to be Solved	Algorithm(s)	Objective	Process	Limitations
[198]	• Network congestion; • Inefficient bandwidth utilization; • Increased Latency.	DQN	To dynamically adjust traffic routes to enhance network performance metrics, by minimizing congestion, optimizing path selection for data flow, and improving the overall QoS in DCNs.	• **State:** the model is trained using the network's traffic patterns and structure by simulating various traffic conditions. States include link utilization, queue lengths, and traffic demands; • **Action:** path selection, load balancing, and re-routing; • **Reward:** rewards are calculated based on throughput, latency, and load balancing.	• Complexity and overhead; • Scalability issues; • Dependency on accurate data.
[208]	• Explosive growth of data traffic; • Limitations of MEC server resources; • Appropriate selection of MEC servers; • Efficient resource allocation strategies.	DQN	To optimize resource allocation in MEC servers to reduce total energy consumption and network latency, thereby improving computational efficiency.	• **State:** information about user data volume, available computational resources on the MEC server, and the state of the communication channel; • **Action:** it is related to resource allocation strategies and transmission power allocation.; • **Reward:** it is designed to be negatively related to energy consumption and positively related to the volume of processed tasks and reduced latency.	• Computational complexity; • Exploration vs. exploitation; • Sensitivity to weight updates; • Dependence on accurate simulations.
[209]	• Dynamic node positions; • Variable channel states; • Inefficiencies in CSMA/CA protocols.	Q-Learning	To optimize the contention window size adaptively to reduce collisions and latency in the network.	• **State:** states are defined as the observed collision probabilities in the network; • **Action:** actions correspond to adjustments in the contention window size; • **Reward:** based on the reduction in collision rates and latency.	• Complexity of real-time learning; • Scalability; • Dependence on accurate real-time data.
[210]	• Energy consumption; • Network longevity; • Dynamic network conditions.	Q-Learning	To optimize routing to enhance EE and network longevity by reducing transmission delays and increasing throughput.	• **State:** energy levels of nodes, packet delivery ratios, and packet consistency; • **Action:** selection of optimal routes for data transmission; • **Reward:** based on EE, reliability, and reduced transmission delays.	• Real-time learning complexity; • Scalability; • Overhead with multiple agents.
[211]	• Congestion in random access channel of LTE-Advanced networks; • Inefficiency in resource allocation; • QoS requirements.	Q-Learning	To optimize the random access channel by dynamically adjusting the allocation of preambles between H2H and M2M devices, reducing congestion and improving access success rates.	• **State:** number of preambles allocated to M2M devices; • **Action:** adjustments in the number of preambles allocated to each type of devices; • **Reward:** difference between actual H2H blocking probability and a threshold value.	• Real-time learning complexity; • Need for re-learning; • Scalability.
[212]	• Efficient resource allocation; • Traffic management; • Latency reduction.	DQN	To optimize the deployment of controllers in an SDVN, adjusting their number based on real-time traffic conditions to reduce latency and improve overall network performance.	• **State:** traffic metrics, (e.g., vehicle density, speed, and current network load), and network conditions, which include information on current network performance, such as delay and packet loss rates; • **Action:** action is related to adjusting the number of active controllers in the network to match current traffic demands; • **Reward:** based on improvements in end-to-end delay and packet loss, encouraging configurations that optimize network performance.	• Computational overhead; • Adaptation delays; • Scalability.
[215]	• Efficiently allocating radio resources between uRLLC-eMBB slices; • Balancing the trade-off between throughput and latency; • Scalability of resource allocation in RAN slicing.	DQN	To maximize eMBB throughput while ensuring that uRLLC latency remains below a predefined threshold, using a hierarchical RL approach for inter-slice and intra-slice resource allocation.	• **State:** generated from training data on queue occupancy and channel quality for uRLLC and eMBB slices, representing the current state of the network; • **Action:** RB allocation ratios between slices (main-agent) and among user groups within slices (sub-agents); • **Reward:** designed to increase throughput and reduce latency, penalizing delay threshold violations.	• Higher complexity with larger state/action spaces; • Performance depends on training data quality and exploration; • Scalability challenges with more slices or groups.
[216]	• High packet loss and interference in LoRa systems; • Inefficient resource allocation in large-scale IoT networks; • Challenges in balancing EE, transmission rate, and latency.	DQN	To optimize resource allocation in LoRa networks by minimizing transmission power, maximizing throughput, and reducing latency, using DQN integrated with IR-LEOA for parameter tuning.	• **State:** generated from data on transmission power, spreading factor, and channel conditions in the LoRa network; • **Action:** selection of optimal resource allocation parameters (e.g., power, channel, spreading factor); • **Reward:** encourages reduced power usage, higher throughput, and lower latency.	• High computational cost and dimensionality challenges during training; • Scalability issues with extremely large IoT deployments; • Performance relies heavily on the quality of training data.

The study in [152] presents an SVM) approach to enhance spectrum management and security in CRNs by classifying legitimate users and Malicious Users (MUs) based on energy detection reports. While this method improves spectrum utilization and network security, its effectiveness depends heavily on high-quality training data. Incomplete datasets may fail to capture sophisticated MU behaviors, reducing detection accuracy. Additionally, the approach faces scalability issues as larger networks increase data volume, slowing real-time classification and decision-making. Computational demands

further complicate implementation, particularly in dynamic environments where generalization to varied conditions is essential. Meanwhile, [165] explores the use of DNN and DQN to address interference, computational complexity, and EE in Sparse Code Multiple Access (SCMA) systems. The DNN achieves a 90% reduction in BER under high SNR conditions by decoding distorted signals, while the DQN dynamically optimizes codebooks to enhance throughput and reduce latency. However, both models face challenges with scalability and computational complexity, limiting their suitability for low-latency applications and requiring further refinement to adapt effectively to real-world conditions.

In [168], a deep residual LSTM model is performed to enhance mobility management in 5G networks by predicting user mobility patterns, achieving a packet loss rate of 0.9% and a handover success rate of 95.1%. While effective, the model faces scalability and computational demands in dynamic environments. Authors in [170] propose a dual optimal channel allocation algorithm to improve resource allocation in multi-hop Device to Device (D2D) communications. Leveraging Unmanned Aerial Vehicles (UAVs) as dynamic relays and ANN for trajectory prediction, it minimizes interference and maximizes throughput, achieving an average D2D throughput higher than other multi-hop D2D communication schemes in the literature. The system improves the churn rate and D2D coverage significantly. However, limitations include scalability issues in handling large-scale dynamic environments and the dependency on accurate user predictions. Future work suggests exploring RL to optimize UAV trajectories further.

In the paper presented in [181] addresses efficient resource allocation in proactive vehicular networks with limited Channel State Information (CSI) and high decision-making latency. It proposes a DQN model that optimizes downlink data transmission by defining network states through an occupation matrix of Resource Unit (RU) and intelligently allocating Resource Blocks (RBs), achieving a 98% of data transmission success rate at 40% resource load. Challenges include scalability as the action space grows, reliance on accurate reward signals, and difficulties adapting to dynamic interference patterns in vehicular networks. In [187], Q-learning optimizes channel access probabilities for License-Assisted Access (LAA) and WLAN, balancing throughput and fairness via a proportional fair utility function. However, modeling interactions, such as varying packet lengths and traffic patterns, increase state-space complexity and computational demands. Achieving a proper balance between exploration and exploitation is also critical, as poor tuning can lead to inefficient spectrum usage, especially under dynamic conditions.

The approach in [192] uses a DQN to optimize task offloading in vehicular edge computing, reducing latency and energy consumption by modeling states with sub-task dependencies, execution status, and resource availability, and deciding whether to process tasks locally or offload them. Rewards minimize latency and energy use, achieving significant improvements over traditional methods. Challenges include the complexity of state and action spaces as sub-tasks and resources grow, convergence issues in dynamic environments, and limited generalization to new scenarios with varying topologies or traffic patterns. The authors in [193] employ Q-Learning to optimize transmission power and relay selection in dual-media cooperative communication, maximizing mutual information under mixed fading and impulse noise. While effective, scalability challenges arise with increased relay algorithms and power levels, and balancing exploration and exploitation is critical to avoid sub-optimal relay choices. Dependence on accurate reward signals is also a limitation, as errors in mutual information feedback can hinder optimization in fluctuating channel conditions.

The paper in [194] investigates optimizing computation task offloading in dynamic edge computing environments using DQN and Q-Learning algorithms. The DQN leverages a NN to approximate Q-values, factoring in task demands, edge resources, and network conditions to decide task allocation, while the Q-Learning algorithm iteratively refines strategies based on immediate and future rewards. Both methods reduce latency and improve resource efficiency, but the DQN struggles with scalability due to complex state and action spaces, and Q-Learning may face slower convergence and limited adaptability to fluctuating environments. The authors in [197] propose a Q-Learning-based routing protocol for Vehicular Ad hoc Networks (VANETs) to optimize latency, throughput, and packet delivery. By evaluating road and vehicle conditions, the protocol selects optimal paths for forwarding packets, achieving improved performance over traditional methods. However, the initial exploration phase may cause temporary latency increases in dynamic traffic conditions, and scaling the network raises computational demands, posing challenges for real-time applications in large, dynamic VANETs.

The study in [198] explores a DQN approach to optimize traffic routing in DCNs, targeting congestion reduction, path selection, and improved QoS. By evaluating states such as link utilization and queue lengths, the model selects paths and balances loads, achieving better throughput, bandwidth utilization, and round-trip time compared to different benchmarks. However, the complexity and computational overhead of real-time decision-making pose challenges as network scale and traffic loads grow, potentially delaying routing adjustments. Additionally, the model's reliance on accurate traffic data means prediction errors can lead to inefficient routing, exacerbating congestion during peak usage. The article in [208] proposes a DQN-based method for optimizing resource allocation in MEC servers to reduce energy consumption and network latency. While effective at managing resources and transmission power, the approach struggles with high computational demands as task and user numbers increase, which can delay real-time decisions. The exploration-exploitation trade-off also affects early performance, and sensitivity to weight updates in dynamic network conditions complicates achieving consistent results, highlighting scalability and stability challenges in MEC systems.

In the research by [209], Q-Learning is applied to optimize contention window size in Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA) protocols, addressing challenges like dynamic node positions, variable channel

states, and traditional inefficiencies. By adapting the contention window based on collision probabilities, the approach enhances throughput and reduces packet loss in dense, dynamic wireless environments. However, real-time learning introduces computational overhead, especially as the network scales, and inaccuracies in channel state information can lead to sub-optimal adjustments and worsening performance. Scalability is also a concern, as managing a larger state-action space in bigger networks complicates maintaining low latency and high throughput. In [210], Q-Learning optimizes routing in WSNs, focusing on EE and network longevity by selecting routes based on node energy, packet delivery ratios, and consistency. This method improves delivery rates, latency, energy consumption, and network lifetime. Nonetheless, real-time learning in dynamic environments increases computational overhead and delays, while scalability issues arise with larger networks due to the growing state-action space. Additionally, coordinating multiple agents introduces resource-intensive communication, reducing efficiency in large-scale deployments.

In [211], the authors use Q-Learning to reduce congestion in the random access channel of Long-Term Evolution Advanced (LTE-A) networks by dynamically allocating preambles between Machine-to-Machine (M2M) and Human-to-Human (H2H) devices. This adaptive approach improves access success rates under varying traffic conditions, achieving high success probabilities for both device types, even during high loads. However, challenges include the computational complexity of real-time learning, which can slow responsiveness, and the overhead of frequent re-learning to handle traffic fluctuations, potentially causing temporary performance drops. Scalability also poses difficulties as network size and device numbers increase, making it harder to maintain low latency and success rates. The paper in [212] explores a DQN approach to optimize controller deployment in Software Defined Vehicular Networks (SDVNs). By dynamically adjusting the number of active controllers based on real-time traffic, the system reduces end-to-end delay and packet loss, particularly under fluctuating vehicular density and speed. Challenges include the computational demands of real-time operation in large networks, adaptation delays during controller adjustments, and scalability concerns in managing an increasing number of controllers. Despite these issues, the DQN-based system outperforms static architectures, enhancing latency and reliability in dynamic vehicular environments.

The study in [215] introduces a hierarchical RL framework for resource allocation in RAN slicing, focusing on ultra-Reliable Low Latency Communications (uRLLC) and enhanced Mobile Broadband (eMBB). The approach integrates MARL, specifically DQN, with sub-agents addressing intra-slice allocation and a main agent handling inter-slice resources. The model reduces uRLLC latency to below 1 ms while maintaining eMBB throughput degradation at a maximum of 4%. Challenges include managing the complexity of state-action spaces and ensuring scalability across larger networks, while future works involve refining the RL algorithm to accommodate additional network slices and traffic conditions. Finally, the paper in [216] proposes a DQN-based approach to optimize resource allocation in Low-Power Wide-Area Network (LPWAN) by learning the optimal configuration of transmission power, spreading factors, and channels. The deep adaptive RL dynamically adjusts these parameters to minimize energy consumption and latency while maximizing throughput. It is combined with the Integrated Remora with Lotus Effect Optimization Algorithm (IR-LEOA) to improve convergence and scalability in high-dimensional optimization tasks. Results demonstrate that the suggested IR-LEOA-based resource allocation over Long Range (LoRa) achieved higher EE compared to traditional methods, while future scope include strengthening security, enhancing scalability and flexibility in ever-more complicated IoT contexts, and validate the algorithm in real-world circumstances.

The qualitative analysis of the 27 surveyed key papers, previously summarized and presented in Table XI (from 1 to 4), reveals common themes and significant contributions to resource allocation optimization in wireless communication systems through the use of ML algorithms. Among the most commonly used algorithms are Q-Learning, DQN, and ANNs, primarily applied in scenarios involving traffic management, resource allocation in mobile networks, and performance optimization in dynamic and highly variable environments.

A common pattern across these studies is the use of RL-based algorithms, such as Q-Learning and DQN, to optimize real-time decisions, including spectrum allocation, route selection, and energy management. However, the limitations of these algorithms are closely related to computational complexity and the need for real-time data. Scalability is a recurring concern, especially in large-scale networks where the number of states and actions grows exponentially, slowing down learning and making adaptation to new conditions more difficult. Additionally, the reliance on accurate data for training these algorithms is critical; any inaccuracies can lead to sub-optimal decisions, harming network performance.

Another common challenge is balancing exploration and exploitation, particularly in dynamic environments where network conditions change rapidly. In many cases, algorithms need to be frequently adjusted or retrained to maintain effectiveness, which can introduce additional computational costs and adaptation periods during which network performance may be compromised.

In terms of contributions to the scientific community, these studies demonstrate that while ML algorithms offer a powerful tool for optimizing wireless communication networks, there is a clear need to develop methods that can handle the complexity and variability of real-time network conditions more efficiently. Hybrid solutions that combine ML algorithms with heuristics or rule-based methods may be a promising direction to overcome current limitations.

B. A Study of Simulation Parameters in ML-Driven Wireless Communication Systems

One of the critical challenges in resource allocation optimization for wireless communication systems using ML algorithms lies in creating accurate and realistic simulation scenarios. The inherent complexity of wireless environments,

characterized by factors such as multi-path propagation, varying signal interference, dynamic user behavior, and fluctuating traffic loads, poses significant difficulties for simulation. Simulating a wireless communication environment is challenging because real-world channel conditions' highly variable and dynamic nature cannot be fully captured, leading to potential discrepancies between the proposed solution's performance in virtual and real-world scenarios. For instance, accurately modeling the impact of multi-path fading and user mobility patterns in a simulated environment can be highly challenging, often leading to discrepancies between simulated and real-world performance.

In this section, we delve into an in-depth examination of the 27 articles selected from our study as detailed in Section VI, which focuses on the resource allocation optimization in wireless communication systems using ML algorithms. The primary goal of this analysis is to extract detailed information regarding the proposed scenarios and the simulation and experimentation parameters provided by the authors to generate the test environments.

Table XII offers a comprehensive overview of each paper, highlighting the detailed descriptions of the proposed scenarios and the simulation parameters used. Moreover, the table provides insightful considerations on possible improvements and adaptations to these scenarios. These recommendations aim to bring the test and experimental conditions closer to real-world environments, thereby enhancing the reliability and relevance of the results obtained from the ML-based solutions for resource allocation optimization. By incorporating more variables and conditions found in real-world scenarios, these improvements can help in achieving more accurate and practical outcomes.

In summary, the analysis of Table XII (Part 1 and Part 2) emphasizes the importance of bridging the gap between simulated and real-world environments. By refining simulation parameters and incorporating real-world complexities, the research community can develop more robust and reliable ML solutions that perform well in practical applications.

Recognizing and standardizing essential parameters for simulating scenarios in wireless communication systems to achieve conditions that closely resemble real-world situations is inherently challenging due to the vast array of applications and diverse scenarios, as highlighted by the 27 articles analyzed in Table XII (Part 1 and Part 2). Each study presents distinct contexts, ranging from UDNs and cognitive radio systems to vehicular ad hoc networks and heterogeneous environments, each possessing unique characteristics and requirements. This diversity necessitates a tailored approach to parameter selection, as simulations must accurately capture specific scenario dynamics, including interference patterns, user mobility, environmental conditions, and technology-specific nuances. The absence of a universal parameter set is problematic because it demands that researchers conduct thorough and meticulous analyses to identify critical elements for each specific scenario, ensuring the simulation's fidelity and applicability to real-world situations. This complexity underscores the need for a flexible yet rigorous framework capable of adapting to the varied demands of different wireless communication contexts, allowing simulations to provide meaningful insights that enhance the understanding and deployment of communication technologies in practical settings. As such, achieving realistic simulations requires a comprehensive understanding of the intricate interactions within each scenario, highlighting the importance of detailed and precise modeling to bridge the gap between theoretical and practical applications.

The comprehensive analysis of the papers described in Table XII (Part 1 and Part 2) also reveals several recurring gaps in simulating wireless communication systems and offers targeted recommendations for improvement. For instance, UDNs in [105] lack critical components such as dynamic interference modeling and user mobility patterns, which are vital for accurately simulating dense urban environments where building obstructions and multi-path propagation are prevalent. Similarly, in the C-RAN scenarios, such as [117], the absence of realistic traffic models and inter-cell interference management hinders the applicability of findings for real-world deployments. Articles like [120], focused on heterogeneous networks, emphasize the need for dynamic channel state feedback and adaptive resource allocation to manage real-time interference and load balancing. The vehicular networks discussed in [197] would benefit significantly from the integration of dynamic traffic models and V2X communication strategies to better simulate real-time vehicle interactions.

Furthermore, in CRNs, such as in [152], incorporating adaptive spectrum access and environmental noise considerations would enhance the simulation of spectrum-sharing scenarios. The multi-technology study in [148] can be improved by incorporating frequency hopping and real-time signal quality assessments to manage interference across different bands better. Across these studies, common themes of improvement include advanced environmental modeling, real-time feedback loops, and adaptive resource management. In UAV-assisted multi-hop D2D communications as described in [170], adding dynamic energy constraints, real-time obstacle modeling, and adaptive UAV trajectory planning would significantly enhance the fidelity of simulations in environments with 3D mobility. For RAN slicing scenarios such as in [215], incorporating heterogeneous gNodeB deployments, realistic user mobility models, and detailed latency thresholds for uRLLC traffic could better represent the complexities of modern networks. The LPWAN simulations discussed in [216] would benefit from modeling environmental conditions like shadowing, fading, and interference patterns, as well as introducing stochastic traffic profiles and adaptive spreading factors for capturing operational variabilities in dense IoT networks.

The considerations previously described highlight the recurring need for simulations to incorporate real-time adaptive mechanisms, comprehensive environmental modeling, and advanced traffic dynamics to bridge the gap between theoretical models and practical deployments. Following this, to address these gaps, simulations should leverage ML techniques for real-time adaptive decision-making, enabling networks to adjust to changing conditions dynamically.

TABLE XII
TABLE OF DETAILED DESCRIPTION OF THE SIMULATION ENVIRONMENT AND CONSIDERATIONS TO ENHANCE A MORE REALISTIC SCENARIO. (PART 1)

Ref.	Detailed Scenario Description	Given Simulation Parameters	Considerations for a more Realistic Scenario
[105]	**UDN:** A two-tier downlink UDN structure with one macro-cell BS and multiple dense FBSs. The FBSs are clustered dynamically to enhance network capacity and user QoS.	Carrier frequency, bandwidth per RB, number of FBSs, maximum transmitted power of FBS, FUEs served by each FBS, thermal noise power, and BER.	Lacks dynamic interference patterns, user mobility models, channel state information, and environmental factors such as shadowing and fading.
[114]	**Hybrid FiWi network architecture**: Combining optical and wireless networks, with components such as ONUs, OLTs, and WRs distributed in clusters to optimize resource management and balance traffic loads.	FiWi network area, total number of clusters, number of OLTs, ONUs, and WRs in the network, transmission range of WR, maximum handling capacity of the ONUs, and randomly load generated at WR.	Incorporate dynamic interference patterns, user mobility, realistic traffic loads, advanced CSI, energy constraints, scalability, propagation models, and application-specific QoS metrics.
[117]	**C-RAN:** Multiple RRHs with varying numbers of UEs (1 to 1000) and high and low throughput UEs. Utilizes LSTM for throughput prediction in different urban scenarios like offices, residential, and markets.	Number of UEs per RRH, number of high and low throughput UE, and number of application types of UEs.	Needs more realistic traffic models, inter-cell interference management, mobility patterns, and heterogeneous network configurations.
[119]	**Femtocell Network:** Simulation of FAPs with multiple UEs per femtocell and varying system bandwidth.	Information related to scenario area, number of FAPs, number of UEs per FAP, system bandwidth, maximum UE transmit power, semi-persistent factor, path loss compensation factor, thermal noise density, and path loss model.	Should include user distribution models, inter-femtocell interference, and advanced propagation models (fading, shadowing).
[120]	**Heterogeneous Wireless Network:** Consists of macro cells and small cells with a focus on resource allocation optimization through allocation schemes like SVM and ANN.	Simulation area, number, and positioning of macro and small cells, transmit power, antenna gain, available bandwidth, noise power spectral density, bias values for range expansion of all cells, and shadow fading feature.	Requires modeling of dynamic interference, user mobility, real-time channel state feedback, and advanced resource allocation strategies.
[138]	**Industrial Wireless Network:** Pick and place operations for industrial robots focusing on energy consumption optimization and real-time resource allocation.	Speed and acceleration limitations of pick and place operation, sequence of pick and place robot motions, distances between the points, number of repetitions.	Lacks real-time feedback control, sensor accuracy, and environmental interference considerations such as signal attenuation and noise.
[145]	**C-RAN for Energy Efficiency:** Multi-cell deployment with centralized baseband processing for RRH-UE interference management, focused on reducing computing resource consumption.	Number of cells of C-RAN, distribution model of UEs per cell, bandwidth, carrier frequency, coverage radius of RRH, maximum RRH transmit power, multi-path fading, distance-dependent path loss RRH-UE, interference threshold, noise power density, constant factor and constant computing resource variables to calculate computing resource requirement of each UE in BBU pool, and limitation of MOPTS for each BBU.	Needs real-time computational load balancing, adaptive interference management, and consideration of environmental effects.
[146]	**Mesh Network Scenario:** Focuses on inter-node distances and transmission power levels for path loss and shadowing models.	Number of network nodes, inter-node distances, transmission power level (transmission power consumption and output antenna power for each power level), path-loss value model, and shadowing model.	Should include node failure and recovery processes, real-time traffic management, and environmental impacts on signal propagation.
[148]	**Multi-Technology Network:** Measures of channels in FM broadcasting, E-GSM, DCS, and UHF television with details on frequency, signal bandwidth, and gain.	Measures of channels in different technologies (FM broadcasting, E-GSM (900) Downlink, DCS (1800) Downlink, and UHF television), which includes: start, center, and stop frequencies; signal bandwidth; gain; decimation rate; and sampled bandwidth.	Requires frequency hopping, co-channel interference management, and real-time signal quality assessment.
[152]	**Cognitive Radio Network:** Focus on spectrum sensing with secondary and malicious users, emphasizing security and interference management.	Bandwidth, sensing period, number of legitimate secondary users, number of malicious users, probability of primary users, channel model, attack type in the network, and sensing type.	Should incorporate dynamic security threats, adaptive spectrum access, and environmental noise variations.
[165]	**SCMA-based Network:** SNR and pilot length considerations in a SCMA network for optimizing resource allocation.	SNR, pilot lengths, SCMA codebook size, number of signal streams, number of retransmissions, PER constraint, mean packet arrival rate, data rate, length of each packet, time delay constraint, number of resources, noise model, values of different types of loss factors (congestion, path, blocking, and interference loss), and fading channel model.	Lacks dynamic channel conditions, adaptive modulation, and real-time error correction mechanisms.
[168]	**5G Networks:** Mobility management using a deep residual LSTM model integrated into the 5G infrastructure, involving BSs, user devices, and a core network, to improve resource allocation and handover efficiency by predicting user mobility patterns.	Number of BSs, area covered, number of users, simulation duration, user mobility model, mobility model random waypoint, minimum and maximum speed, and pause time range.	Include other models, including vehicular mobility, environmental factors such as shadowing, seasonal variations, channel conditions (e.g., multi-path fading), and dynamic user behaviors like varying speeds and irregular pause times.
[170]	**UAV-Assisted Multi-Hop D2D Scheme for 5G and Beyond:** UAVs act as dynamic relays within zones and sub-zones, to optimize resource allocation and throughput for simultaneous uplink and downlink transmissions by predicting UAV trajectories based on user activity and minimizing interference.	Zone and sub-zone areas, number of UAVs, UAV height and wireless range of UAV, dense and rare zone populations, variance, mean and shadowing for LOS and NLOS, numerology, maximum resource blocks available, simulation time, packet size, required data rate, noise PSD, total transmission power of BS, D2D-Tx, and UAV.	Account for realistic UAV energy constraints, dynamic obstacles affecting UAV paths, refined shadowing models incorporating urban morphology, mobility patterns of users, non-uniform noise power density, stochastic packet arrival rates, and weather-induced variations in wireless link quality.

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Comprehensive environmental modeling, including considerations for shadowing, fading, and interference patterns, should become standard practice to better represent the complex interactions within wireless networks. Additionally, incorporating multi-layer network strategies, such as cross-layer optimization and integrated resource management, can offer a holistic approach to simulations. By implementing these enhancements, simulations can yield results that more closely mirror real-world conditions, thereby improving their practical relevance and reliability. These improvements not only increase the realism of simulations but also enhance their predictive power for network performance, aiding in

TABLE XII
(Continued) TABLE OF DETAILED DESCRIPTION OF THE SIMULATION ENVIRONMENT AND CONSIDERATIONS TO ENHANCE A MORE REALISTIC SCENARIO. (PART 2)

Ref.	Detailed Scenario Description	Given Simulation Parameters	Considerations for a more Realistic Scenario
[181]	**LTE-A Networks:** Adaptive resource allocation with considerations for varying user demands and network conditions.	Environment length and width, number of RBs and RUs, number of RBs in RRUI, maximum RUs occupied by each AP, vehicle and AP density, vehicle speed, transmit power of each RU in each AP, path-loss factor, maximum SIR threshold, transmission rate requirement, probability of selecting RB randomly to form RRUI, and high-quality RB to form RRUI.	Needs real-time interference management, adaptive user prioritization, and traffic load balancing.
[187]	**LTE and Wi-Fi Coexistence:** Examines RL for resource allocation between LTE BSs and Wi-Fi APs, focusing on packet data and transmission rates.	Number of LTE BSs and Wi-Fi APs, number of users per LTE BS/Wi-Fi AP, probability to user transmit data, Wi-Fi and LTE packet data, SIFS, DIFS, RTS, CTS, ACK, time slot, and LTE and Wi-Fi transmission rates.	Requires real-time traffic management, spectrum-sharing strategies, and latency-sensitive application integration.
[192]	**Edge Computing in Vehicular Networks:** Task offloading between vehicles and edge/cloud servers with consideration of CPU clock speeds and transmission rates.	Input/output data size and computation size of each sub-task, edge transmission and remote cloud transmission rate of uplink and downlink, CPU clock speed of vehicle computation module, edge server, cloud server, transmission power, and execution power of vehicle computation.	Should incorporate vehicular mobility models, dynamic network topologies, and environmental impacts on computational loads.
[193	**Hybrid Wired and Wireless Communication:** Mixed media communication with adaptive strategies for resource allocation optimization, considering power line and wireless channels.	Simulation method, total system power, link distance normalization, distance attenuation factor, average noise power of the channel, BernoulliGaussian model impulse noise, parameters of the power line and wireless channel fading, power line and wireless channel SIC capabilities, and outage probability threshold of the system.	Needs realistic noise models, interference management, and environmental effects on signal quality.
[194]	**MEC:** Focus on energy-efficient communication with task distribution across devices and edge servers.	Number of devices, number of tasks, system bandwidth, transmission power, background noise, data size, number of CPU cycles per bit, computation capability of each device, energy consumption per CPU cycle, computation and storage capability of the edge server.	Requires adaptive resource allocation, real-time energy consumption modeling, and user mobility considerations.
[197]	**VANETs:** Intelligent routing protocols for communication between vehicles and RSUs.	Number of vehicles, network size, road network, road segments, vehicle speed minimum, data packets size, control packets size, communication range vehicles, and communication model.	Should include dynamic traffic models, real-time vehicle densities, and V2X communication strategies.
[198]	**DCNs:** resource allocation optimization in data centers with communication mode and network topology considerations.	Communication mode in a data center, network topology, type of switches, link bandwidth in the topology, and weight factors related to average occupied bandwidth of available path.	Needs real-time traffic patterns, network reconfiguration, and environmental impacts on data transmission.
[208]	**MEC Systems:** Task distribution between base stations and mobile devices with a focus on computing power and channel gain.	BS coverage, system bandwidth, number of MEC servers, number of mobile devices, wireless channel gain, gaussian channel noise, task data volume, user transmission bandwidth, MEC server computing power, and mobile device computing power.	Requires user mobility patterns, adaptive scheduling, and environmental factors affecting network performance.
[209]	**IoT Networks:** Multiple access points with a focus on communication protocols and traffic types in IoT networks.	Simulation scenarios, network size, number of APs, channel frequency, transmission rate, traffic type, mobility model, movement speed, packet size, and time slot.	Should include real-time data exchange, adaptive protocols, and environmental impacts on device interactions.
[210]	**Sensor Networks:** Energy-efficient routing in sensor networks using RL techniques.	Network area, total number of sensor nodes, distribution model of malicious nodes, time taken for simulation, type of antenna, type of channel, and control packet size.	Needs dynamic environmental conditions, energy consumption models, and interference from external networks.
[211]	**Heterogeneous Networks:** Analysis of H2H and M2M traffic using Q-Learning for preamble assignment and resource allocation optimization.	Network type, sub-frame duration, simulation duration, number of preambles, number of RBs, H2H, traffic arrival rate, M2M traffic arrival rate, number of RBs used by H2H, number of RBs used by M2M, M2M message size, and M2M modulation.	Lacks realistic traffic dynamics, interference management, and adaptive resource allocation strategies.
[212]	**Vehicular Networks with OpenFlow:** Simulation of vehicle densities, speeds, and RSU ranges in an OpenFlow-controlled vehicular network.	Vehicles density, vehicles' speed, RSU range, number of RSUs, bandwidth, OpenFlow version, simulation area, simulation time, propagation model, traffic type, packet size, and periodicity.	Should incorporate real-time vehicle dynamics, adaptive routing, and traffic management strategies.
[215]	**RAN Slicing:** gNodeB managing eMBB and uRLLC slices, using multiple RL agents to optimize RB allocation, balancing eMBB throughput with delay requirements of uRLLC traffic.	Number of gNodeB, user groups and slices, MCS indexes, packet size, delay threshold of uRLLC, arrival traffic distribution per user.	Introduce heterogeneous gNodeBs, user mobility models, traffic patterns with bursty arrivals, MCS transitions based on SINR, varying delay thresholds for uRLLC, and packet loss due to interference.
[216]	**LoRa-based LPWAN System:** LoRa nodes distributed across a grid, with a central gateway managing resource allocation, using DARL model to optimize parameters such as transmission power, spreading factor, and channel to improve EE, throughput, and latency.	Grid area size, number of LoRa nodes, gateway placement, spreading factor, transmission power, channel allocation, number of iterations, receiver sensitivity, wireless channel conditions (fading, co-spreading factor interference, and temporal collisions), and data transmission rate.	Integrate environmental factors like urban shadowing and fading, dynamic interference models, realistic traffic loads including bursty and periodic patterns, user mobility, variations in gateway placement, energy constraints of LoRa devices, and stochastic channel conditions for both uplink and downlink transmissions.

the development of robust and efficient communication systems.

C. Reproducibility Challenges in ML: A Review of Parameters Documentation Across Algorithms

The reporting of algorithmic parameters in ML papers is critical to ensure transparency, enable reproducibility of results, and facilitate fair comparisons by allowing researchers to accurately evaluate the influence of specific configurations on performance and adapt them to different scenarios or datasets. Consequently, such reporting practices can provide a solid foundation for advancing research and validating innovative solutions.

To evaluate the transparency and reproducibility of the leading ML algorithms proposed in the 27 main works selected in our survey, specific key aspects were identified and tailored to the characteristics of each ML technique analyzed. Figure 8 provides an overview of these 27 studies, highlighting the level of detail reported for the key parameters of the proposed ML algorithms. This classification offers valuable insights into how well these works support reproducibility.

Figure 8 presents a mapping of the papers selected in this section, analyzing the documentation of algorithm parameterization in the proposed ML solutions. It depicts a matrix where the rows represent the key parameters that are essential to document the various algorithms surveyed, positioned on the right side. The columns correspond to the selected papers and the algorithms they employ, as shown at the top. The intersections of rows and columns, represented by small colored circles, indicate the relationship between each paper and a specific algorithm parameter under evaluation. The level of documentation for these parameters is conveyed through the colors of the circles: (i) red if the parameter was not reported; (ii) orange if the parameter was partially or incompletely addressed; (iii) green if the parameter was well-documented; and (iv) white, if the parameter was not applicable to the analysis. This visualization provides a clear overview of how thoroughly each paper addresses essential algorithmic parameters.

According to the taxonomy presented in our survey in Figure 6, the leading algorithm within the UL paradigm is K-Means. For this algorithm, two articles were selected for detailed analysis based on the exclusion criteria outlined in Section VI [105], [114]. The evaluation focused on three essential parameters, as presented in Figure 8: the initialization method for cluster centroids, the distance metric used to assign points to clusters, and the criteria for determining convergence. These parameters are critical for reproducing the algorithm because initialization affects clustering stability and outcomes, the distance metric defines the similarity measure central to clustering performance, and the convergence criteria determine the stopping condition, ensuring CE and accuracy.

Figure 8 also presents an analysis of these essential parameters in the selected articles, [105] and [114], revealing both strengths and limitations. In [105], the authors provide detailed documentation of the distance metric (Euclidean) and introduce a novel dynamic centroid adjustment method tailored to UDN scenarios, but they lack specificity regarding the initialization process and convergence criteria. Similarly, [114] reports the use of Euclidean distance clearly but provides incomplete information on both the initialization strategy and the stopping conditions for convergence, omitting key details such as a movement threshold or a maximum number of iterations for stopping. The lack of detailed reporting on these critical parameters significantly impacts the reproducibility and reliability of K-Means implementations, leading to inconsistent results and limiting its applicability in diverse scenarios.

As indicated by the taxonomy of ML techniques summarized in Figure 6, SVM is identified as one of the leading algorithms within the SL paradigm. For this algorithm, according to Figure 8, three key parameters were evaluated: kernel configuration, regularization parameter (C), and feature scaling. The kernel configuration determines how the algorithm maps input data into higher-dimensional spaces to improve separability, making it a fundamental aspect of the SVM model. The regularization parameter controls the trade-off between maximizing the decision margin and minimizing classification errors, which directly influences the model's performance. Feature scaling is essential to ensure that the SVM is not disproportionately influenced by features with larger numerical ranges, maintaining consistency in classification outcomes. Based on the paper selection criteria for assessing reproducibility of ML solutions, three studies were analyzed [120], [145], [152].

In [120], the kernel configuration is documented, specifying linear and second-order polynomial kernels, but fails to include details on the regularization parameter (C), which is crucial for balancing margin maximization and error minimization. Additionally, the article does not address feature scaling, leading to potential inconsistencies in model performance. Similarly, [145] specifies a Gaussian kernel function but omits details on the regularization parameter and feature scaling, further limiting reproducibility. In [152], no information is provided regarding kernel configuration, regularization parameters, or feature scaling, making replication particularly challenging. While some of these studies provide partial details, the lack of comprehensive reporting on these essential parameters can hinder the reproducibility of SVM implementations and impede fair comparisons between studies. Future works should prioritize detailed reporting of all critical SVM parameters to support the scientific community in replicating, validating, and building upon existing research.

In addition to SVM, three neural network variants—ANN, DNN, and LSTM—were also categorized as leading SL algorithms, as determined by the taxonomy in Section IV and summarized in Figure 6. In general, DNNs typically features a greater number of hidden layers compared to ANNs, while LSTMs incorporates memory for sequential data processing, along with a few specific variations. For these NN algorithms, five key parameters were identified to assess their reporting in the reviewed studies, aiming to facilitate the reproduction of the proposed methods: network architecture, main functions, optimization parameters, training parameters, and LSTM parameters. Network architecture pertains to the description of neural network structures, such as the number of layers and neurons per layer. The main functions include activation and loss functions specified in the studies. Optimization parameters focus on details about the optimization algorithm and learning rate. Training parameters cover aspects such as the number of epochs and batch size used. Finally, LSTM parameters are specific to LSTM architectures, detailing the number of LSTM cells per layer and whether they operate uni-directionally or bi-directionally. For our in-depth analysis, 5 ANN articles, 1 DNN article, and 3 LSTM articles were selected, as illustrated in Figure 8.

The analysis of the nine articles reveals significant variability in reporting the key parameters necessary for reproducing neural network-based SL algorithms. For network architecture,

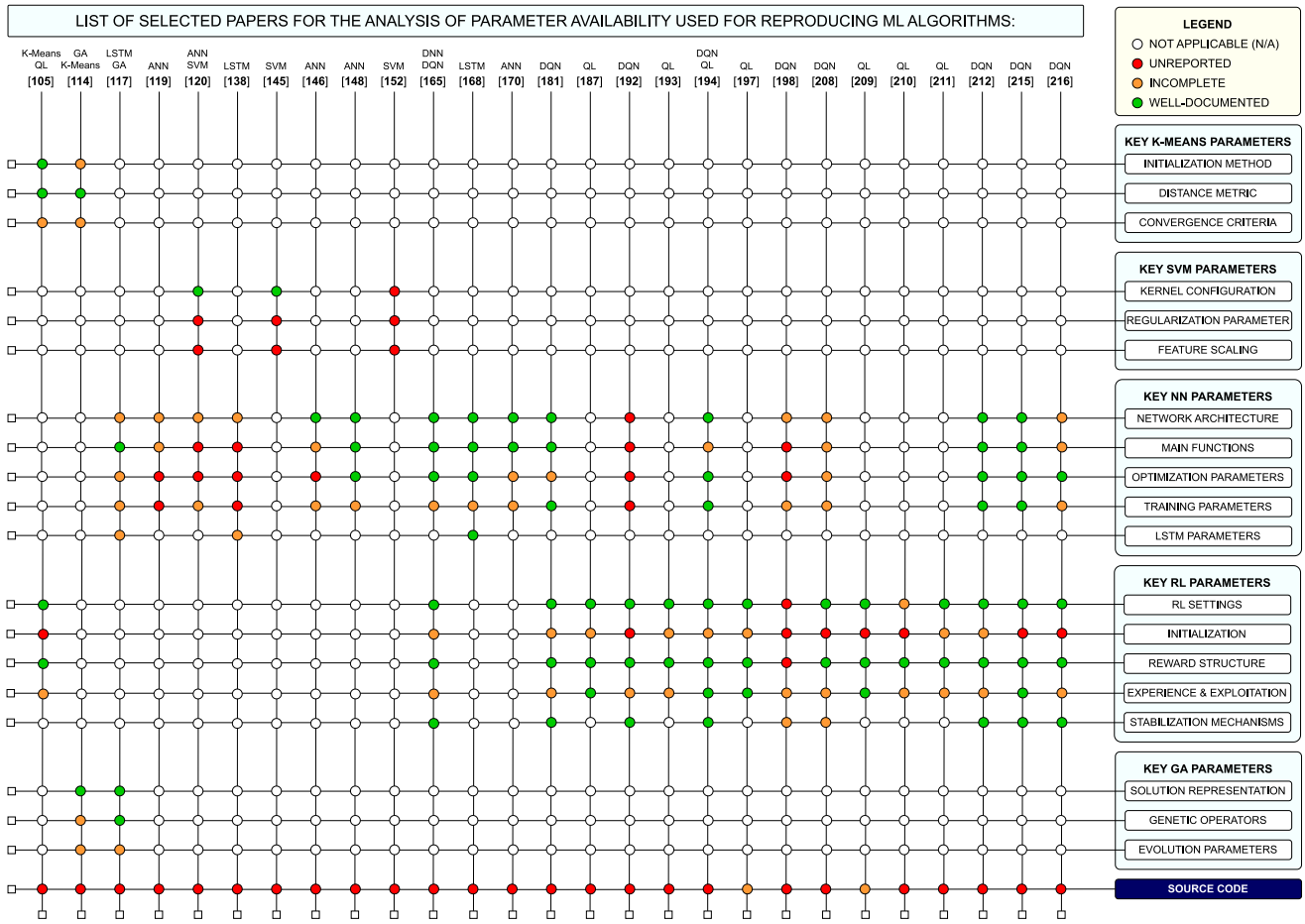


Fig. 8. Mapping of selected papers and classification of information level provided regarding key parameters used in proposed ML algorithms.

five articles—[146], [148], [165], [170], and [168]—are well-documented, providing explicit details about the number of layers and neurons, essential for replicating the model structure. The remaining articles, including [117], [119], [120], and [138], are classified as incomplete, as they mention the architecture but lack critical specifics, such as the exact number of layers or neurons.

For main functions, five articles—[117], [148], [165], [170], and [168]—are well-documented, specifying commonly used activation functions like ReLU, Sigmoid, and Tanh, and loss functions such as mean squared error and cross-entropy. These were the most frequently cited functions among the reviewed papers. Two articles, [146] and [119], are classified as incomplete, providing some information but omitting certain details, potentially regarding the activation or loss functions. The remaining two, [120] and [138], do not report these parameters at all.

Optimization parameters, including the optimizer and learning rate, show significant variability. Three articles—[148], [165], and [168]—are well-documented, specifying widely used optimizers such as Adam, SGD, and AdaGrad. Nesterov and the Adaptive Gradient Algorithm (AdaGrad) were also mentioned, though less frequently. Two articles, [117] and [170], are incomplete, providing details about the optimization parameters but omitting important specifics regarding the optimizer type, learning rate, or related configurations. The remaining four articles—[119], [120], [138], and [146]—are unreported, providing no information about optimization parameters.

For training parameters, which include batch size and epochs, seven articles—[117], [120], [146], [148], [165], [170], and [168]—are incomplete, generally mentioning one aspect, such as the number of epochs, but omitting others like batch size or other relevant training details. Two articles, [119] and [138], are unreported, offering no information on these parameters. Training parameters were the least reported among all categories, which significantly hampers reproducibility. A lack of details on batch size and epochs affects the model's convergence process and computational stability, as these parameters directly influence how the network updates weights during training and how efficiently it utilizes available computational resources. Without this information, it becomes challenging to replicate the training conditions and assess the scalability and performance of the proposed models.

LSTM parameters, which apply only to LSTM-based models, are well-documented in [168], providing comprehensive details on the number of LSTM cells, bi-directionality, and residual connections. Articles [117] and [138] are classified as incomplete, mentioning LSTM usage but lacking specifics about the architecture or configuration. These reporting gaps,

particularly in optimization and training parameters, highlight the importance of consistent documentation to enable reproducibility and foster fair comparisons across studies.

In addition to the previously discussed key NN parameters, most of the selected and analyzed SL-based articles fail to provide details on weight initialization, stopping criteria, or regularization techniques, which are crucial for model performance and reproducibility. For example, [146] mentions early stopping with cross-validation to prevent overfitting, while [148] discusses weight initialization and dropout but omits details on stopping criteria. The absence of comprehensive reporting on these aspects hinders the ability to replicate models effectively and assess their robustness under different conditions since they are crucial for ensuring that the model is well-regularized and that training stops appropriately when performance stabilizes.

When analyzing state-of-the-art studies on RL, the taxonomy in our survey identified Q-Learning and DQN as the leading algorithms, as shown in Figure 6. To ensure transparency, reproducibility, and comparability, the following key RL parameters were deemed essential: RL settings, initialization, reward structure, experience and exploration, and stabilization mechanisms. RL settings assess whether elements like state and action spaces, learning rate, and discount factor are thoroughly detailed. Initialization for Q-Learning focuses on Q-value setup, ensuring convergence stability, while for DQN, it refers to network weight initialization, crucial for effective learning. Reward structure evaluates how well-defined systems incentivize optimal agent behavior. Experience and exploration parameters include strategies like ϵ-greedy, initial and final epsilon values, and decay rate, which help balance the trade-off between exploration and exploitation during learning. For DQN specifically, the key NN parameters previously defined and stabilization mechanisms were also considered. In this context, NN parameters are critical to estimate the Q-value function effectively. At the same time, stabilization mechanisms refer to techniques designed to enhance the stability and convergence in DQN using NNs to approximate the Q function and to mitigate instabilities that arise in this context. Examples include replay buffers, target network updates, soft or hard updates, and gradient clipping. Lastly, 8 studies on Q-Learning and 9 on DQN were selected for evaluating the documentation of these parameters, totaling 16 papers, as both RL algorithms are addressed in [194]. The analysis of parameter documentation in these papers is summarized in Figure 8.

Based on Figure 8, the analysis of Q-Learning studies reveals that RL settings are well-documented in 14 of the 16 articles, with most providing comprehensive details on states and action spaces, learning rate, and discount factor, explaining their selection and impact on learning dynamics. The remaining papers, [198] and [210], are unreported and incomplete, respectively, lacking detailed explanations on how the learning rate and discount factor were selected or adjusted.

Initialization parameters were not fully detailed across all studies, with articles either not reporting them or addressing them incompletely. For Q-Learning, 5 articles—[187], [193], [194], [197], and [211]—provided partial documentation, while the 3 remaining ones omitted these parameters entirely, often neglecting details about Q-value initialization methods, such as randomization or strategies to avoid local optima, which are critical for stability and reproducibility. For DQN, 4 papers failed to document initialization parameters, and the remaining 5—[165], [181], [194], [212], and [216]—offered limited descriptions, such as stating that weights were randomly initialized but without specifying the random seed. The random seed ensures reproducibility by determining the sequence of random numbers used in processes like weight initialization, directly impacting convergence time and final performance. Its omission hinders faithful reproduction and consistent comparison of results.

Reward structures are well-documented across the vast majority of studies, clearly outlining how rewards guide agents toward objectives like EE, throughput, or fairness. For experience and exploration parameters, 5 papers—[187], [194], [197], [209], and [215]—provided detailed documentation of strategies such as ϵ-greedy and epsilon decay. The remaining papers offered partial documentation, often omitting key details, such as decay rates or tuning methods, which impacts reproducibility and consistent learning outcomes.

Regarding the parameters specific to DQN, stabilization mechanisms were well-documented in 7 of the 9 papers for this algorithm. The remaining 2 papers—[198] and [208]—although providing information on stabilization mechanisms, did not fully clarify certain additional details, such as replay buffer size, sampling method, or target network update frequency. Additionally, regarding the documentation of key NN parameters for DQN algorithms, [212] and [215] provided comprehensive documentation overall, while [165], [181], and [194] addressed most parameters but lacked details on training parameters, optimization parameters, and loss and activation functions, respectively. In [208] and [216], most parameters were partially documented, whereas [192] and [198] largely left them unreported. Among the commonly used models, notable examples include optimizers such as Gradient Descent (GD) [194] and Adam [212], along with activation functions like Tanh, linear, and ReLU [165], [181], [212]. Furthermore, MSE emerged as the most frequently used loss function across the surveyed DQN-based papers [194], [212].

Finally, as highlighted in our survey taxonomy presented in Figure 6, GA was identified as a leading evolutionary algorithm within the "Others" category. For GA, three essential types of parameters were analyzed: solution representation, genetic operators, and evolution parameters. Solution representation defines how potential solutions, or chromosomes, are encoded, which is critical for efficiently navigating the search space. Genetic operators, including selection, crossover, and mutation, govern how new solutions are generated and evaluated to improve the quality of the population iteratively. Finally, evolution parameters, such as population size, termination criteria, and additional strategies like elitism, guide the algorithm's overall convergence and performance [256]. Based on the criteria established for analyzing reproducibility in ML solutions, two studies employing GA were selected for review [114], [117], as summarized in Figure 8.

The first article, [114], applies GA for traffic offloading in FiWi networks. The solution representation is well-documented, with chromosomes clearly encoding traffic offloading decisions and supported by a diagram illustrating the methodology. Genetic operators are classified as incomplete due to insufficient details on mutation implementation despite mentions of selection and crossover. Evolution parameters are also incomplete, with limited information on population size and iterations and no mention of termination criteria or elitism. The second article, [117], employs GA for dynamic resource allocation in C-RAN. The solution representation is well-documented, with chromosomes encoding resource allocation decisions and supported by a flowchart for clarity. Genetic operators are also well-documented, offering detailed explanations of selection, crossover, and mutation. However, evolution parameters remain incomplete, lacking details on population size, termination criteria, and advanced strategies. These gaps hinder reproducibility and limit reliable comparisons across applications.

The vast majority of the 27 surveyed papers do not provide or share the source code of the proposed ML algorithms. This lack of accessibility represents a significant challenge for the scientific community, as it hampers the verification of implementation details and the reproduction of results, both of which are fundamental for scientific validation and advancing the state-of-the-art. Figure 8 also highlights this limitation, showing that only a small fraction of the studies address this aspect. In [209], the data used is available upon request from the corresponding authors, while [197] mentions the possibility of making the code accessible. However, the widespread absence of source code severely limits other researchers' ability to replicate and improve upon the studied works, ultimately compromising scientific transparency and collaboration.

Based on a comprehensive analysis of the reproducibility potential of the 27 key papers selected in our survey, several recurring gaps, critical areas for improvement, and future directions to address this prevalent limitation within the scientific research community have been identified. These avenues for enhancement, along with insightful recommendations, will be explored in greater depth in Section VII-A.

D. Summary and Lessons Learned

Section VI of this survey provides a qualitative analysis of key papers on ML-driven resource allocation for wireless communications, addressing three critical areas. First, it explores leading ML algorithms, their adaptability, scalability, and efficacy in optimizing aspects like EE, latency, and throughput, demonstrating their versatility and suitability for specific challenges within wireless systems.

The second focus is on simulation parameters, emphasizing the need to align setups with real-world conditions for meaningful evaluations. It underscores the critical role of realistic datasets, robust evaluation metrics, and well-defined training configurations, highlighting the persistent challenge of bridging the gap between simulation and practice due to the variability of channel conditions and the dynamic nature of wireless environments.

The third aspect examines reproducibility challenges, identifying gaps in transparency and standardization in documenting algorithm parameters. Addressing these issues through detailed reporting and open-access repositories is essential to foster comparability and reliability in research outcomes.

Several lessons emerge from this analysis. The selection of ML algorithms must be context-aware, considering network conditions and optimization goals. Hybrid and adaptive models show promise for evolving challenges, as optimizing multiple KPIs often involves trade-offs. For instance, RL excels in dynamic scenarios like power control, spectrum sharing, and load balancing due to its adaptability to varying channel conditions. Meanwhile, SL is ideal for tasks with abundant labeled data, such as traffic prediction, and UL is effective for clustering problems like user grouping or anomaly detection. Evolutionary algorithms, with their strength in exploring complex search spaces, are valuable for global optimization tasks, including resource scheduling and network topology design. Collaborative approaches between different ML paradigms can harness complementary strengths, enabling robust solutions for the dynamic demands of wireless systems.

Furthermore, ensuring simulation fidelity through diverse datasets, stochastic modeling, and standardized benchmarks is critical for robust evaluations. Reproducibility demands detailed guidelines for parameter documentation and sharing, which are crucial for credibility in ML research. By synthesizing these lessons, this survey highlights the transformative potential of ML in wireless networks and the critical pathways for future research. A unified effort toward standardizing evaluation practices, enhancing simulation fidelity, and data sharing will be pivotal for advancing ML applications in wireless communications.

VII. Limitations, Challenges, and Future Directions

Based on the results obtained through the extraction and synthesis of data collected from the selected works for our survey, various limitations, challenges, and future research directions can be delineated. Accordingly, the following subsections establish connections between these observations, considering four distinct aspects: transparency in ML algorithms, available technological resources, interdependence of wireless communications and ML employment, and future trends. Lastly, a final subsection provides the concluding remarks based on the discussions related to these different aspects.

The first aspect pertains to the lack of pertinent information in works involving the use of ML algorithms, with a focus on potential enhancements for the provision of such content in a more detailed manner. The objective of this approach is to enrich the scientific community and, consequently, facilitate reproducibility, comparisons, updates, and enhancements related to these works, contributing to the advancement of research in critical resource allocation optimization for wireless communications with the assistance of AI.

The second aspect, on the other hand, relates to current technologies and available devices to design efficient ML algorithms for optimizing resources in wireless communications. This approach may bring forth pertinent discussions concerning the adoption of new emerging technologies, the feasibility of implementing proposals in a real-world environment, and the coexistence of multiple scenarios with distinct requirements being served by a unique telecommunications system with dynamic resource allocation optimization.

The third aspect, related to future trends, is inclined to delve more profoundly into the impact of future mobile networks in the context of employing ML algorithms for the critical resource allocation optimization. Some of the key topics are associated with the B5G scenarios and upcoming 6G applications and how the future of these telecommunications systems can wield influence on the utilization of AI and ML, such as in the selection of more efficient learning algorithms aligned with new enabling technologies and networks.

Finally, the fourth aspect, related to the interdependence of wireless communications and ML employment, emphasizes that while resource allocation optimization and advancements in communication technologies benefit from ML-driven improvements, they also play a crucial role in enabling the efficient training, deployment, and real-time operation of learning models.

A. *Transparency in ML Algorithms*

Given the exponential growth of works and articles related to ML across different fields, there is a growing concern within the scientific community regarding the reproducibility and transparency of research papers proposing ML solutions. To address this challenge, some initiatives are emerging and being developed to improve these aspects.

However, there is still no universal or officially mandated template currently implemented for reporting ML algorithms in published scientific articles. Many researchers adopt guidelines as a matter of good practice or to comply with specific requirements set by journals or conferences, but these remain non-regulatory.

The issues highlighted in the survey pertaining to the lack of comprehensive details on the ML algorithms used, including hyper-parameters and the absence of comprehensive performance evaluation metrics in many of the surveyed papers, are of paramount importance and deserve attention. These limitations underscore critical areas for improvement in the research on resource allocation optimization in wireless communications using ML.

Firstly, the inadequacy in reporting algorithmic specifics, such as hyper-parameters (e.g., number, type, and feature of layers, number of neurons, optimization algorithms, and learning rates), impedes the reproducibility of the research. Reproducibility is a cornerstone of scientific inquiry, and without clear information on how the ML models were designed and trained, it becomes challenging for other researchers to build upon and validate the findings. This lack of transparency hinders the progression of knowledge in the field and raises concerns about the rigor and credibility of the research.

Secondly, the deficiency in comprehensive performance evaluation metrics is equally significant. Inadequate reporting of loss functions used during training, metrics employed for predictive performance assessment, and the absence of information on how these metrics were calculated and interpreted can entail a greater challenge in validating and generalizing the proposed solutions. Sound evaluation metrics are essential for objectively gauging the effectiveness and efficiency of the ML algorithms in optimizing resources for wireless communications. Without these metrics, it is challenging to draw meaningful comparisons between different studies or to assess the practical applicability of the proposed methodologies.

In this context, the absence of benchmarks represents a significant challenge to be overcome in the current research landscape, underscoring the need for a collaborative effort within the scientific community to establish standardized metrics and datasets for ML research. The establishment and use of standardized benchmarks, particularly metrics for datasets, are crucial for evaluating and comparing the performance of various ML algorithms. Currently, suppose a researcher develops a new algorithm. In that case, it can be exceedingly difficult to compare its effectiveness with others without access to the same datasets that were used in previous studies. This lack of standardized benchmarks hinders the ability to conduct fair and meaningful comparisons, thereby impeding the advancement of the field. The development of universally accessible and widely accepted benchmark datasets is essential to foster transparent, reproducible research. Such benchmarks would not only facilitate the comparison of different ML algorithms but also encourage the development of more robust and efficient models.

According to Section V-B, common data types are provided in the datasets available by the authors of the surveyed papers, such as (i) network traffic data, which captures various metrics such as throughput, latency, jitter, and packet loss, essential for QoS analysis and traffic prediction; (ii) user and device preferences, that includes user ratings, device classification metrics, and usage patterns, which are used for content caching, user profiling, and resource management; (iii) energy and computational resource utilization metrics, focusing on CPU, Random Access Memory (RAM), and power usage metrics, crucial for optimizing energy efficiency and resource allocation; and (iv) mobility patterns and positioning data, which provides information on device movement and location, vital for optimizing network planning and resource distribution in mobile networks.

Based on this information, several benchmarks could be developed to standardize and enhance comparability across different studies in ML-driven wireless communication research. In this context, we suggest (i) a unified traffic prediction benchmark, (ii) QoS metrics repository, (iii) EE benchmark suite, (iv) content caching and distribution benchmark, and (v) mobility and positioning dataset.

A unified traffic prediction benchmark could be established by combining datasets such as the Milan Cellular Traffic [126], Deutsche Telekom Network Traffic Datasets [139], and

Abilene Network [185]. This benchmark would provide a comprehensive dataset encompassing diverse traffic patterns and conditions from various networks, offering a common ground for evaluating traffic prediction models. This benchmark would enable consistent comparisons across studies and provide a baseline for algorithm performance.

Similarly, a QoS metrics repository could be developed using datasets like the Deutsche Telekom Network Traffic Datasets [139] and WLAN Throughput [214], providing standardized network performance metrics for evaluating QoS optimization models, including latency, throughput, and jitter, collected from different network environments. This repository would facilitate the evaluation of QoS optimization models and allow researchers to test their algorithms under standardized conditions.

An EE benchmark suite could combine a collection of datasets focused on energy consumption and power management metrics for various network devices and configurations, such as the Squid Cache and Nginx Proxy Datasets [125], and GWAT-13 Materna [128]. This suite would enable researchers to benchmark their energy optimization strategies against standardized data, fostering innovation in energy-efficient communications.

To address user preferences and content delivery optimization, a content caching and distribution benchmark could be created, resulting in a dataset (e.g., MovieLens Datasets [102], [203]) that captures user preferences and content demand across different contexts and network conditions. This benchmark would provide a standardized framework for evaluating caching strategies and content delivery optimizations.

Mobility and positioning dataset benchmarks could be developed using, for example, the HSM [131] and SUMO-based [143] datasets, including diverse mobility patterns and positioning data from various environments. This dataset would allow researchers to test their models' adaptability and performance in dynamic mobile environments.

In light of these observations, we can note a pressing need for developing standardized benchmarks and datasets that cover a wide range of scenarios and conditions in wireless communications. Such benchmarks would enable more consistent comparisons and evaluations of ML models, fostering greater transparency and reproducibility. Furthermore, encouraging the creation and sharing of open-access databases would significantly enhance transparency and reproducibility in the field. Researchers should be incentivized to share detailed information about their datasets and methodologies, allowing others to replicate and validate their findings.

Another promising trend is the integration of emerging technologies, such as FL, which allows for decentralized data processing and model training across multiple devices and can facilitate the collection and analysis of diverse datasets while preserving user data privacy. This approach can create more diverse and representative datasets, capturing the heterogeneity of real-world environments without compromising data security.

Establishing standardized protocols for reporting dataset usage, sources, and pre-processing steps would improve clarity and transparency in research publications. These protocols should include guidelines for documenting dataset characteristics, ensuring that researchers provide comprehensive information about their data sources.

Finally, the cross-domain collaboration between academia and industry can lead to the development of more representative datasets that reflect current trends and challenges in wireless communications. Industry partners can provide real-world data and insights, enriching the research community's understanding of practical applications.

As analyzed in Section VI-C, beyond the lack of details concerning the datasets used in the proposed studies, there is also a significant gap regarding essential information about the parameters of the ML algorithms necessary for faithfully reproducing the works found in the literature. To address this challenge, as done in this study, it is advisable to propose the definition and standardization of the key parameters for each ML algorithm that should be reported to enable easy reproduction. In this regard, Section VI-C also offers suggestions for some parameters considered critical for the leading ML algorithms highlighted in our survey. Furthermore, in cases where processes involve randomness, such as the initialization of weights in ML algorithms (e.g., DQN) in some cases, it is crucial to provide the random seed or relevant information regarding the values used to achieve results similar to those presented in the proposed works.

Additionally, it is worth noting that the studies generally do not provide the source code for the algorithms. Although there may be concerns about proprietary information, publishing these codes—at least in terms of parameter values, methodology, and datasets, as previously discussed—would not only enhance transparency in the use of AI, particularly in ML, within the scientific community but also promote collaborative information and knowledge sharing. This collaboration is vital, as the advancement of these algorithms largely depends on the cumulative experience and data shared within the community. Moreover, establishing a culture of transparency and reproducibility can accelerate innovation and improve the reliability of research findings in the field of ML, ultimately benefiting both the academic community and industry practitioners.

By addressing the limitations previously identified and embracing new methodologies and standards, the research community can advance toward more robust, transparent, and impactful solutions in the domain of wireless communications. Enhanced transparency in reporting algorithmic details and the utilization of well-defined, widely accepted performance evaluation metrics are crucial steps in fostering a more robust and credible body of research. By resolving these issues, future work in this domain can build upon a stronger foundation, facilitating the development of more effective and efficient solutions with clear, practical implications for the telecommunications industry.

B. Available Technological Resources

The issues identified here are of considerable significance and warrant addressing, as they underscore critical challenges in the field of resource allocation optimization for wireless

communications employing ML algorithms. Additionally, they delineate potential pathways for future research endeavors, particularly in the context of the forthcoming B5G and 6G technologies.

The first issue pertains to the limitation of presently available hardware resources and the quest for more efficient technologies. This limitation bears immediate importance, as the efficacy of ML algorithms for resource allocation optimization in wireless communications is inherently dependent on hardware capabilities. The expectation that 6G technology will usher in new advancements and efficiencies is an imperative consideration. As such, addressing this limitation is pivotal for realizing the full potential of ML in optimizing critical resources. The shift toward 6G holds the promise of substantially improving the performance of these algorithms by enabling new technologies (e.g., Terahertz (THz), molecular, and quantum communications), thus mitigating the hardware-related constraints faced in the current scenario [257].

In the next section, the second issue relates to the prevalent reliance on simulation-based validation due to the scarcity of state-of-the-art equipment for real-world implementation. The availability of cutting-edge hardware for practical deployment is an essential step in ensuring the viability of ML-based resource allocation optimization solutions. The impending developments in B5G and 6G technologies are expected to offer improved infrastructure [258], reducing the dependency on simulations and enabling direct experimentation in real-world settings. This transition is pivotal for validating the efficacy and reliability of ML algorithms under authentic conditions.

The third issue highlights the fragmentation in resource allocation optimization solutions, as discussed in Section III-G, wherein existing works tend to specialize in one or two of the five pre-defined resources without adaptability for diverse applications. Achieving adaptability and dynamic resource allocation optimization, tailored to the specific requirements of various applications, is a critical future objective. The emergence of B5G and 6G networks can be instrumental in fostering such adaptability, as they are envisioned to provide flexible, application-aware network architectures [259]. These advancements can empower ML algorithms to dynamically adjust their resource allocation optimization focus in response to diverse application demands, thus enhancing the agility and versatility of telecommunications systems.

In conclusion, the significance of addressing these challenges lies in the potential for advancing the capabilities of ML algorithms in optimizing critical resources for wireless communications. The transition to B5G and 6G technologies holds promise in mitigating current limitations and ushering in an era of enhanced hardware, real-world validation, and adaptive resource allocation optimization, ultimately contributing to the evolution of future telecommunications networks. Therefore, in the following section, some of the main technological trends of 6G will be detailed, and consequently, how they may impact overcoming the previously discussed challenges related to the limitations of available technological resources.

C. Interdependence of Wireless Communications and ML Employment

The interdependence between wireless communications and ML employment is evident, as the evolution of communication technologies not only benefits from ML-driven optimizations but also plays a crucial role in enabling the efficient training, deployment, and operation of advanced learning models. Therefore, in addition to focusing on ML applications for resource allocation optimization in wireless communications, thereby enabling the optimal performance of future-generation network services, it is equally important to consider how advancements in wireless communications can support and enhance the development and deployment of ML techniques, especially in the era of distributed data and computation. This interdependence becomes especially critical with advanced ML models, such as Large Language Models (LLMs) and Generative Artificial Intelligence (GenAI), which demand substantial computational power for processing large data volumes, efficient data transfer, and real-time responsiveness [260]. Meeting these requirements enables these models to provide instant and interactive responses in applications like virtual assistants, natural language translation, and content personalization, allowing them to address user needs and create seamless, efficient experiences immediately. Therefore, resource allocation optimization in wireless networks is foundational in enabling these models to function effectively in decentralized settings, such as edge and fog computing environments.

For instance, in distributed GenAI applications, where model computations are divided across multiple edge nodes to reduce latency and improve responsiveness, effective allocation of network resources, like bandwidth and power, is essential for seamless model execution [261]. By optimizing bandwidth allocation, these systems can handle the significant data exchanges between nodes, ensuring minimal delays and maintaining the QoS necessary for real-time interactions with GenAI applications. In the case of LLMs, which may rely on high-speed data access across distributed sites, the availability of low-latency paths and computational resources close to end-users enables quicker response times, making interactions more seamless and reducing the dependency on centralized cloud infrastructure [262].

Additionally, efficient power management within the network infrastructure allows these computationally intensive applications to function sustainably on mobile devices or remote edge nodes, which often have limited energy resources. This is especially relevant for AI-driven real-time language translation application or personalized GenAI assistants, which require frequent updates and consistent computational support. NTM further ensures that critical data flows, such as model parameters and inference results, remain stable across the communication channels, reducing the risk of data bottlenecks that could disrupt the performance of distributed ML models.

By leveraging resource allocation optimization strategies, wireless networks can effectively support the growing demands of decentralized ML applications, facilitating the deployment of more advanced, reliable, and responsive AI services directly to end-users. This interplay between wireless

communications and ML fosters a more efficient ecosystem for the continuous evolution of AI applications, ensuring that resource constraints are minimized and enabling a broader integration of AI functionalities across diverse devices and use cases. This advancement will help achieve one of the technological trends of future mobile networks: creating intelligent networks with AI and ML integrated not only across all network layers but also within user-edge devices.

D. Future Trends

The limitations and challenges raised in this survey hold particular relevance within the context of diverse usage scenarios of 5G, such as eMBB, uRLLC, and massive Machine Type Communications (mMTC). Furthermore, the emergence of 6G technology introduces exciting possibilities for reshaping the landscape of telecommunications. Although 6G services have yet to be standardized, studies indicate that the trend in future mobile networks promises an extension of existing 5G services, albeit with even more challenging requirements, which will dramatically increase the complexity and demands on telecommunications networks. Consequently, these services are being referred to as ultra-Reliable Low Latency Communications+ (uRLLC+), enhanced Mobile Broadband+ (eMBB+), and massive Machine Type Communications+ (mMTC+) [263]. In addition to these, other perspectives are expected to be incorporated, such as AI and sensing [264].

Addressing these challenges within these specific contexts is vital for the evolution of telecommunications networks compared to the current state of affairs [265]. In this context, dynamic resource allocation optimization related to transmission power, time, frequency, computational capacity, and space is crucial to meet future applications' requirements with distinct goals in a unique wireless communications system. Although all of these resources influence each of the previously mentioned scenarios, according to each application or demand, there are resources capable of more strongly impacting the expected outcomes.

In the 5G landscape, eMBB aims to deliver high-speed connectivity for applications such as high-definition video streaming and augmented reality [266]. In order to meet this demand, optimization of frequency and space resources are consequently mandatory to increase throughput, and an efficient CRM is relevant to ensure the performance of data-intensive applications.

uRLLC, on the other hand, focuses on ultra-reliable, low-latency connections essential for critical services like autonomous vehicles and remote surgeries [267]. Latency and CRM are indispensable to be optimized in uRLLC, given the primary focus of this scenario on ensuring communications with extremely low latency through optimization of time resources. For this, efficient allocation of computational capacity resources is also vital to maintain low latency in real-time critical communications.

mMTC targets the connectivity of a massive number of low-power IoT devices [268]. This scenario can be obtained by efficiently allocating resources such as transmission power, frequency, computational capacity, and space, consequently optimizing PowCon, NTM, and CRM network KPIs. PowCon is imperative for low-power IoT devices, ensuring communication effectiveness and battery life. Efficient CRM is vital for processing data from a large number of IoT devices. Meanwhile, optimized NTM ensures effective user and network traffic management, considering the high density of connected devices.

The challenge lies in creating a system that can dynamically adapt its resource allocation to suit the unique requirements of each scenario. This adaptability is vital because, in today's networks, solutions of resource allocation optimization tend to specialize in one of these areas and may not perform optimally when confronted with different usage scenarios.

The transition to 6G introduces a novel dimension to this adaptability. While 5G was revolutionary, 6G is envisioned to be transformative. It promises to offer intelligent, application-aware network architectures that can dynamically reconfigure themselves based on real-time requirements [269]. For example, in scenarios where ultra-low latency is critical, such as uRLLC for remote surgeries, the network can allocate resources and prioritize data accordingly. Conversely, when dealing with a scenario that requires massive connectivity for an IoT infrastructure, like mMTC, the system can efficiently manage the influx of connections without compromising performance [270].

The dynamism and adaptability of 6G networks can truly revolutionize telecommunications. It allows for the optimal utilization of resources, mitigating the limitations of existing hardware and reducing reliance on simulations. This adaptability ensures that the network is always fine-tuned for the specific demands of each application, optimizing performance and efficiency. It also opens the door to new, unimagined applications and services that can thrive in a flexible, agile, and responsive telecommunications ecosystem [265].

In emerging 6G networks, several key services are anticipated to define the next generation of mobile communications, extending and enhancing the foundational capabilities of 5G. As previously mentioned, although not yet standardized or defined, several essential types of services for 6G are already being discussed within the scientific community. These include uRLLC+, which builds upon ultra-reliable, low-latency communication to deliver even faster response times and higher reliability for critical applications; eMBB+, which advances enhanced mobile broadband to support massive data throughput and ultra-high-definition experiences across more devices; and mMTC+, which scales massive machine-type communication, enabling connectivity across an even greater number of IoT devices with improved efficiency [263], [264].

Additionally, AI is expected to be deeply integrated into network management and service optimization, leveraging ML to predict and adapt to user and network demands in real-time. Sensing, another prospective service, will add environmental awareness to the network, allowing devices to gather and respond to spatial, tactile, and contextual data [264]. Together, these services are set to vastly expand the potential of mobile networks, enabling diverse applications to coexist seamlessly and efficiently [271]. This transformative shift is poised to

redefine the telecommunications landscape, ushering in an era of innovation and connectivity that is both versatile and visionary.

The evolution of 5G services, such as eMBB+, uRLLC+, and mMTC+, into 6G networks is just the beginning of what the next-generation mobile technology will offer. Beyond simply enhancing and improving the existing services, 6G will introduce new services, key technologies, and use cases that promise to revolutionize telecommunications. Although 6G is still in the early stages of standardization, some emerging services and technologies are becoming clear, and they will bring with them new challenges and opportunities, particularly in the context of resource allocation optimization.

For instance, the evolution toward 6G networks introduces novel communication technologies that promise unprecedented capabilities and new avenues for resource allocation optimization. Molecular communication leverages biological molecules to transmit information, potentially enabling efficient, ultra-small-scale communication in biomedical and nanotechnology applications. THz communication, operating at extremely high frequencies, offers high data rates and low latency, ideal for data-intensive applications but requiring sophisticated resource allocation to overcome propagation challenges [272]. Visible Light Communication (VLC) uses light to transmit data, making it ideal for environments with heavy radio-frequency interference, although resource allocation optimization involves adapting to lighting conditions and line-of-sight limitations [273]. Quantum communication, which employs quantum states for secure information exchange, enhances the security and reliability of 6G networks but demands resource management that can account for complex quantum protocols [274]. Holographic communication enables immersive 3D image transmission, revolutionizing remote interactions but necessitating dynamic resource distribution to handle large bandwidth requirements [275]. Lastly, haptic communication facilitates real-time transmission of tactile feedback, critical for applications like remote surgeries, where precise, low-latency allocation of network resources is essential [276].

These advanced communication technologies are set to significantly enhance network performance in terms of latency, speed, bandwidth, and security. However, to fully leverage these improvements, a more robust and efficient solution for resource allocation optimization is essential. As each technology introduces unique demands on network infrastructure, from maintaining ultra-low latency for haptic communication to managing the high bandwidth requirements of holographic transmission, ML-based optimization solutions must evolve to dynamically adapt resource distribution in real-time. By developing more sophisticated algorithms, 6G networks can efficiently balance resources to meet the stringent requirements of these novel communication modalities, ensuring seamless, reliable connectivity across diverse applications.

In addition, as the vision for 6G networks materializes, several transformative technological trends are emerging, each accompanied by unique challenges for the application of ML in resource allocation optimization. Notable among these trends are RIS, Extended Reality (XR), Integrated Sensing And Communication (ISAC), large-scale automation, quantum computing, and 3D coverage networks [277], [278]. While these innovations promise substantial advances in network capabilities, they also present new complexities for ML algorithms tasked with dynamically optimizing resource distribution across increasingly intricate and high-stakes environments.

RIS involve engineered surfaces that can alter electromagnetic waves to enhance signal paths, thereby improving network coverage and signal strength [279]. However, RIS systems introduce significant challenges for ML in resource allocation, as the dynamic nature of reconfigurable surfaces requires algorithms capable of real-time adaptation to shifting signal environments. Developing models that can effectively and efficiently manage RIS behavior across diverse settings while minimizing resource waste represents a core challenge in this domain.

XR, which includes Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR), demands exceptionally high data rates and ultra-low latency to provide seamless user experiences [280]. ML-based resource allocation faces the challenge of accurately predicting user demand, which fluctuates with varying levels of XR immersion. This calls for ML models that can rapidly allocate resources based on complex and unpredictable usage patterns, as delays or bandwidth issues could disrupt the immersive quality, undermining the user experience.

ISAC seeks to unify communication and sensing functions within 6G, enabling applications such as autonomous vehicles and remote environmental monitoring [281]. However, the dual requirements of high precision in sensing and efficient data communication present challenges for ML models in balancing these competing priorities. Achieving an optimal trade-off between sensing accuracy and data transmission efficiency demands highly adaptive algorithms that can manage resources in real-time, as any imbalance could compromise system performance or even safety in critical applications.

Large-scale automation, vital for smart cities, smart factories, and autonomous systems, introduces new challenges for ML in managing network resources [282]. As automation increases, so does the scale and complexity of data traffic, requiring ML algorithms that can analyze vast data volumes and make rapid, context-sensitive decisions. This level of scalability remains a significant hurdle for ML models, as they must not only process data efficiently but also ensure optimal resource distribution across various automated systems, each with unique demands and operational constraints.

Quantum computing stands as a cornerstone of 6G, promising unprecedented computational power that can accelerate ML processing times and improve complex network simulations [283], [284]. However, the challenge lies in developing quantum-ready ML algorithms that can take full advantage of quantum hardware's potential for faster processing and complex problem-solving. Transitioning current ML models to be quantum-compatible and scalable across real-world applications remains a formidable obstacle, especially as 6G

networks demand more sophisticated optimization solutions in real-time.

Finally, 3D coverage networks aim to extend connectivity to three-dimensional spaces, including high-rise urban areas, aerial drones, and aerospace applications [285]. This increased dimensionality in coverage creates substantial complexity for ML-based resource allocation, as it requires algorithms that can adapt to both horizontal and vertical demands across a range of mobility scenarios. Efficiently managing resources for such dynamic, multi-dimensional networks is challenging, particularly in ensuring connectivity stability and optimal allocation across all spatial dimensions.

Together, these 6G technological trends highlight both the potential and the challenges of leveraging ML for resource allocation optimization. As these trends evolve, the ML models supporting them must become increasingly adaptable, efficient, and capable of balancing a diverse array of requirements to fully realize the promise of 6G technology.

With the advent of future mobile networks, numerous benefits are anticipated, such as increased ease and the ability to reduce the processing time of ML algorithms and enhance the computational capacity of wireless communications systems. These advancements are exceedingly important for executing tasks of resource allocation optimization with excellence. They may even alter the focus of selecting the most commonly used algorithms at present, given that many of these choices take into account their degree of complexity. A clear example, as seen in the results obtained in our survey and discussed in the previous section, is the current extensive exploration of RL algorithms capable of handling discrete action spaces more effectively, simplifying action selection when compared to environments with continuous action spaces [105], [165], [210].

Considering technological advancements, potential enhancement can facilitate continuous action space RL algorithms more readily, such as DDPG and Deep Reinforcement Learning (DRL) in general [181], [198], [212]. In this case, power management, for instance, can be computed not only in levels (e.g., low, medium, and high) but also through the selection of optimal and specific numerical values, exponentially improving the efficiency in resource allocation optimization and the results achieved.

To condense, incorporating considerations of resource allocation optimization into each of these scenarios and applications is pivotal in ensuring the performance, efficiency, and adaptability of both B5G and 6G networks. Through such optimization, the full potential of these technologies can be harnessed, ultimately shaping the future of telecommunications and ushering in an era of unprecedented connectivity and innovation.

E. Concluding Remarks

This section highlights key insights derived from the analysis of limitations, challenges, and future directions for ML-driven resource allocation optimization in wireless communications. The concluding remarks provide a comprehensive perspective on the discussed constraints, technological considerations, and emerging opportunities, offering valuable guidance for both researchers and practitioners in the field.

The lack of comprehensive reporting on algorithmic parameters, performance metrics, and benchmarks hinders reproducibility and comparability in research. Addressing these gaps through standardized benchmarks, datasets, metrics, and reporting protocols is essential to foster transparency and collaboration within the scientific community. Furthermore, publishing open-access code and datasets can significantly enhance knowledge sharing and innovation.

The reliance on simulation-based validation due to hardware limitations underscores the need for emerging technologies like 6G to enable real-world implementation. Advancements in hardware and enabling technologies, such as THz and quantum communications, promise to support more complex algorithms, like DDPG and other continuous-action space RL models, improving their practical applicability. Future trends, including RIS, XR, and ISAC, highlight the growing complexity of use cases in 6G, demanding adaptive ML models capable of dynamic resource allocation across evolving scenarios.

Ultimately, optimizing ML solutions for resource allocation in future networks requires addressing transparency, fostering collaboration, and leveraging advanced technologies to unlock the full potential of 6G. Based on the lessons learned throughout our survey, particularly regarding the limitations, challenges, and future trends highlighted in this section, we present Figure 9, which illustrates a roadmap of future directions for optimizing critical resources in wireless communications using ML algorithms. This roadmap provides a comprehensive guide for advancing scalable, efficient, and adaptable telecommunications systems.

As illustrated in the roadmap, enhanced ML solutions are critical for achieving dynamic and efficient resource management in evolving 6G networks. A key factor for advancing these solutions, as discussed in the previous section and highlighted in the figure, is the importance of greater detail in ML algorithm publications to promote transparency and reproducibility. Providing thorough information on the dataset utilized, standardizing essential algorithm parameters, and specifying the parameters of the network simulation environment—including scenario limitations—fosters a comprehensive understanding of the work's methodological foundation. Such detailed reporting enables researchers to faithfully replicate experiments, which in turn enriches the intellectual foundation of the scientific community and supports the iterative improvement of future work. Through these practices, the field can collectively build on a more transparent and reproducible body of ML-based resource allocation research, driving forward robust advancements in wireless communications.

The advancements in mobile communication networks, beyond 5G and into 6G, will be driven by cutting-edge technological trends, such as RIS, XR, ISAC, large-scale automation, quantum computing, and 3D coverage networks, alongside emerging communication technologies like molecular, quantum, THz, hologram, haptic, and VLC. Together, these innovations promise to elevate the capacity for new hardware



Fig. 9. Roadmap of future directions for resource allocation optimization in wireless communications.

integration and novel communication methods, laying the groundwork for high-performance computing systems that can support more complex and efficient ML solutions for resource allocation optimization. Unlike current 5G networks, which are constrained by more limited data rates, latency, and computational power, these advancements will enable enhanced ML algorithms to operate within more dynamic and data-intensive environments, optimizing resource allocation in real-time with greater precision.

For instance, quantum technology, beyond enabling new applications in 6G, such as ultra-secure real-time communication between autonomous vehicles, will advance quantum computing capabilities, allowing increasingly faster ML algorithm processing times and empowering the deployment of

increasingly complex algorithms to operate effectively, thereby enhancing resource allocation optimization in wireless communications and improving network performance [286].

However, the increased capabilities also introduce new challenges, as the diversity and intensity of resource demands across applications like XR and holographic communication will require ML solutions to be both adaptive and scalable to manage unprecedented volumes and varieties of data. By leveraging high-performance computing and innovative hardware, the next generation of wireless communication networks will enable ML-based resource allocation to reach new heights in efficiency, paving the way for future advancements in dynamic and intelligent network management.

Additionally, future wireless communication networks bring, alongside emerging technological trends and communication technologies, new applications and use cases with increasingly challenging requirements. Among these, we can highlight, for example, autonomous transport and delivery, remote surgeries, holographic meetings, and 3D environmental monitoring [287].

Unlike the key services in 5G, whose requirements are often more segmented and specific—such as the emphasis on latency in uRLLC or bandwidth in eMBB—the new use cases in 6G bring a substantial increase in complexity, further elevating the demands of 5G services (e.g., uRLLC+, eMBB+ and mMTC+), and combining them with new types of services, such as AI and sensing. These applications require a broader set of distinct services and requirements and consequently demand optimization across nearly all KPIs (e.g., PowCon, throughput, latency, NTM, and CRM) to ensure perfect execution.

Considering 5G networks, the remote control of robots in automated factories exemplifies a use case for the uRLLC service, where ultra-low latency communication is essential to ensure real-time responsiveness and precision [288], [289]. In automated manufacturing, the coordination of robotic actions is crucial to maintaining seamless production, as any delay could disrupt operations and affect quality. uRLLC provides the necessary ultra-reliable, low-latency communication to execute commands instantaneously, making latency the primary KPI in this scenario. Additionally, NTM plays a vital role in prioritizing critical control data, ensuring uninterrupted operations. Meanwhile, PowCon and CRM are of lesser concern, as the factory environment can accommodate higher energy consumption and localized computational resources.

For the eMBB service in 5G, 4K live streaming serves as a fitting application, particularly in large public gatherings such as concerts or festivals where high-definition coverage is in demand both on-site and remotely [290], [291]. eMBB enables the high throughput required to deliver uninterrupted, high-quality video to a large audience, ensuring a seamless and immersive experience. In this context, throughput is the primary KPI, as it ensures the continuous transmission of 4K content without buffering or interruptions. NTM is also crucial for managing substantial data flow and maintaining stable connections in congested settings, while latency and CRM are of secondary importance, given the emphasis on data capacity and quality over real-time responsiveness or CE.

Finally, considering mMTC in fifth-generation mobile networks, environmental monitoring with IoT sensors in smart farming exemplifies a valuable use case [292], [293]. In agricultural settings, IoT sensors are distributed over large areas to track soil conditions, temperature, humidity, and other environmental factors. These low-cost, resource-constrained sensors depend on mMTC to support a high density of connections while conserving energy for autonomous, long-term operations in remote locations. For this use case, PowCon is a critical KPI, as prolonged functionality without maintenance is essential. NTM is equally important to prevent network congestion and ensure timely data transmission, while CRM supports the efficient management of limited computational resources within an IoT network, facilitating data processing at the edge and reducing transmission demands.

As previously discussed in this section, to meet future 6G applications, sixth-generation network use cases are expected to integrate multiple services, thereby requiring the simultaneous and efficient fulfillment of multiple performance requirements. Some examples of 6G use cases illustrated in Figure 9 include remote surgery with real-time feedback, collaborative autonomous vehicles, and interactive meetings.

Remote surgeries with real-time haptic feedback, enabled by advanced 6G capabilities, require ultra-responsive, precise control and instant feedback for tactile sensation, allowing surgeons to perceive textures and resistance through robotic interfaces [294], [295]. This use case necessitates extremely low latency to ensure immediate response, high throughput to support 8K or higher video resolution, and reliable connectivity for uninterrupted performance, along with sensing capabilities to capture environmental and tactile data and AI to support data processing and decision-making. Among the KPIs analyzed, PowCon is essential to maintain continuous and efficient device operation without frequent maintenance or battery replacement. Latency is also critical, as real-time response is indispensable to surgical precision. Throughput is vital to enable seamless high-definition video and haptic data transmission, while CRM is important for handling the intensive computational demands of real-time data processing. Finally, NTM is necessary to prioritize essential control and feedback data, ensuring uninterrupted surgical procedures.

In addition, in densely populated urban areas, collaborative autonomous vehicles are expected to interact with each other and with city infrastructure to enhance traffic flow, reduce accidents, and improve safety [296], [297]. Enabled by 6G networks, these vehicles communicate in real-time, exchanging high-definition environmental data and sensory inputs. This use case depends on ultra-low latency for synchronized actions, high throughput to handle substantial data volumes, and sensing to identify objects in the environment while AI processes this data to make instantaneous decisions. Among the KPIs for autonomous vehicles, PowCon is essential to maximize operational time for vehicles and urban sensors without frequent recharging. Latency is critical to ensure real-time responses among vehicles, preventing accidents and facilitating coordination. Throughput supports the seamless transfer of high-quality video and data between vehicles and infrastructure, while CRM enables efficient data processing

within each vehicle, reducing dependency on cloud resources. Finally, NTM prioritizes communication between vehicles and with infrastructure, minimizing potential delays from traffic congestion.

Lastly, 6G-enabled interactive holographic meetings represent an innovative 6G use case, allowing participants to appear as life-sized holograms in real-time, with facial expressions and gestures captured and transmitted for a highly immersive experience [298], [299]. This use case demands ultra-low latency to synchronize gestures and expressions, high throughput to transmit high-resolution holographic images, and sensing to capture environmental data, with AI enhancing hologram quality. In this context, PowCon is essential for prolonged use of augmented reality and holographic devices without frequent recharging. Latency is critical to synchronize gestures and expressions in real-time, preserving the natural flow of interaction. Throughput is required to transmit high-quality holographic images and real-time sensor data, ensuring an immersive experience. CRM is important for handling complex data processing across local and remote devices, maintaining the fluidity of the holographic interaction. Finally, NTM is necessary to prioritize holographic and sensor data, particularly when many users are connected simultaneously, preventing potential network congestion.

The examples of 5G use cases, when compared with those of 6G previously discussed and summarized in Figure 9, as well as the types of services and requirements associated with each use case, highlight the exponential increase in complexity needed to realize future wireless network applications. Consequently, we can also observe that, to overcome the challenges posed by these use cases and maximize the potential of emerging technologies, dynamic resource allocation optimization in mobile communication networks stands as a crucial pillar to be explored. Moreover, ML solutions can significantly contribute to achieving the desired outcomes.

In summary, the roadmap outlined here serves as a vital guide for the scientific community, highlighting the need for more detailed reporting of ML algorithms in scientific publications to foster reproducibility and continuous learning. By setting standards for transparency regarding datasets, algorithm parameters, and simulation environments, this roadmap empowers researchers to build on each other's work, enriching collective knowledge and advancing resource allocation optimization methods. Additionally, the integration of new 6G technological trends and communication technologies, such as RIS, THz communication, and quantum computing, will enable the deployment of increasingly sophisticated ML algorithms on advanced hardware platforms. These advancements, however, come with new challenges, as emerging use cases and applications, from autonomous transport to holographic meetings, will demand increasingly complex and efficient resource allocation optimization across multiple performance metrics. This roadmap thus underscores the essential role of robust, adaptive solutions in multiple resource allocation optimization, positioning the wireless communications field to meet the rigorous demands of future networks and drive forward transformative innovations in connectivity.

VIII. Conclusion

In conclusion, this survey delved into the realm of ML applied to the resource allocation optimization in wireless communications. The efficient allocation of critical resources, such as power, time, frequency, computational capacity, and space, and consequently the optimization of various parameters taken into consideration as network KPIs to be analyzed, including PowCon, throughput, latency, NTM, and CRM, was meticulously investigated. The extensive survey encompassed diverse applications, learning paradigms, distributed learning approaches, algorithms, performance evaluation metrics, and gains categorized as EE, SE, LE, PDE, and CE.

The synthesis of data extracted from the surveyed papers yielded invaluable insights and discussions. These included a comprehensive overview of contemporary research, the potential to optimize multiple resources within a wireless network, statistical and qualitative analysis of learning paradigms highlighting the trade-offs observed among KPIs, distributed learning approaches, and performance gains achieved for each network KPI analyzed in the wireless communication systems. A taxonomy of ML algorithms was devised, shedding light on the algorithms most frequently employed for each learning paradigm and elucidating the rationale behind their selection, advantages, and disadvantages in comparison to alternative strategies, in addition to their suitability for different application scenarios.

Our survey also undertook a thorough review of experimentation frameworks, dataset analysis, and the gaps in the parameters reported for the ML algorithms used. The objective was to identify the tools most commonly employed in designing experimentation environments and input data for ML practices in wireless resource allocation optimization while also gaining insights into how the transparency of ML solutions can be enhanced to ensure reproducibility. Such an approach aims to foster a more consistent understanding of existing methodologies, enabling the scientific community to identify gaps or even explore potential standardization methods for datasets, thereby facilitating fair comparisons between different ML techniques. Furthermore, this thorough analysis highlighted the importance of improving experimentation setups to better approximate real-world scenarios, given the inherent sensitivity of wireless networks to environmental conditions.

The previously mentioned findings provided the foundation for an exploration of limitations, challenges, and future directions across various facets. Notably, the impact of transparency in research publications emerged as a key consideration, emphasizing the need for a more detailed exposition of the developed ML algorithms. The availability of technological resources and hardware limitations also emerged as key factors, while future trends contemplated the forthcoming wave of mobile network technologies, novel technologies, and applications.

From this analysis, a roadmap was charted, delineating future directions for resource allocation optimization in wireless communications. The premise of this roadmap underscored the growing role of AI and ML techniques in light of the observed results and performance gains. The

trajectory envisioned that forthcoming mobile telecommunications networks, along with new technologies and hardware, coupled with the push for more transparent research detailing ML strategies, could catalyze the development of increasingly efficient ML algorithms. As hardware constraints potentially wane, this shift may redirect the focus from current prevalent algorithms toward more intricate approaches.

With technological advancements and the implementation of future trends—including quantum communication and computing, transmission at THz frequency scales, ISAC, and 3D coverage networks, among others—and consequently overcoming certain hardware limitations, not only will current 5G services become fully realized, but also the potential services and use cases of 6G will materialize.

In this context, AI, particularly in the domain of ML, will become indispensable for the effective optimization and orchestration of network resources. The dynamic allocation of multiple resource resources, which remains an underexplored avenue in research, holds significant potential to enhance the capabilities of telecommunications systems, enabling the fulfillment of increasingly stringent requirements, such as ultra-low latency, massive device connectivity, and unprecedented levels of EE. These advancements, coupled with the adaptability of next-generation networks, will ensure that society can fully leverage the potential of future applications, fostering innovation and addressing the multifaceted challenges posed by next-generation wireless systems.

In essence, the extensive review undertaken in this technical survey lays the groundwork for a future where AI and ML will play an increasingly pivotal role in resource allocation optimization for wireless communications. As we stand on the cusp of new mobile telecommunications networks and technologies, combined with enhanced transparency in research, we are poised to witness the evolution of highly efficient ML algorithms, ushering in a new era of dynamic resource allocation capable of catering to diverse scenarios and applications. The journey ahead promises innovation, efficiency, and adaptability in the ever-evolving landscape of telecommunications.

List of Acronyms

3C	Communication, Computing and Caching
5G	Fifth Generation of Wireless Communication Technology
6G	Sixth Generation of Wireless Communication Technology
A3C	Asynchronous Advantage Actor-Critic
A3FN	Asynchronous Federated Framework Network
ABR	Adaptive Bitrate
ACRL	Actor-Critic Reinforcement Learning
ACDRL	Actor-Critic Deep Reinforcement Learning
ACK	Acknowledgement
AdaGrad	Adaptive Gradient Algorithm
AI	Artificial Intelligence
AMAE	Asymmetric Mean Absolute Error
AMC	Automatic Modulation Classifier
ANFIS	Adaptive Neuro Fuzzy Inference System
ANN	Artificial Neural Network
AP	Access Point
API	Application Programming Interface
APSNR	Average Peak Signal-to-Noise Ratio
AR	Augmented Reality
ARIMA	Auto-Regressive Integrated Moving Average
AVC	Advanced Video Coding
B5G	Beyond fifth Generation of Wireless Communication Technology
BBU	Baseband Unit
BER	Bit Error Rate
B-KNN	Bagging with K-Nearest Neighbors
BLER	Block Error Rate
BPNN	Back-Propagation Neural Network
bps	bits per second
BS	Base Station
BW	Bandwidth
CART	Classification And Regression Tree
CC	Cloud Computing
CDF	Cumulative Distribution Function
CE	Computational Efficiency
CH-FBS	Cluster Head - Femto-cell Base Station
CIFAR	Canadian Institute for Advanced Research
CNN	Convolutional Neural Network
ConvLSTM	Convolutional Long Short-Term Memory
CPU	Central Processing Unit
CR	Cognitive Radio
C-RAN	Cloud-Radio Access Network
CRL	Collaborative Reinforcement Learning
CRM	Computational Resource Management
CRN	Cognitive Radio Network
CSI	Channel State Information
CTS	Clear to Send
CSMA/CA	Carrier Sense Multiple Access/Collision Avoidance
CUDA	Compute Unified Device Architecture
D2D	Device to Device
DAG	Directed Acyclic Graph
DARL	Deep Adaptive Reinforcement Learning
DASH	Dynamic Adaptive Streaming over HTTP
DB	Database
DCGAN	Deep Convolutional Generative Adversarial Network
DCN	Data Center Network
DCS	Digital Cellular System
DDPG	Deep Deterministic Policy Gradient
DIFS	Distributed Interframe Space
DL	Deep Learning
DL-ENN	Dynamic Learning-Enabled Neural Network
DNN	Deep Neural Network
DDQN	Double Deep Q-Network
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
DSRC	Dedicated Short Range Communications
EDA	Estimation of Distribution Algorithm

E-GSM	Extended Global System for Mobile Communications
EE	Energy Efficiency
eMBB	enhanced Mobile Broadband
eMBB+	enhanced Mobile Broadband+
ENN-CAWOA	ELMAN Neural Network-Chaotic Adaptive Whale Optimization Algorithm
ESN	Echo State Network
EUTRAN	Evolved Universal Terrestrial Radio Access Network
EXP4	Exploration and Exploitation using Expert Advice
FAP	Femtocell Access Point
FBS	Femto-cell Base Station
FedAvg	Federated Averaging
FEEL	Federated Edge Learning
FiWi	Fiber Wireless
FL	Federated Learning
FL-GR	Federated Learning with Gradient Recycling
FM	Frequency Modulation
FMCW	Frequency Modulated Continuous Wave
FNN	Feedforward Neural Network
F-RAN	Fog-computing-based Radio Access Network
F-RCNN	Faster-Region Convolutional Neural Network
FRST-SVM	Fuzzy Rough Set Theory - Support Vector Machine
FUE	Femto-cell User Equipment
FPGA	Field-Programmable Gate Array
FTN	Faster than Nyquist
FW	Framework
GA	Genetic Algorithm
GAN	Generative Adversarial Network
GBR	Gradient Boosting Regressor
GD	Gradient Descent
GenAI	Generative Artificial Intelligence
GERAN	GSM/EDGE radio access network
GPU	Graphics Processing Unit
GSM	Global System for Mobile Communication
GRU	Gated Recurrent Unit
H2H	Human-to-Human
HEVC	High Efficiency Video Coding
HSM	Hydraulic System Monitoring
HTTP	Hypertext Transfer Protocol
ID3	Iterative Dichotomizer 3
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
IESO	Independent Electricity System Operator
IoRT	Internet of Robotic Things
IoT	Internet of Things
IoUAV	Internet of Unmanned Aerial Vehicle
IoV	Internet of Vehicles
IR-LEOA	Integrated Remora with Lotus Effect Optimization Algorithm
IRS	Intelligent Reflecting Surface
ISAC	Integrated Sensing And Communication
IT	Information Technology
K-GMM	K-Means Gaussian Mixture Model
K-NN	K-Nearest Neighbors
KPI	Key Performance Indicator
LAA	License-Assisted Access
LabVIEW	Laboratory Virtual Instrument Engineering Workbench
LASSO	Least Absolute Shrinkage and Selection Operator
LDA	Linear Discriminant Analysis
LE	Latency Efficiency
LLM	Large Language Model
LMS	Least Mean Square
LoRa	Long Range
LoRaSim	Long Range Simulator
LoRaWAN	Long Range Wide Area Network
LOS	Line of Sight
LPWAN	Low-Power Wide-Area Network
LSTM	Long Short-Term Memory
LTE	Long-Term Evolution
LTE-A	Long-Term Evolution Advanced
M2M	Machine-to-Machine
MA	Memetic Algorithm
MAE	Mean Absolute Error
MANET	Mobile Ad Hoc Network
mAP	mean Average Precision
MAPE	Mean Average Percentage Error
MARL	Multi-Agent Reinforcement Learning
MATLAB	Matrix Laboratory
MC-NOMA	Multi-Carrier Non-Orthogonal Multiple Access
MCS	Modulation and Coding Scheme
MEC	Mobile Edge Computing
MGO	Mountain Gazelle Optimization
MIMO	Multiple-Input Multiple-Output
MISO-NOMA	Multiple-Input Single-Output - Non-Orthogonal Multiple Access
MIP	Mixed Integer Programming
ML	Machine Learning
MLP	Multilayer Perceptron
mMTC	massive Machine Type Communications
mMTC+	massive Machine Type Communications+
mmWave	millimiter Wave
MNIST	Mixed National Institute of Standards and Technology
MNO	Mobile Network Operator
MTC	Machine Type Communications
MOPTS	Million Operations Per Time Slot
MOS	Mean Opinion Score
MPE	Mean Percent Error
MR	Mixed Reality
MRT	Maximum Rate Transmission
MSAD	Mean Squared Absolute Difference
MSD	Mean Square Deviation
MSE	Mean Squared Error
MU	Malicious User
NAF	Normalized Advantage Function
NS	Network Simulator

NS-2	Network Simulator-2
NS-3	Network Simulator-3
NARNN	Non-linear Auto-Regressive Neural Network
NOA	Number of Optimizations per Article
NOMA	Non-Orthogonal Multiple Access
NFV	Network Functions Virtualization
NIST	National Institute of Standards and Technology
NLOS	Non-Line of Sight
NMSE	Normalized Mean Square Error
NN	Neural Network
NSGA-II	Non-dominated Sorting Genetic Algorithm II
NTM	Network Traffic Management
ODL	OpenDayLight
OFDM	Orthogonal Frequency-Division Multiplexing
OLT	Optical Line Terminal
ONU	Optical Network Unit
O-RAN	Open-Radio Access Network
PCA	Principal Component Analysis
PDE	Packet Delivery Efficiency
PDR	Packet Drop Ratio
PER	Packet Error Rate
PLR	Packet Loss Ratio
PowCon	Power Consumption
PR	Percentage Rate
PSD	Power Spectral Density
PSO	Particle Swarm Optimization
PU	Primary User
QoE	Quality of Experience
QoS	Quality of Service
RAM	Random Access Memory
RAN	Radio Access Network
RB	Resource Block
RBF	Radial Basis Function
ReLU	Rectified Linear Unit
RIS	Reconfigurable Intelligent Surface
RL	Reinforcement Learning
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
RQS	Rank based Quality Score
RRH	Remote Radio Head
RRM	Radio Resource Management
RRUI	Radio Resource Utilization Information
RSU	Road Side Unit
RTS	Request to Send
RTT	Round-Trip Time
RU	Resource Unit
SACRL	Soft Actor-Critic Reinforcement Learning
SARSA	State-Action-Reward-State-Action
SATCOM	Satellite Communications
SCMA	Sparse Code Multiple Access
SCN	Small Cell Network
SDN	Software-Defined Network
SD-IoUAV	Software Defined Internet of UAV
SDVN	Software Defined Vehicular Network
SDR	Software-defined Radio
SE	Spectral Efficiency
SER	Symbol Error Rate
SGD	Stochastic Gradient Descent
SINR	Signal-to-Interference-plus-Noise Ratio
Sklearn	Scikit-learn
SIC	Self-Interference Cancellation
SIFS	Short Interframe Space
SimDB	Simulation Database
SIR	Signal-to-Interference Ratio
SL	Supervised Learning
SMS	Short Message Service
SNDLib	Survivable Network Design Library
SNDZoo	Survivable Network Design Zoo
SNR	Signal-to-Noise Ratio
SOM	Self-Organizing Map
SS	Spectral Sensing
SSIM	Structural Similarity Index Measure
STAC	Sequential Twin-Actor Critic
STD	STandard Deviation
STK	Satellite Tool Kit
SU	Secondary User
SUMO	Simulation of Urban MObility
SVD	Singular Value Decomposition
SVDM	Software-defined Vehicular Networks
SVM	Support Vector Machine
SVR	Support Vector Regression
SW	Software
SWIPT	Simultaneous Wireless Information and Power Transfer
TDMA	Time Division Multiple Access
TF	TensorFlow
THz	Terahertz
TL	Transfer Learning
TNA	Total Number of Articles
UAV	Unmanned Aerial Vehicle
UCI	University of California Irvine
UDN	Ultra-Dense Network
UE	User Equipment
UHD	Ultra High Definition
UHF	Ultra High Frequency
UL	Unsupervised Learning
uRLLC	ultra-Reliable Low Latency Communications
uRLLC+	ultra-Reliable Low Latency Communications+
USRP	Universal Software Radio Peripheral
V2X	Vehicle-to-Everything
VANET	Vehicular Ad hoc Network
VLC	Visible Light Communication
VM	Virtual Machine
VNF	Virtual Network Function
VoIP	Voice over Internet Protocol
VQM	Video Quality Metric
VR	Virtual Reality
XR	Extended Reality
WBAN	Wireless Body Area Network
WLAN	Wireless Local Area Network
WR	Wireless Router
WSN	Wireless Sensor Network

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